

Unit - I

Fundamentals of graphs

Graphs - Introduction - isomorphism -
 Subgraphs - Walks - Paths - Circuits -
 Connectedness - Components - Euler graphs -
 Hamiltonian Paths & Circuits.

Graph

A Graph $G = (V, E)$ consists of a set of objects $V = \{v_1, v_2, \dots, v_n, \dots\}$ called vertices & another set $E = \{e_1, e_2, \dots\}$ whose elements are called edges such that each edge e_k is identified with an unordered pair of vertices (v_i, v_j) of vertices.

The vertices v_i & v_j associated with edge e_k are called the end vertices of e_k .

Self loop

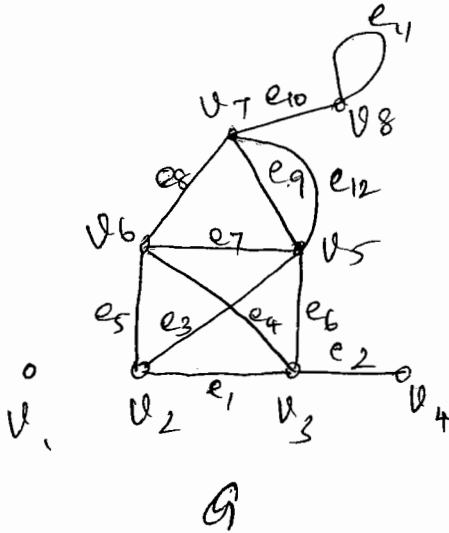
An edge to be associated with a vertex pair (v_i, v_i) .

i.e. An edge having the same vertex as both end vertices is called a self loop.

Simple Graph

A graph that has neither self loops nor parallel edges is called a simple graph.

Example: 1



In this example,

* The edges e_9 & e_{12} are Multiple edges.

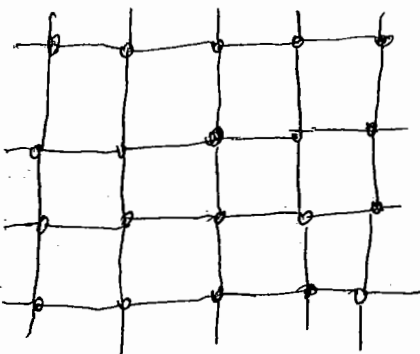
* The edge e_{11} is called as a self loop of this graph.

Finite & infinite graphs:

A graph with finite no. of vertices as well as finite no. of edges is called a finite graph otherwise it is an infinite graph.

Example: 2

* The above mentioned graph is finite



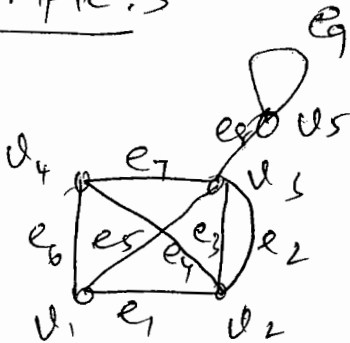
is known as infinite graph.

Incidence & adjacent:

* If a vertex v_i is an end vertex of some edge e_j then v_i & e_j are said to be incident to each other.

+ Two non $\parallel$ edges are said to be adjacent if they are incident on a common vertex.

Example: 3



In this graph,

* The edges e_6 & e_7 are incident with the vertex v_4

* Also the edges e_6 & e_7 are adjacent.

* The edge e_8 is incident with v_3 also with v_5 .

Degree of a vertex:

The no. of edges incident on a vertex v_i with self loops counted twice, is called the degree of the vertex v_i . & it is denoted by $d(v_i)$

from Example ①

$$d(v_1) = 0, \quad d(v_2) = 3, \quad d(v_3) = 4$$

Isolated vertex

A vertex having no incident edge is called as an isolated vertex.

i.e., A vertex with degree zero is called as an isolated vertex.

* In Example 1

$$d(v_1) = 0$$

$\therefore v_1$ is said to be an isolated vertex.

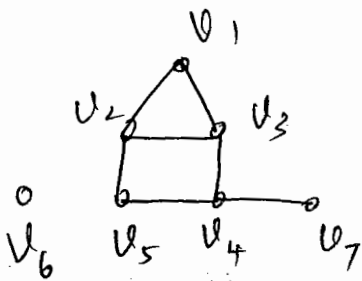
Pendant vertex:

A vertex of degree one is called a pendant vertex.

Null graph

A Graph without any edges is called as a null graph.

Example:



In this graph,

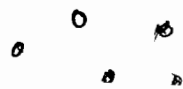
$$* d(v_6) = 0$$

$\therefore v_6$ is isolated vertex

$$* d(v_7) = 1$$

$\therefore v_7$ is pendant vertex.

For Null graph:

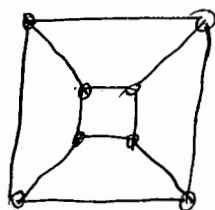


In this graph, we have

Regular graph

A graph in which all vertices are of equal degree is called a regular graph.

Example



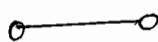
is 3-regular graph
Since each vertex is of degree-3.

Complete graph:

A simple graph in which there exists an edge between every pair of vertices is called a complete graph.

The complete graph of n vertices is denoted by K_n

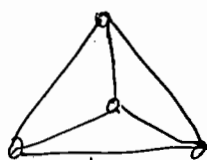
Examples



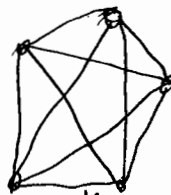
K_2



K_3



K_4



K_5

Complete graphs of two, three, four & five vertices.

Note

* We study only finite graphs

* $n(G)$ denotes the no. of vertices of G .

(or) n

* m (or) $m(G)$ denotes the no. of edges of G .

*

Trivial & non trivial graphs

A graph with just one vertex is called trivial & all other graphs are non trivial.

i.e., for trivial graph $|V(G)| = 1$

& for non trivial graph $|V(G)| \geq 2$.

Note: Throughout we consider,

(1) $\delta(G)$ denotes the minimum degree of vertices of G .

(2) $\Delta(G)$ denotes the maximum degree of vertices of G .

Theorem: 1

In a graph (undirected) G , the sum of the degrees of the vertices of a graph is equal to twice the number of edges.

$$\text{i.e., } \sum d(v_i) = 2e$$

Proof

* let G be a graph with e edges & n vertices $v_1, v_2, \dots, v_n$.

* let $e_k \in (v_i, v_j)$ & hence e_k is counted twice

* Each contributes two degrees.

* The sum of the degrees of all vertices in G is twice the no. of edges in G .

$$\therefore \sum_{i=1}^n d(v_i) = 2e$$

Theorem: 2

P.T The no- of odd degree vertices in a graph is always even.

Proof:

* let $V_1 = \{ \text{Set of odd degree vertices in } G \}$
 & $V_2 = \{ \text{Set of even degree vertices in } G \}$

* By theorem, we know that

$$\sum_{i=1}^n d(v_i) = 2e$$

$$\sum_{v_i \in V_1} d(v_i) + \sum_{v_j \in V_2} d(v_j) = 2e$$

where $V_1 \cup V_2 = V$

$$\sum_{v_i \in V_1} d(v_i) = 2e - \sum_{v_j \in V_2} d(v_j)$$

$$= \text{even}^{\text{no}} - \text{even no-}$$

$$= \text{even}$$

$$\therefore \sum_{v_i \in V_1} d(v_i) = \text{even}$$

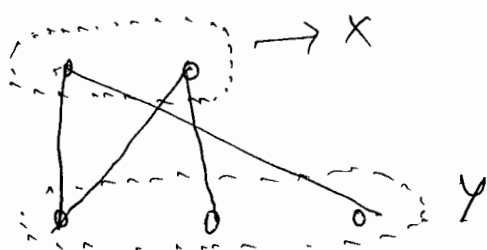
H.P.

Bipartite graph

A Bipartite graph is one whose vertex set can be partitioned into two subsets X & Y , so that each edge has one end in X & one end in Y .

Such a partition (X, Y) is called a bipartition of the graph.

Example



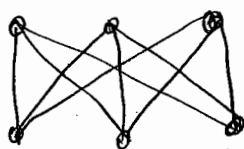
$$\text{Since } X \cup Y = V(G) \\ X \cap Y = \phi$$

Complete Bipartite graph

A complete bipartite graph is a simple ~~graph~~ bipartite graph with bipartition (X, Y) in which each vertex of X is joined to each vertex of Y .

If $|X|=m$ & $|Y|=n$ such a graph is denoted by $K_{m,n}$.

Example

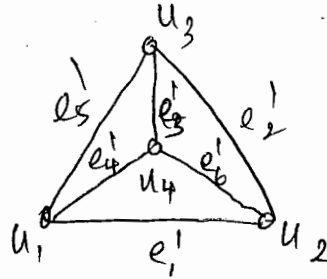
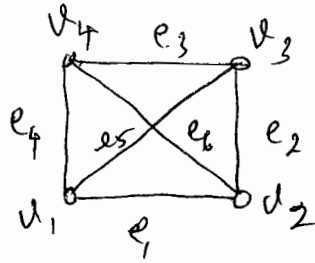


It is denoted by $K_{3,3}$

Identical Graphs:

Two graphs G & H are said to be identical, if $V(G) = V(H)$ & $E(G) = E(H)$ & it is written as $G = H$

Example:



Isomorphism

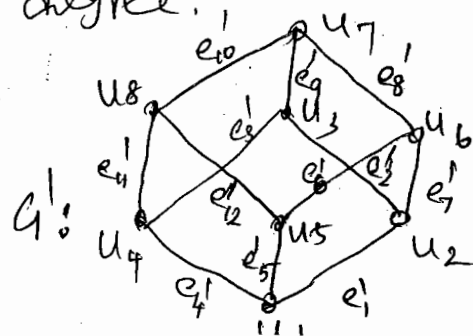
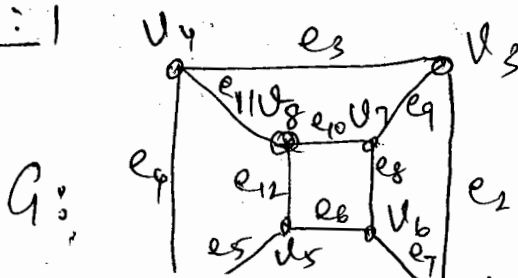
Two graphs G & G' are said to be isomorphic if there is a 1-1 correspondence between their vertices & edges such that the incidence relationship is preserved.

i.e., If ① They have same no/- of vertices & edges

② Incidence & adjacency relationship will be preserved in G' as in G

③ An equal no/- of vertices with given degree.

Example: 1



Let $\phi: G \rightarrow G'$ be the mapping s.t.

$$\phi(u_1) = u_1, \phi(u_2) = u_2, \phi(u_3) = u_3, \dots, \phi(u_8) = u_8$$

$$\& \phi(e_1) = e_1, \phi(e_2) = e_2, \dots, \phi(e_{12}) = e_{12}$$

$$i) |V(G)| = |V(G')|$$

$$ii) |E(G)| = |E(G')|$$

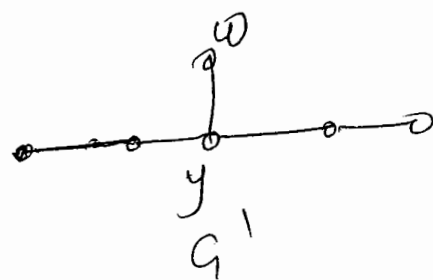
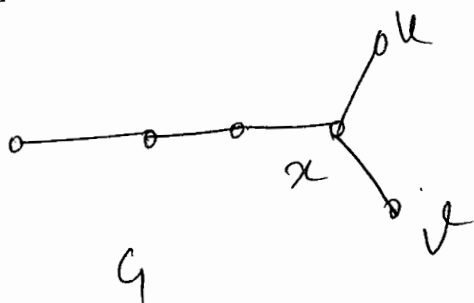
iii) Incidence & adjacency relationship preserved.

iv) An equal no- of vertices with a given degree.

$\therefore G$ & G' are isomorphic.

$$\text{i.e., } G \cong G'$$

Example: 2



$$i) |V(G)| = |V(G')|, \text{ ii) } |E(G)| = |E(G')|$$

iii) Incidence & adjacency relationship is not there. Since there is only one pendant vertex w , adjacent to y in G' , while in G there

are two pendant vertices u & v are adjacent to x in G .

Subgraph

A graph H is a subgraph of G written as $H \subseteq G$ if $V(H) \subseteq V(G)$ & $E(H) \subseteq E(G)$ & each edge of H has the same end vertices in H as in G .

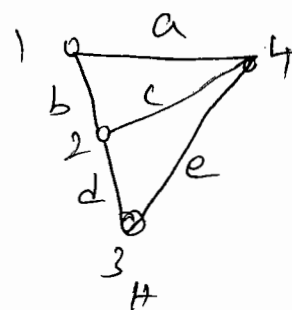
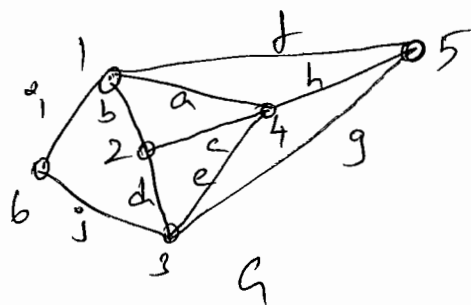
Proper Subgraph:

When $H \subseteq G$ but $H \neq G$, & we call H as a proper subgraph of G .

Supergraph

If H is a subgraph of G then G is a Supergraph of H .

Example



Here H is a subgraph of G

also H is a proper subgraph of G .
& G is a supergraph of H .

Note:

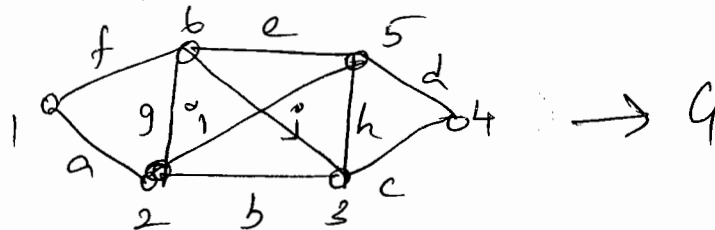
- ① Every graph is its own subgraph
- ② A subgraph of a subgraph of G is a subgraph of G
- ③ A single vertex in a graph G is a subgraph

④ A single edge in G , together with its end vertices is also a subgraph of G .

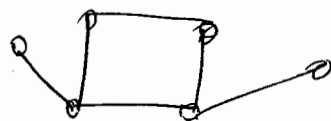
Spanning Subgraph:

A Spanning Subgraph of G is a subgraph H with $V(H) = V(G)$

Example



$\& H:$



$\rightarrow$ Spanning Subgraph

underlying simple graph

By deleting all the self loop & parallel (or) Multiple edges but one link joining every pair of adjacent vertices, we obtain a simple Spanning Subgraph of G , & it is called as the underlying simple graph of G .

Disjoint Subgraphs

Let $G_1, \& G_2$ be the subgraphs of G . We say that $G_1, \& G_2$ are disjoint if they have no vertex in common

$$\text{i.e., } V(G_1) \cap V(G_2) = \phi$$

Edge disjoint Subgraphs:-

Let G_1 & G_2 be subgraphs of G . We say that G_1 & G_2 are edge-disjoint if they have no edge in common.

$$\text{i.e., } E(G_1) \cap E(G_2) = \phi$$

Note:

If $V(G_1) \cap V(G_2) = \phi$ then $E(G_1) \cap E(G_2) = \phi$.
Hence if G_1 & G_2 are disjoint then G_1 & G_2 are edge disjoint but not conversely.

Operations on graphs

Union of two graphs

Let $G_1 = (V_1, E_1)$ & $G_2 = (V_2, E_2)$ be the graphs. Then the union of two graphs G_1 & G_2 is denoted by $G_1 \cup G_2 = G_3$ with $V(G_1) \cup V(G_2)$ & $E(G_1) \cup E(G_2)$
i.e., $V_1 \cup V_2$ & $E_1 \cup E_2$

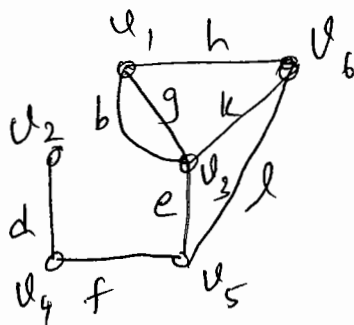
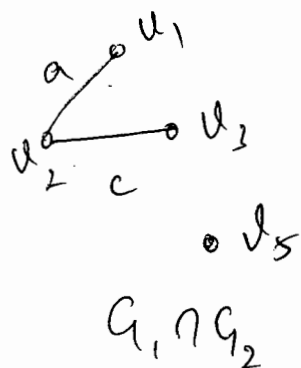
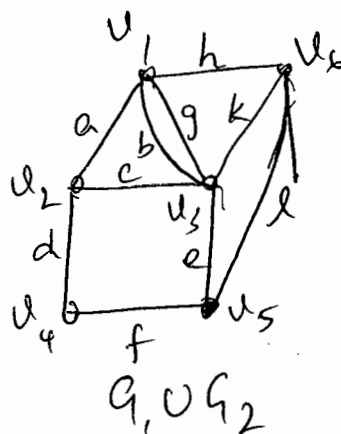
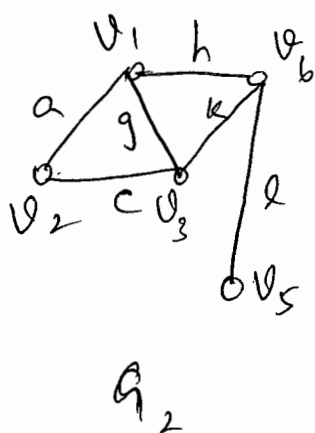
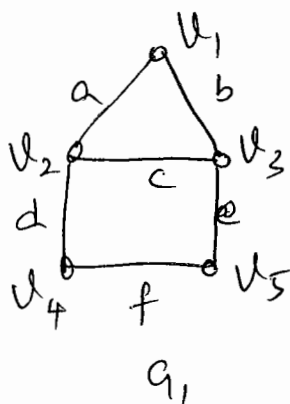
Intersection of two graphs:-

The intersection of two graphs $G_1 = (V_1, E_1)$ & $G_2 = (V_2, E_2)$ is denoted by $G_3 = G_1 \cap G_2$ with $V_1 \cap V_2$ & $E_1 \cap E_2$.

Ring Sum of two Graphs

The ring sum of two graphs $G_1 = (V_1, E_1)$ & $G_2 = (V_2, E_2)$ is denoted by $G_1 \oplus G_2$ & it is a graph consisting of the vertex set $V_1 \cup V_2$ & of the edges that are either in G_1 or G_2 but not in both.

Examples



Note:

① If G_1 & G_2 are disjoint Then

i) $G_1 \oplus G_2 = G_1 \cup G_2$

ii) $G_1 \cap G_2$ is a null graph.

② For any graph G , $G \oplus G =$ a null graph.

walk

A walk is defined as a finite alternating sequence of vertices & edges beginning & ending with vertices, such that no edge is incident with the vertices preceding & following it. No edge appears more than once in a walk. But a vertex may appear more than once.

Note: A walk is also referred to as an edge trail or a chain.

Terminal vertices:

A walk begins & ends with the vertices. Such vertices are called as terminal vertices.

Closed walk:-

A walk which begins & ends at the same vertex is called closed walk.

Open walk:-

A walk which is not closed is called an open walk.

i.e., If the terminal vertices are different that walk is called as open walk.

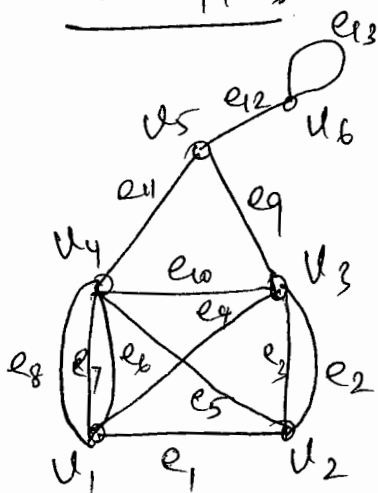
Path:

In an open walk if no vertex appears more than once then that open walk is said to be a path.

Circuit(or) cycle

In a closed walk if no vertex appears more than once then that closed walk is called as a circuit(or) cycle.

Example:



walk: $v_1, e_1, v_2, e_5, v_4, e_{10}, v_3, e_4, v_1, e_7, v_4$

open walk: $v_1, e_1, v_2, e_5, v_4, e_{10}, v_3, e_4, v_1, e_7, v_4$

closed walk: $v_1, e_1, v_2, e_5, v_4, e_{10}, v_3, e_4, v_1, e_7, v_4, e_8, v_1$

Path: $v_1, e_1, v_2, e_5, v_4, e_{10}, v_3$

Circuit: $v_1, e_1, v_2, e_5, v_4, e_{10}, v_3, e_4, v_1$

length of the Path:

The no. of edges in the path is called length of the path.

Note:

Every path is a walk.

Connectedness

(11)

Connected Graph

A Graph G is said to be connected if there is atleast one path between every pair of vertices in G .

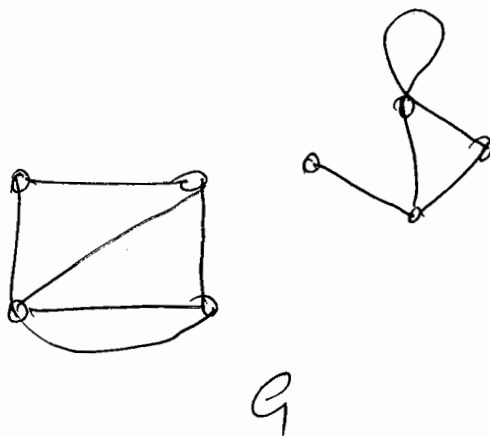
disconnected Graph :

A Graph G is not connected is said to be disconnected.

Components

A disconnected Graph consists two or more connected graphs. Each of these connected subgraphs is called a component.

Example:



Since the Graph G is disconnected & it has 2 components

Note:

- ① If a Graph G has exactly one component then G is connected otherwise G is disconnected
- ② The no. of components in a disconnected

Graph is denoted by $w(G)$

③ For connected graph $w(G) = 1$
& for disconnected graph $w(G) \geq 2$

④ The complement of a connected graph is not connected

Theorem:
In a graph G , any $u-v$ walk contains a $u-v$ path.

Proof
* we prove the result by induction on the length of the walk.

* let $P(n)$ denotes the stt.

* Any walk of length 0 (or) 1 is obviously a path

* $P(1)$ is true & let us assume that $P(k)$ is true.

* let $u = v_0, v_1, \dots, v_k = v$ be the $u-v$ walk contains $u-v$ path

* To prove: $P(k+1)$ is true.

i.e., $u = v_0, v_1, \dots, v_k, v_{k+1} = v$ be the $u-v$ walk contains $u-v$ path.

* If all the vertices of the walk are distinct, then the proof is complete.

* If not, then there exists $i \neq j$

* Consider $u = v_0, v_1, v_2, \dots, v_i, v_{i+1}, \dots, v_{k+1} = v$
 is a $u-v$ walk of length less than $k+1$

By induction hypothesis,
 Any $(u-v)$ walk contains a (u,v) path.

Theorem:

A graph G is disconnected if and only if
 its vertex set V can be partitioned into two
 non empty, disjoint subsets V_1 & V_2 such that there
 exists no edge in G whose one end vertex is in
 subset V_1 & the other in subset V_2 .

Proof:

- * let us suppose that such a partition exists
- * Consider two arbitrary vertices u & v of G
 s.t. $u \in V_1$ & $v \in V_2$
- * No path can exist between the vertices u & v
- * otherwise, there would be at least one edge whose one end vertex would be in V_1 & other in V_2 which is a contradiction
- * If a partition exists, then G is disconnected.

* Conversely, assume that G be a disconnected graph.

- * Consider a vertex u in G .
- * let V_1 be the set of all vertices that are connected by paths to u .
- * Since G is disconnected.
- * V_1 does not include all vertices of G .
- * The remaining vertices will form a set V_2 .
- * Hence the partition.

Theorem:

If a graph has exactly two vertices of odd degree, there must be a path joining these two vertices.

Proof:-

* let u & v be the two odd degree vertices of G .

* Case-1 : Suppose G is connected.

* Then by the definition of Connectedness, there exists a $u-v$ path.

* Case-2 : Suppose G is disconnected.

Subcase-1:-

* let u & v belongs to the same component of G

* Since each component is connected & hence the result.

* Subcase - 2
 * Suppose $u \notin V$ belongs to two different components of G .
 i.e., $u \in V_1$ & $v \in V_2$ Which is a contradiction.

* If a graph has exactly two vertices of odd degree then there must be a path joining these two vertices.

Theorem:
 A simple graph with n vertices & k components can have at most $(n-k)(n-k+1)/2$ edges.

Proof:

* Let G be a simple graph with n vertices
 * & let $G_1, G_2, \dots, G_w$ be the components of G

* Let the no. of vertices in each of the w components of G be $n_1, n_2, \dots, n_w$.

* Here, $n_1 + n_2 + \dots + n_{i-1} + n_i + n_{i+1} + \dots + n_w = n$

* Here, $n_i = n - (n_1 + n_2 + \dots + n_{i-1} + n_{i+1} + \dots + n_w)$

also $\sum_{i=1}^w n_i = n_1 + n_2 + n_3 + \dots + n_{i-1} + n_i + n_{i+1} + \dots + n_w$

$\sum_{i=1}^w (n_i - 1) = n_1 - 1 + n_2 - 1 + \dots + n_{i-1} - 1 + n_i - 1 + n_{i+1} - 1 + \dots + n_w - 1$
 $= n_1 + n_2 + \dots + n_{i-1} + n_i + n_{i+1} + \dots + n_w - w$

$$* \sum_{i=1}^{\omega} (n_i - 1) = n - \omega \quad \text{--- (1)}$$

$$* \text{ clearly } n_i \leq n - \omega$$

$$n_i \leq n - \omega + 1$$

* Since G is a simple graph & each G_i is connected for all i ,

$$\therefore m(G_i) \leq \frac{n_i(n_i - 1)}{2}, \quad 1 \leq i \leq \omega$$

$$\text{Since } m(G) = m(G_1) + m(G_2) + \dots + m(G_\omega)$$

$$* m(G) \leq \sum_{i=1}^{\omega} \frac{n_i(n_i - 1)}{2}$$

$$\leq \frac{(n - \omega + 1)}{2} \sum_{i=1}^{\omega} (n_i - 1)$$

$$m(G) \leq \frac{(n - \omega + 1)(n - \omega)}{2} \quad (\text{from (1)})$$

* Thus a simple graph with n vertices & ω components can have at most $\frac{(n - \omega)(n - \omega + 1)}{2}$ edges.

Euler graphs

* Euler line:

A closed walk in a graph contains all the edges of the graph, then the walk is called an Euler line.

* Euler graph (Eulerian)

A graph which contains an Euler line is known as Euler graph.

Theorem:

A connected graph G is Eulerian iff all vertices of G are of even degree.

Proof:

* Necessary part:

* Suppose G is Eulerian.

* Then G contains an Euler line.

* Let C be an Euler line of G with origin & terminus u .

* Each time a vertex v occurs as an internal vertex of C .

* Two of the edges incident with v are accounted

(i.e., one going ~~from~~ the vertex & another leaving)

- * Since an Euler line contains every edge of G , $d(v)$ is even for all $v \neq u$.
- * Similarly, C starts and ends at u , $d(u)$ is also even.
- * Thus all vertices of G are of even degree.
- * Proof of Sufficient Part:
 - * Suppose that G is connected & its vertices all have ~~even~~ degree.
 - * If there is more than one vertex in G , then each vertex must have degree greater than 0.
 - * Begin at a vertex v .
 - * Clearly v will be on at least one cycle.
 - * Since G is connected, there must be an edge (v, v_1) for some vertex $v_1 \neq v$.
 - * Since v_1 has even degree greater than 0, there is an edge (v_1, v_2) where $v_2 \neq v_1$.
 - * These two edges make a trail from v to v_2 .
 - * Continue this ^{walk} (trail), leaving each vertex on an edge that was not previously used, until returning to v .
 - * This is always possible because v is on a cycle.
 - * The circuit formed by this

- * If C_1 covers all the edges of G , then the proof is complete.
- * Otherwise, remove all the edges that contribute to C_1 from G , leaving the graph G_0 .
- * The remaining vertices are still even, and since G is connected there is some vertex u in both G_0 & C_1 .
- * Repeat same process as before, beginning at u .
- * The new circuit, C_2 can be added to C_1 by starting at v , moving along C_1 to u , travelling around C_2 back to u and then along the remaining of C_1 back to v .
- * Repeat this process, adding each new circuit found to create a larger circuit.
- * Since G is finite, this process must end at some point, & the resulting circuit will be an Eulerian circuit.

Unicursal Graph:

An open walk which includes all edges of a graph exactly once is called as a unicursal line (or) an open Euler line.

A connected graph that has a unicursal line will be called as unicursal graph.

Note:

In an unicursal graph by adding an edge between the initial & final vertices we shall get an Euler graph.

Theorem:

A Connected graph G is unicursal iff it has exactly two vertices of odd degree.

Proof:

- * Suppose that G is unicursal.
- * Then G contains an Open Euler line.
- * By Theorem, each vertex other than origin & terminus of this walk has even degree.

Conversely,

- * Assume that G is a connected graph with at most two vertices of odd degree.
- * Let u & v be the two odd degree vertices of G .

* Let $G+e$ denotes the graph obtained from G by the addition of a new edge e joining u & v .

* Clearly, each vertex of $G+e$ has

* $\therefore G + e$ has an Euler walk C with
 $C = u e_1 v_1 e_2 v_2 \dots e_k v_k e_{k+1} u \in u$.

* The walk C is an Euler walk of G

* H.P

Theorem:

In a connected graph G with exactly $2k$ odd vertices, there exist k edge-disjoint subgraphs such that they together contain all edges of G & that each is a unicursal graph.

Proof:

* let the odd vertices of the given graph G be named as $v_1, v_2, \dots, v_k$, $w_1, w_2, w_3, \dots, w_k$ in any arbitrary order.

* Add k edges to G between the vertex pairs $(v_1, w_1), (v_2, w_2), \dots, (v_k, w_k)$ to form a new graph G' .

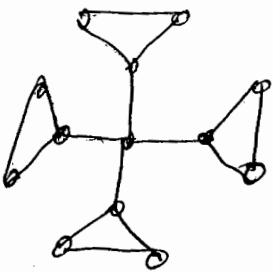
* By adding k edges between all the pairs, we will get k cycles.

* Since every vertex of G' is of even degree.

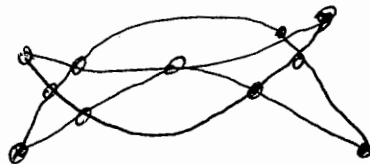
we can find an Euler line P .

- * Now if we remove from P the k edges we just added (no two of these edges are incident on the same vertex).
- * Now P will be split into k walks, each of which is a unicursal line.
- * The first removal will leave a single unicursal line.
- * The second removal will split that into two unicursal lines.
- * and each successive removal will split a unicursal line into two unicursal lines, until there are k of them.
- * Hence proved.

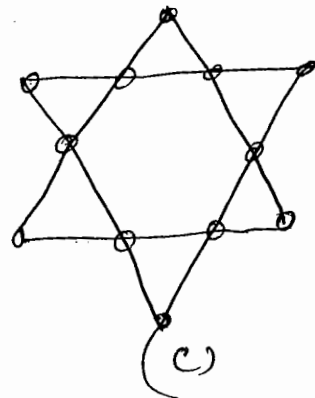
Examples for Euler graph:



(1)

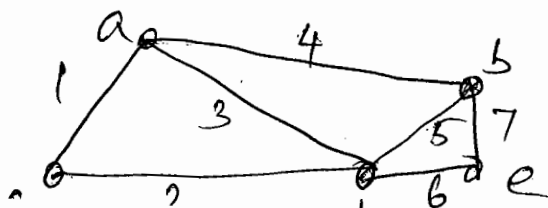


(b)



(c)

Examples for unicursal graph



Decomposition:

A decomposition of a graph G is a set of subgraphs $H_1, H_2, \dots, H_k$ that partition the edges of G .

$$\text{i.e. } G = \bigcup_{i=1}^k H_i$$

$$\& \text{ } E(H_i) \cap E(H_j) = \emptyset \text{ for all } i \neq j$$

Theorem:

A connected graph G is an Euler graph iff it can be decomposed into circuits.

Proof:

* Suppose G can be decomposed into circuit.

* G is a union of edge disjoint circuits.

* Since the degree of every vertex in a circuit is two the degree of every vertex in G is even.

* By theorem, G is Eulerian.

Conversely,

* Assume G is an Eulerian

* By theorem, each vertex of G has

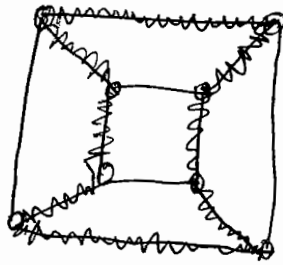
- * Consider a vertex v_1 .
- * There are at least two edges incident at v_1 .
- * let one of these edges be between v_1 & v_2
- * Since v_2 is also of even degree, it must have another edge between v_2 & v_3 .
- * Proceeding in this way, we arrive the vertex v_1 , this forming a circuit, called C .
- * let us remove C from G .
- * All vertices in the remaining graph must also be of even degree.
- * From the remaining graph remove another circuit in exactly the same way as we removed C in G .
- * Continue this process until no edges are left.
- * Hence the theorem.

Hamiltonian Paths & Circuits

Hamiltonian circuit:

~~Another~~ A Hamiltonian circuit in a connected graph is defined as a closed walk that traverses every edge of the graph exactly once.

Example



→ Hamiltonian circuit exist

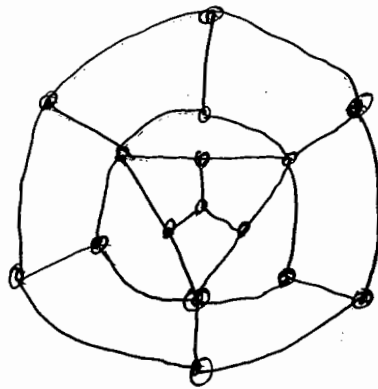
→ Hamiltonian Graph

Hamiltonian Path:

The removal any one edge from a hamiltonian circuit we will get a path which is known as hamiltonian path.

i.e., "An open walk in a connected graph which covers all the vertices of G exactly once."

Example:



Graph with
hamiltonian path
& without
hamiltonian
circuit.

Note:

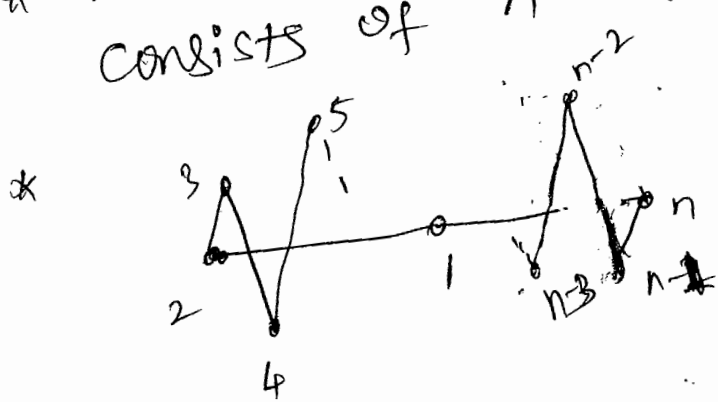
- ① A Hamiltonian graph of n vertices consists of exactly n edges.
- ② A Hamiltonian path of n vertices consists of exactly $(n-1)$ edges.

③ A graph is hamiltonian if it contains a hamilton cycle.

Theorem: In a complete graph with n vertices there are $(n-1)/2$ edge-disjoint hamiltonian circuits, if n is an odd number ≥ 3 .

Proof:
 * let G be a complete graph with n vertices & $\frac{n(n-1)}{2}$ edges.

* let C_n be a hamiltonian circuit in G consists of n edges.



* let us consider a hamiltonian circuit with n vertices & n edges which is given above

* keeping the vertices fixed on a circle, rotate the polygonal pattern clockwise by

$$\frac{360}{n-1}, 2 \times \frac{360}{n-1}, 3 \times \frac{360}{n-1} \dots \frac{(n-1)}{2} \times \frac{360}{(n-1)}$$

degrees.

* Each rotation produces a Hamiltonian circuit that has no edge in common with any of the previous ones.

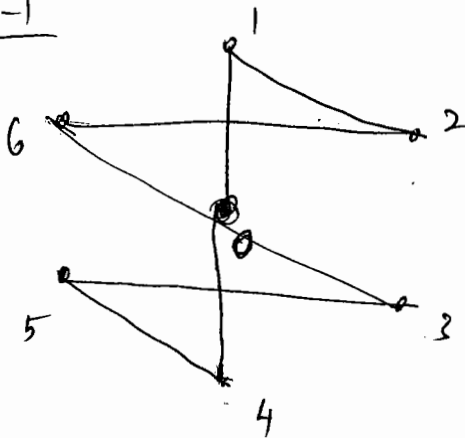
* Thus we have $\frac{(n-1)}{2}$ edge disjoint Hamiltonian circuits.

Note:

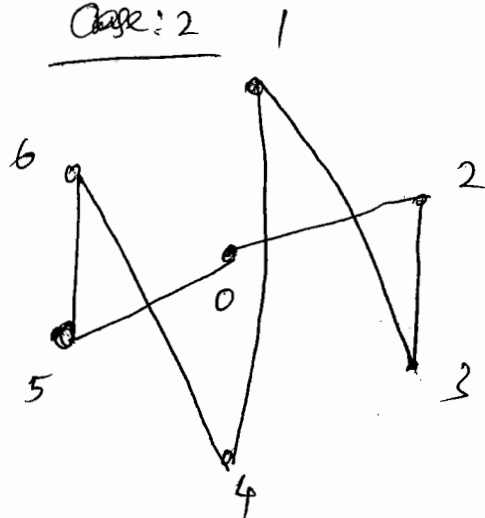
For Example

Let us consider a complete graph with 7 vertices

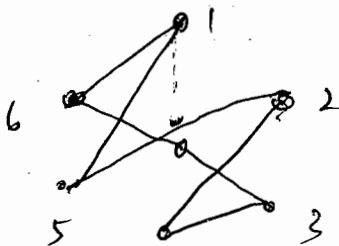
Case-1



Case: 2

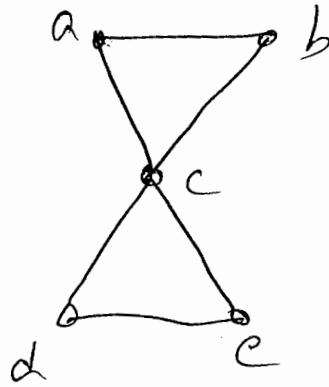


Case: 3



$\therefore$ for $n=7$ vertices
There will be $\frac{7-1}{2}=3$
Edge disjoint Hamiltonian
circuits.

Arbitrarily traceable graphs:



"G" - Euler Graph

In an Euler graph,

For an arbitrary vertex v ,

Starting from the vertex v and tracing all the edges exactly once but we unable to give a closed walk such graph is known as arbitrarily traceable graph.

Example In the above graph "G"
from c , If we start tracing
through a & we unable to trace all
the edges in a closed walk

$\therefore$ This graph is arbitrarily traceable
from c .

Note:

- ① Every Eulerian is not Hamiltonian graph
- ② Every Hamiltonian is not an Eulerian graph
- ③ The removal of one vertex from a complete graph remains complete.
- removal of any vertex in a regular

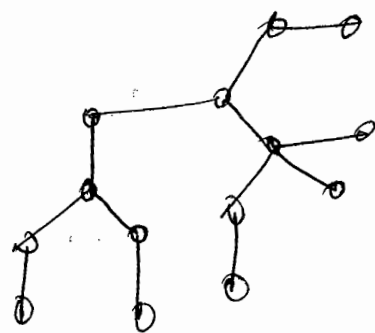
Syllabus: -

Trees, Fundamental circuits & Cutsets

Trees - properties of trees - distance & centers
 in tree - Rooted & binary trees - Spanning
 trees - Spanning trees in a weighted graph.
 Cutsets - properties of cutset - fundamental
 circuits & cutsets.

Tree A connected graph without any circuits
 is said to be a tree

Example:



tree

Note:

① Every tree is a
 simple graph

② A subgraph
 of a tree is
 also a tree.

③ The sorting of mail according to zip code
 & the sorting of punched cards are done
 according to a tree. Such kind of tree
 is known as "decision tree (or) sorting tree"

Properties of trees

Theorem: 1

P.T In a tree any two vertices are connected by a unique path.

Proof

- * let T be a tree
- * By definition, T is connected
∴ There exist atleast one path between every pair of vertices.
- * Suppose that, There exist two distinct uv paths P_1 & P_2 in T
- * $P_1 \cup P_2$ will contain a circuit
∴ $P_1 \cup P_2 - e$ is connected.
- * Which is a contradiction for a tree.
- * any two vertices are connected by a unique path.

Theorem - 2

It in a Connected Graph G there is one and only one path between every pair of vertices, then G is a tree.

Proof:

- * let G be a connected graph
- * By definition of Connectedness, any two vertices of G are connected by a path.
- * Assume G has one & only path between every pair of vertices.
- * If G contains no cycle then the proof is complete.
- * Suppose G contains a cycle C .
- * Then there is at least one pair of vertices $u \& v$ has two distinct paths which is a contradiction
- * G is a tree.

Theorem: 3

P.T. A tree with n vertices has $(n-1)$ edges.

Proof:

- * By induction hypothesis we will prove it.
- * Let $P(n)$ denotes the statement i.e., "A tree with n vertices has

* for $n=1$,

we have a tree with 1 vertex &
0 edge.

" "

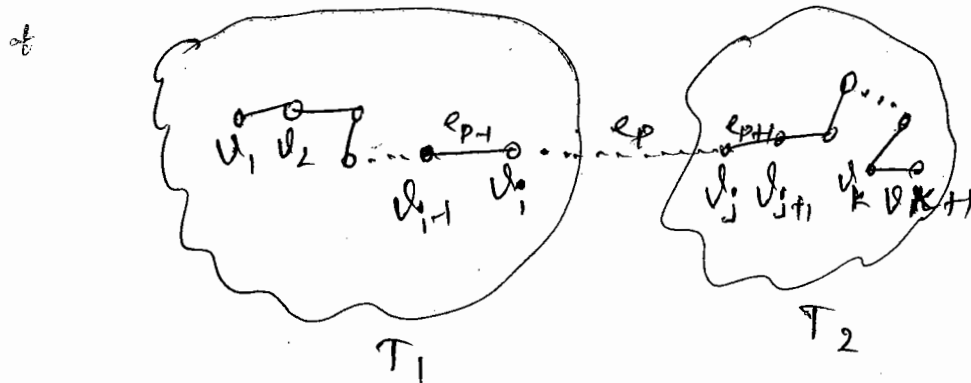
$\therefore P(1)$ is true.

* let us assume that $P(k)$ is true.

i.e., a tree with k vertices has $k-1$ edges.

* To Prove: $P(k+1)$ is true.

i.e., A tree with $k+1$ vertices has k edges.



Now
 T be a tree with $k+1$ vertices
& e_p is an edge $\in (v_i, v_j)$

* Remove $e_p \in (v_i, v_j)$ then the graph
is disconnected into two components
namely T_1 & T_2

* No/- of vertices in $T_1 = k_1$

No/- of edges in $T_1 = k_1 - 1$

(Since $k_1 < k$)

No/- of vertices in $T_2 = k_2$ where $k_2 < k$

No/- of edges in $T_2 = k_2 - 1$

2. Total No/- of Edges in T

$$= \text{No/- of Edges in } T_1 + \text{no/- of edges in } T_2$$

$$+ 1 \text{ (edge } e_p \in (v_i, v_j))$$

$$= k_1 + k_2 + 1$$

$$= k_1 + k_2 + 1$$

$$\left(\because k_1 + k_2 = k + 1 \text{ vertices} \right)$$

$$= k + 1 + 1$$

$$= k \text{ edges}$$

$\therefore P(k+1)$ is true.

$\therefore$ By the Principle of Mathematical Induction,

A tree with n vertices has $(n-1)$ edges.

Theorem: 4

P.T. A Connected Graph with n vertices & $(n-1)$ edges is a tree.

Proof:

* let G be a Connected Graph with n vertices & $(n-1)$ edges.

* To Prove: G is a tree

* Since Every Connected Graph contains a Spanning tree

* G Contains a Spanning tree T .

* Since T is a Subgraph of G

$$\therefore m(T) \leq m(G)$$

where $m(T) \rightarrow$ edges of T

$m(G) \rightarrow$ edges of G .

* Since T is a spanning Subgraph of G ,

$$n(T) = n(G)$$

where $n(T) \rightarrow$ no. of vertices in T

$n(G) \rightarrow$ " " in G .

* Since T is a Spanning tree

$$\therefore m(T) = n(T) - 1$$

$$\begin{aligned} * \text{ Hence } n(T) - 1 &= m(T) \leq m(G) = n(G) - 1 \\ &= n(T) - 1 \end{aligned}$$

* Thus $m(T) = m(G)$

* Therefore $G = T$.

* Hence G is a tree.

Minimally Connected

A connected graph ' G ' is said to be minimally connected graph if removal of any one edge from it disconnects the graph.

Theorem: 5

A graph is a tree if and only if it is minimally connected.

Proof:

- * let G be a tree.
- * By definition, G is connected.
- * To prove: G is ~~not~~ minimally connected.
- * Suppose that G is not minimally connected.
- * Then $\exists$ an edge e s.t. $G-e$ is connected.
- * e is in some circuit which implies G is not a tree.
- * G is minimally connected.

Conversely,

- * let G be minimally connected.
- * To prove: G is a tree.
- * If G cannot have a circuit then G is a tree.
- * otherwise, G contains a cycle C .
- * Remove one of the edges in C such that $G-e$ is connected.

Theorem: 6

P.T. A graph G with n vertices, $(n-1)$ edges & no circuits is connected.

Proof:

* let G be a graph with n vertices, $(n-1)$ edges & no circuits.

* Proof by Contradiction.

* Suppose G is disconnected

* let G consist of two components G_1 & G_2 such that $u \in G_1$ & $v \in G_2$

* Add an edge 'e' between u to v

* Since there was no path between u & v in G .

* Adding e in G did not create a circuit.

* Thus $G+e$ is a connected & acyclic graph with n vertices & n edges.

* which is a contradiction

* G is connected.

Note:

A tree with two or more vertices
... to be nontrivial tree.

Pendant vertices in a tree:

Theorem: 7

Every non-trivial tree has at least two vertices of degree one.

Proof

* let T be a nontrivial tree.

then $d_T(v) \geq 1$ for all $v \in V(T)$

* Also, $\sum_T d_T(v) = 2m(T)$

$$\& m(T) = n(T) - 1$$

$$\begin{aligned} \text{Hence } \sum_T d_T(v) &= 2m(T) \\ &= 2(n(T) - 1) \\ &= 2n(T) - 2 \end{aligned}$$

* If $d_T(v) \geq 2$ for all $v \in V(T)$

$$\begin{aligned} \text{Then } 2n(T) - 2 &= \sum_T d_T(v) \\ &= \{d(v) + d(v) + \dots + d(v)\} \\ &\quad \text{(n(T) times)} \\ &\geq (2 + 2 + \dots + 2) \text{(n(T) times)} \\ 2n(T) - 2 &\geq 2n(T) \end{aligned}$$

which is a contradiction.

* So there is a vertex, say x
such that $d_T(x) = 1$.

* If $d_T(v) \geq 2$ for all $v \neq x$ & $v \in V(T)$

$$\begin{aligned}\text{Then } 2n(T) - 2 &= \sum_{v \in V(T)} d_T(v) \\ &= 1 + \sum_{v \neq x} d_T(v)\end{aligned}$$

$$\geq 1 + (2 + 2 + \dots + 2)(n(T) - 1) \text{ times}$$

$$\geq 1 + 2n(T) - 2$$

$$2n(T) - 2 \geq 2n(T) - 1$$

which is a contradiction.

* So there is another vertex say y ,

$$y \neq x \text{ such that } d_T(y) = 1$$

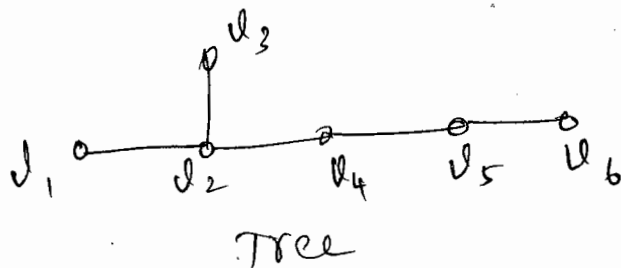
* Hence T has at least two vertices of degree one.

Distance & Centres in a tree

Distance:

In a Connected Graph G , the distance $d(u_i, u_j)$ between two of its vertices u_i & u_j is the length of the shortest path between them.

Example: -1



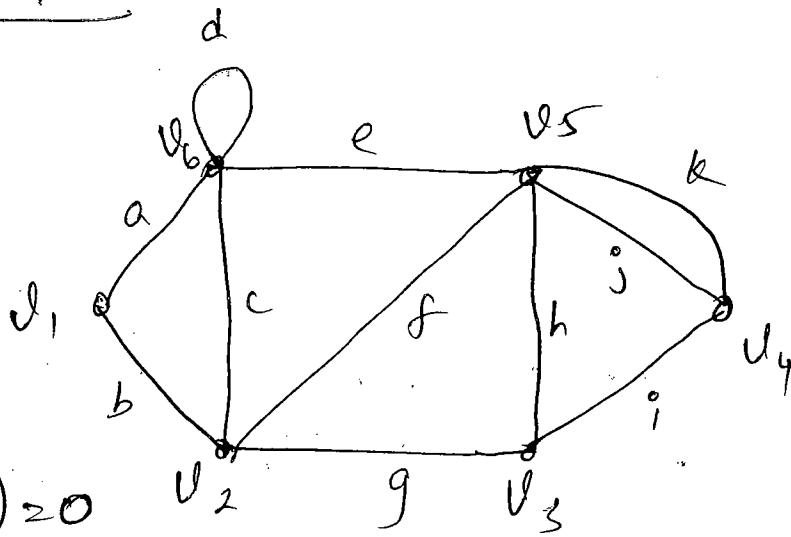
$$\begin{array}{lll} d(u_1, u_2) = 1, & d(u_1, u_5) = 3 & d(u_2, u_4) = 1 \\ d(u_1, u_3) = 2 & d(u_1, u_6) = 4 & d(u_2, u_5) = 2 \\ d(u_1, u_4) = 2 & d(u_2, u_3) = 1 & \text{and so on} \dots \end{array}$$

Note:

① If there is no path connecting u & v in G then

$$d(u, v) = \infty$$

Example: 2



$$d(v_1, v_1) = 0$$

$$d(v_1, v_2) = 1$$

$$d(v_1, v_3) = 2$$

$$d(v_1, v_4) = 3$$

$$d(v_1, v_5) = 2$$

$$d(v_1, v_6) = 1$$

$$d(v_2, v_3) = 1$$

$$d(v_2, v_4) = 2$$

and so on...

Theorem: 8

P.T. The distance between vertices of a connected graph is a metric.

Proof:

* $d(v_i, v_j) \geq 0$ Since by definition

$d(v_i, v_j) = 0$ if & only if $v_i = v_j$

* $d(v_i, v_j) = d(v_j, v_i)$

* Since $d(v_i, v_j)$ is the length of the shortest path between vertices v_i & v_j

* This path cannot be longer than another path between v_i & v_j which goes through

$$* d(u_i, u_j) \leq d(u_i, u_k) + d(u_k, u_j)$$

Hence Proved

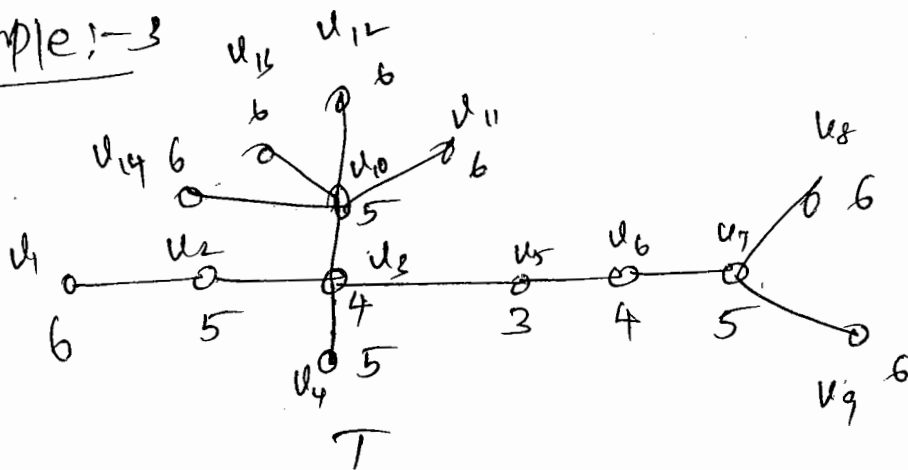
Eccentricity:

The eccentricity $E(u)$ of a vertex u in a graph G is the distance from u to the vertex farthest from u in G .

i.e.,

$$E(u) = \text{Max} \{ d(u, u_j) : u_j \in V(G) \}$$

Example:-3



$$\begin{array}{llll} E(u_1) = 6 & E(u_4) = 5 & E(u_7) = 5 & E(u_{10}) = 5 \\ E(u_2) = 5 & E(u_5) = 3 & E(u_8) = 6 & E(u_{11}) = 6 \\ E(u_3) = 4 & E(u_6) = 4 & E(u_9) = 6 & \text{and so on.} \end{array}$$

Radius

The minimum eccentricity of G is called Radius of G

$$\gamma(G) = \min \{ E(v) : v \in V(G) \}$$

Diameter:

The diameter of G is the maximum eccentricity of G .

2. It is denoted by

$$\text{diam}(G) = \max \{ E(v) : v \in V(G) \}$$

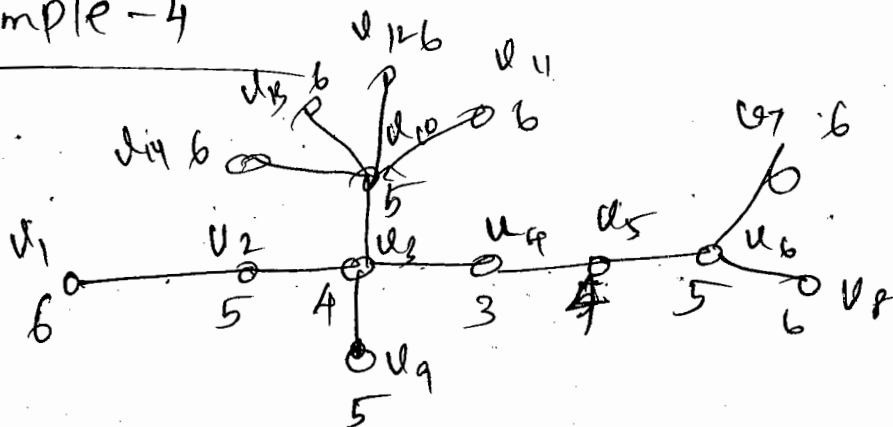
Central Vertex:

A vertex v of G is called central vertex if $E(v) = \gamma(G)$.

Center

The set of all vertices of G is called the center of G .

Example-4



In the above tree T ,

The eccentricities of the corresponding

$$E(v_1) = 6, E(v_2) = 5, E(v_3) = 4, E(v_4) = 3, \\ E(v_5) = 4, E(v_6) = 5, E(v_7) = 6 \dots$$

① Minimum Eccentricity is 3

$$\therefore r(G) = 3$$

② Maximum Eccentricity is 6

$$\text{Diam}(G) = \max(E(v_i)) = 6$$

③ Central vertex:

$$E(v_4) = r(G) = 3$$

$\therefore v_4$ is the central vertex

④

Center

v_4 is the center.

Theorem - 9

P.T. In a simple graph G , $r(G) \leq \text{diam}(G) \leq 2r(G)$

Proof:

* It is obvious from the definition that,

$$r(G) \leq \text{diam}(G)$$

* To prove: $\text{diam}(G) \leq 2r(G) \dots$

* Select vertices u & v in G s.t.

$$d(u, v) \geq \text{diam}(G)$$

* Let w be a central vertex of G

Hence

$$\begin{aligned} \text{diam}(G) = d(u, v) &\leq d(u, w) + d(w, v) \\ &\leq e(w) + e(w) \\ &= 2e(w) = 2r(G) \end{aligned}$$

$$\therefore \text{diam}(G) \leq 2r(G)$$

Examples

① For the complete graph K_n ,

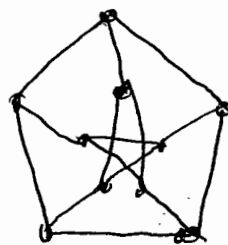
$$r(K_n) = \text{diam}(K_n) = 1$$

② For the complete bipartite graph $K_{m,n}$,

$$r(K_{m,n}) = \text{diam}(K_{m,n}) = 2.$$

③ For the Peterson graph,

$$r(P) = \text{diam}(P) = 2$$



Definition - Subtree

A Connected Subgraph of a tree T is a subtree of T .

Theorem - 10

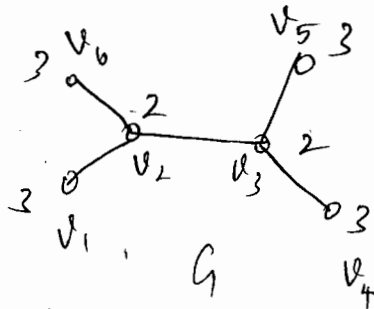
P.T. Every tree has either one or two centers.

(or) Every tree has a center consisting of either a single vertex or two adjacent vertices.

Proof {Main proof is given after these examples}

Explanation with examples:

① A tree with two centers:-

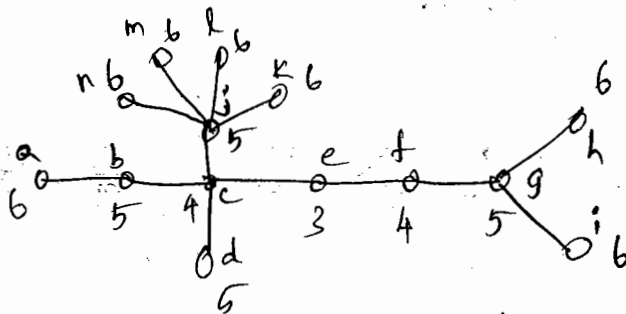


$$\gamma(G) = \min \{EC(G)\} = 2$$

$\therefore$ Centers are $\{v_2, v_3\}$

$\therefore$ This is a tree with two centers.

② A tree with one center:-

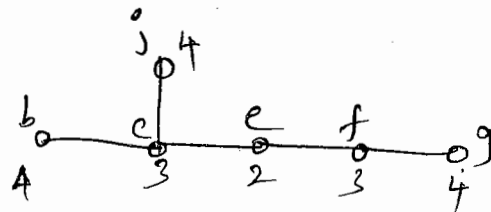


- T

$$\text{In } T, \gamma(T) = \min \{EC(T)\} = 3$$

Center is c.

- * By deleting all the pendant vertices in T , we will get a subtree T' .

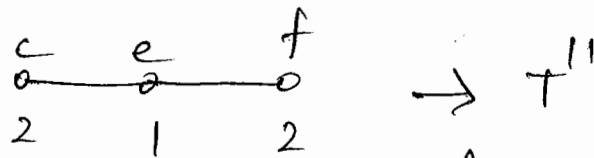


$$\gamma(T') = \min \{E(T')\} = 2$$

T'

Center of $T' = \{e\}$ is the same as the center of T

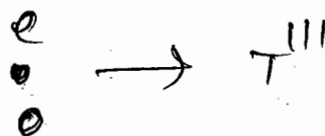
- * Again by deleting all the pendant vertices in T' we will get a subtree T''



$$\gamma(T'') = \min \{E(T'')\} = 1$$

$\therefore$ Center of $T'' = \{e\}$ is as the same as the center of T also center of T' .

- * Continuing the process,



which is also a center

$\therefore$ By observing the above examples,

(17)
by deleting all the pendant vertices continuously, The center remains unchanged.

Main proof of the theorem (Theorem-10)

- * The result is obvious for the tree K_1, K_2 .
- * The vertices of K_1 & K_2 are central vertices.
- * Now let T be a tree with $|T| \geq 3$
- * Then T has at least two pendant vertices.
- * Clearly, the pendant vertices of T cannot be central vertices.
- * Delete all the pendant vertices from T .
- * This results in a subtree T' of T .
- * As any maximum distance path in T from any vertex of T' ends at a pendant vertex of T .
- * The eccentricity of each vertex of T' is one less than the eccentricity of the same vertex in T .

* Hence, the vertices of minimum eccentricity of T' are the same as those of T .

* i.e., T & T' have the same center.

* Repeat the process of deleting the pendant vertices in the successive subtrees of T until the tree reduced to k_1 or k_2 .

* This will always be the case as T is finite.

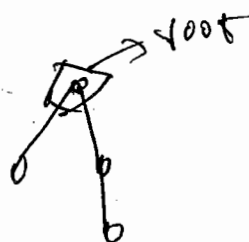
* Hence, the center of T is either a single vertex or a pair of adjacent vertices.

•

Rooted & Binary tree

A tree in which one vertex is distinguished from all the others is called as a rooted tree.

Ex:



Binary tree:

A binary tree is defined as a tree in which there is exactly one vertex of degree two & each of the remaining vertices is of degree one (or) three

Note:

- ① A node with degree two in a binary tree is said to be a root.
- ② Every binary tree is a rooted tree.

Theorem - 1

P.T. The no/- of vertices n in a binary tree is always odd.

Proof

* Let T be a binary tree with n vertices.

* By definition of binary tree, one vertex is of degree two

& the remaining $(n-1)$ vertices is of degree one (or) three

* i.e., remaining $(n-1)$ vertices is of

* By theorem,

The no- of odd degree vertices is even no-.

* $\therefore (n-1)$ is even

& n is odd.

Hence proved.

Theorem: 12

P.T. The no- of pendant vertices in a binary tree is $\frac{n+1}{2}$.

Proof:

* Let P be the no- of pendant vertices in a binary tree T .

* Since P vertices is of degree one
1 vertex is of degree two

& $n - (P+1)$ vertices is of degree three.

* By theorem,

$$\sum d(u_i) = 2e \quad \& \quad e = n-1$$

$$\therefore e = \frac{1}{2} \sum d(u_i) \quad \& \quad e = n-1$$

$$\therefore n-1 = \frac{1}{2} \sum d(u_i)$$

(23)

$$\frac{1}{2} \{ 1 \times p + 3(n-p-1) + 2 \times 1 \} = n-1$$

$$p + 3n - 3p - 3 + 2 = 2n - 2$$

$$-2p + 3n - 1 = 2n - 2$$

$$3n - 1 - 2n + 2 = 2p$$

$$2p = n + 1$$

$$p = \frac{n+1}{2}$$

Note:

- ① A non pendant vertex in a tree is called an internal vertex.

Spanning tree

A spanning subgraph of a graph, which is also a tree is called a spanning tree of the graph.

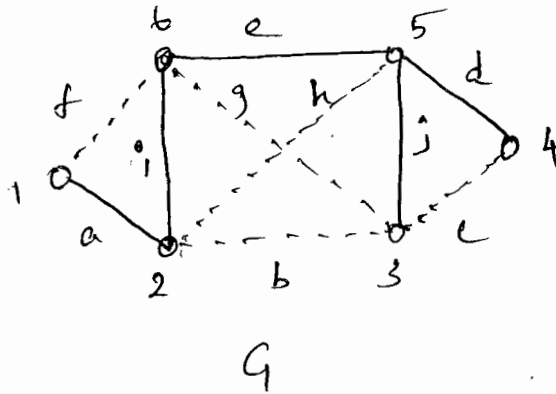
Branch

An edge in a spanning tree T is called a branch in T .

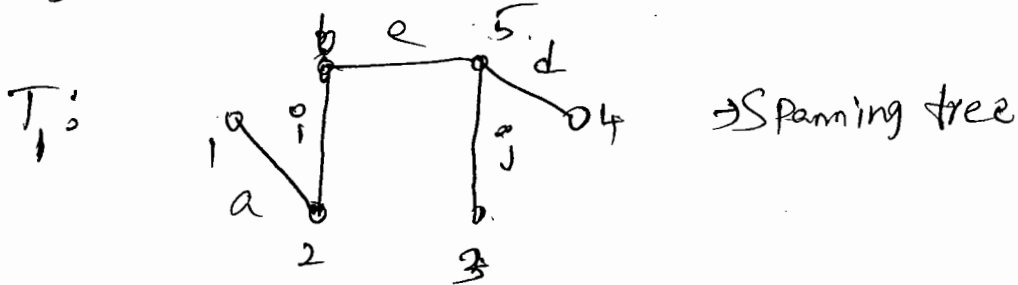
Chord

An edge of G that is not in a spanning tree T is called a chord.

Example - ①



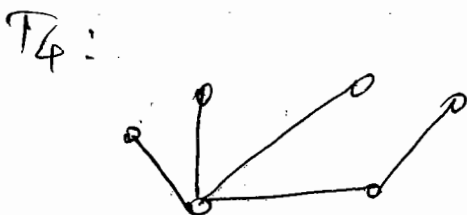
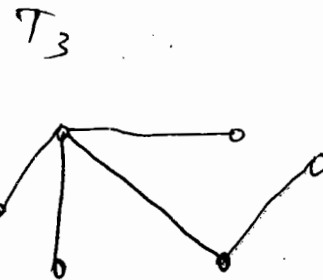
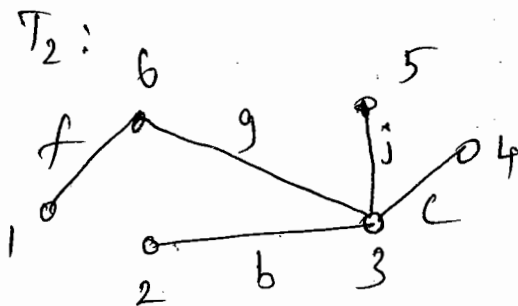
* For this graph G, the tree (spanning tree) is given with dark lines,



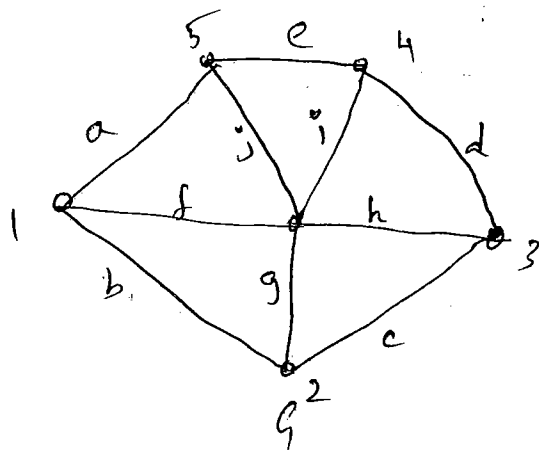
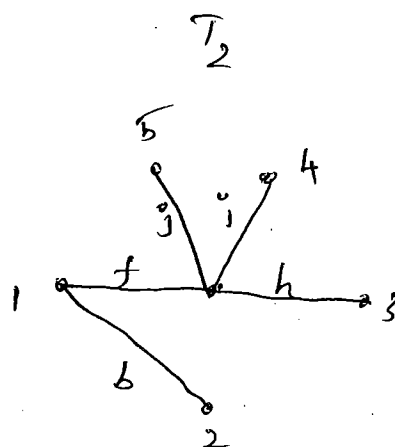
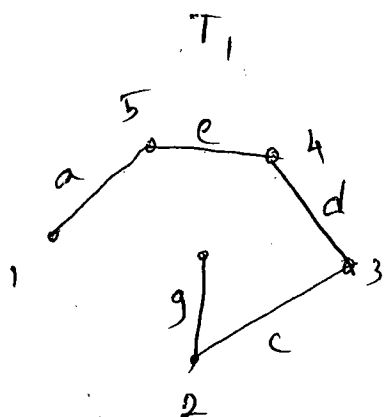
Branches : $\{a, d, e, i, j\}$

Chords : $\{f, g, h, b, c\}$

* Another Spanning tree T_2, T_3 of G.



like this we can generate many spanning tree

Example - 2Spanning trees:For T_1 ,Branches: $\{a, c, d, e, g\}$ Chords: $\{b, f, h, i, j\}$ for T_2 Branches: $\{b, f, h, i, j\}$ Chords: $\{a, c, d, e, g\}$

From these Example,

we know that, $T_1 \cup T_2 = G$ where T_1 & T_2 are spanning tree (distinct) $\therefore T_1$ is the complement of T_2 (or) T_2 is the complement of T_1 called as Co-tree of T .

Note:

- ① Branches & chords are defined only w.r.to a given Spanning tree.
- ② An edge that is a branch of one spanning tree T_1 in a graph G may be a chord w.r.to another spanning tree T_2 .
- ③ $G = T_1 \cup T_2$ where T_1 is a spanning tree, & T_2 is the complement of T_1 in G which is called as cotree of T_1 .
- ④ All spanning trees are the largest trees among all trees of the graph G .
- ⑤ A spanning tree is also called as minimal tree subgraph (or) minimal tree of G .

Theorem: 13

With respect to any of its spanning trees, a connected graph G of n vertices & e edges has $n-1$ tree branches & $e-n+1$ chords.

Proof

- * let G be a Connected graph with n vertices & e edges.
- * let T be a Spanning tree.
- * Since T contains all n vertices of G ,
& T has $(n-1)$ edges.
- * The no/- of chords in G is equal to
$$e - (n-1) = e - n + 1$$

Definition Rank & Nullity

- * The Rank r of an undirected graph is defined as $r = n - k$
where $n \rightarrow$ no/- of vertices
 $k \rightarrow$ no/- of components.
- * The nullity M of an undirected graph is defined as $M = e - n + 1$
 $=$ no/- of chords in G .

Note:

- ① Rank + nullity = e (no/- of edges of G)
- ② The nullity of a graph is also referred as cyclomatic no/- (or) first Betti number.

Theorem - 14

If u & v are non adjacent vertices of a tree T , then $T+uv$ contains a unique cycle.

Proof:

* If P is the unique uv path in T then $P+uv$ is a cycle in $T+uv$.

* It is unique as a path P is unique in T .

Theorem: 15

Every connected graph contains at least one spanning tree.

Proof:

* let G be a connected graph

* let C be the set of all connected spanning subgraphs of G .

* let T be a minimally connected spanning subgraph in C .

* If T does not contain any circuit then T is a tree.

* Suppose T contains circuit.

* Choose an edge " e " in the circuit.

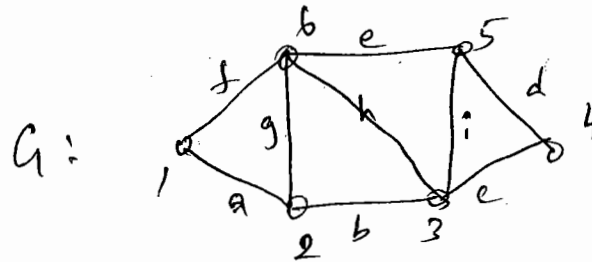
* 'T-e' is connected,
which is a contradiction since T is
minimally connected

* Hence T is a spanning tree.

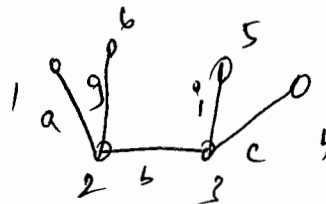
Fundamental Circuit:

Let T be a spanning tree in a
connected graph G. Adding one chord to
T will create exactly one circuit. Such
a circuit, formed by adding a chord to
a spanning tree is called a fundamental
circuit.

Example:



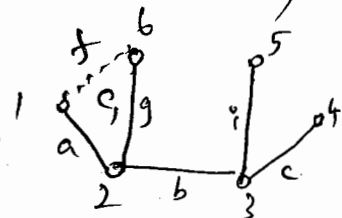
Spanning tree T:



Branches: {a, b, c, g, e}
Chords: {f, d}

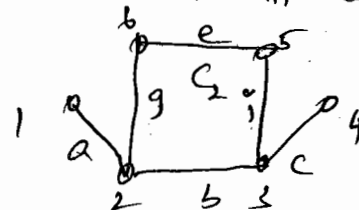
Now,

① T + f (chord) gives a circuit C_1



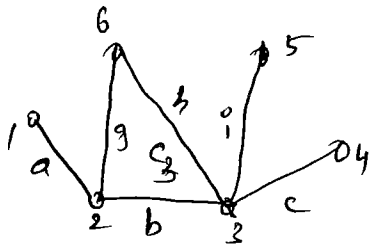
$\therefore C_1$ is a fundamental circuit

② T + e gives another circuit C_2



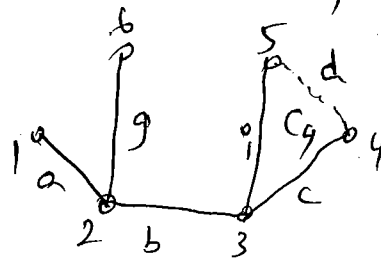
$\therefore C_2$ is a fundamental circuit.

③ $T+h$ creates another circuit C_3



$\therefore C_3$ is a fundamental circuit.

④ $T+d$ creates another circuit C_4



$\therefore C_4$ is a fundamental circuit.

* Since for the given graph G with respect to the spanning tree T , we have 4 chords.

* They are $\{f, e, h, d\}$

* By adding each chord to a spanning tree T we will get different circuits.

* Those circuits are called fundamental circuit with respect to the given spanning tree T .

* No. of fundamental circuits
 $=$ no. of chords in a spanning tree
 $= N = e - n + 1$

Note (i.)

* The above example is nothing but, finding all the fundamental circuits for

Theorem: 16

A connected graph G is a tree if and only if adding an edge between any two vertices in G creates exactly one circuit.

Proof:

- * let G be a connected graph
 $\&$ it is a tree
- * let $u \& v$ be any two vertices of G
- * Since G is a tree,
 then there is a unique path
 between every pair of vertices.
- * let P be the unique uv path
 in G
- * Then $P + uv$ is a cycle in $G + uv$
 which is unique as a path P is
 unique in G .
- * Adding an edge between any
 two vertices in G creates exactly one
 circuit.

Conversely,

- * let G be a connected graph
- * choose any two vertices $u \& v$ in G

- * Then $G+uv$ contains a unique cycle.
- * Delete that edge uv in G then G is acyclic.
- * Hence G is a tree.

Spanning trees in a weighted graph

Weighted graph

A graph G is said to be weighted if there is a real no. associated with each edge of G .

Weight of a Spanning tree:

The weight of a spanning tree T of a graph G is defined as the sum of the weights of all the branches in T .

Shortest Spanning tree:

A spanning tree with smallest weight in a weighted graph is called a shortest spanning tree (or) shortest-distance spanning tree (or) minimal spanning tree.

Algorithm for Shortest Spanning tree:

① Prime Algorithm

Step: 1 Select any vertex

Step: 2 Select an edge with minimum weight connected to that vertex

Step: 3 Select the shortest edge which is connected to any vertex already connected & which doesn't form a circuit

Step: 4 Repeat Step-3 until all vertices have been connected

② Kruskal's Algorithm

Step: 1 Select an edge with minimum weight in the given network.

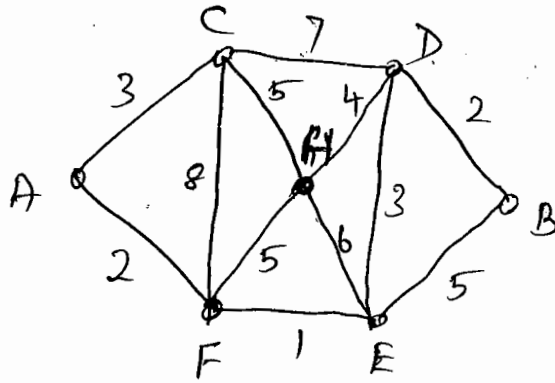
Step: 2 Select the next edge with minimum weight which does not create a circuit.

Step: 3

Repeat Step-2 until all vertices have been connected.

Sung

① Find the Shortest Spanning tree from the given network by using Prim's & Kruskal's algorithm.



Sol

1) Prim's Algorithm approach

Step: 1

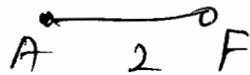
Select any vertex. So choose A.

A •

Step: 2

Select the shortest edge (an edge with minimum weight) connected to A.

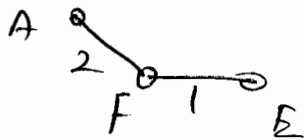
the Shortest edge is $AF = 2$



Step: 3

Select the next Shortest edge connected to the vertices A & F which does n't Create a Circuit.

The Shortest edge is $FE = 1$

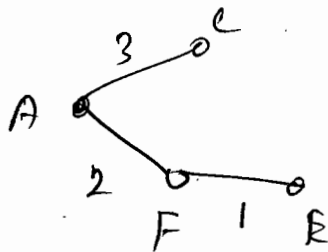


Step: 4

The Next Shortest edge connected to A, F & E which does n't Create a Circuit

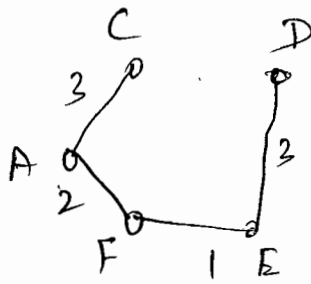
$AC = 3$ & $ED = 3$

Choose any one edge arbitrarily ($AC = 3$)



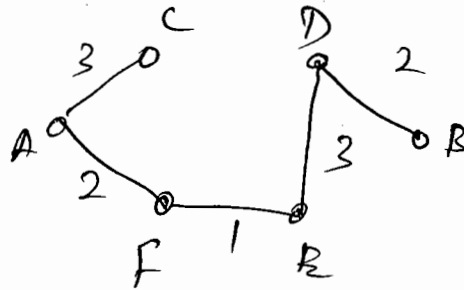
Step: 5

The Next Shortest edge connected to A, C, E & F which does n't Create a Circuit ~~which~~ $ED = 3$



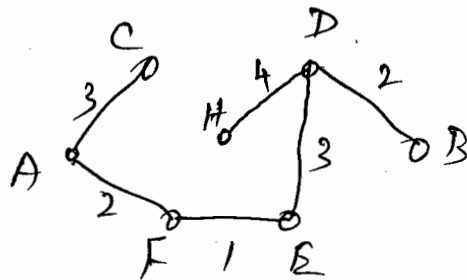
Step: 6

The next Shortest edge Connected to A, C, D, E & F is $DB = 2$ which does not form a Circuit.



Step: 7

The next Shortest edge Connected to A, B, C, D, E, F is $DH = 4$ which does not Create a circuit.



Since we covered all the vertices
 $\therefore$ The Process will ends here.

$\therefore$ Shortest Spanning tree we arrived.

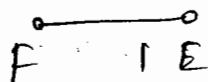
$$\begin{aligned} \text{Shortest Path} &= 1 + 2 + 2 + 3 + 3 + 4 \\ &= 15 \end{aligned}$$

ii) ~~Dir~~ Kruskal's approach

⊕ Step: 1

Select the shortest edge in the given network.

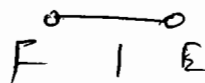
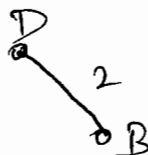
$$FE = 1$$



Step: 2

Select the next shortest edge which does not create a cycle.

$DB = 2$
 $AF = 2$
 (Choose any arbitrary edge of DB & AF)

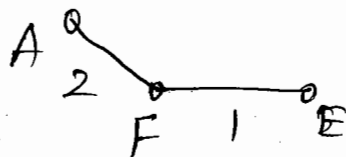
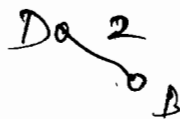


Step: 3

Select the next shortest edge which does not create a cycle

$$AF = 2$$

(Since $DB = 2$ chosen already)

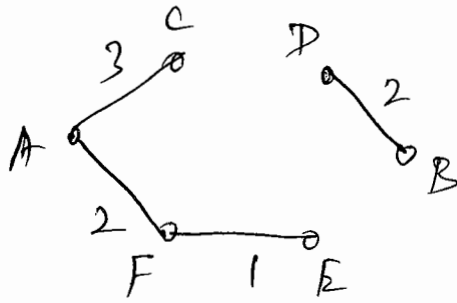


Step: 4

Select the next shortest edge which does not create a cycle

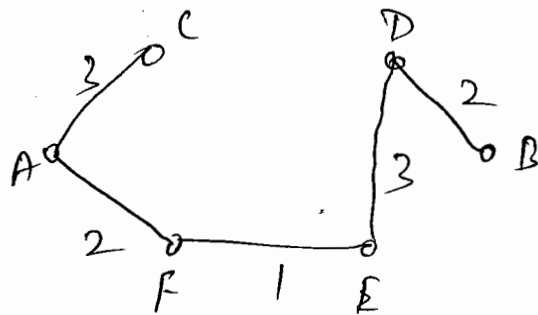
$$ED = 3 \text{ \& } AC = 3$$

Out of $ED=3$ & $AC=3$, Choose any edge as arbitrary
So choose $AC=3$



Step: 5

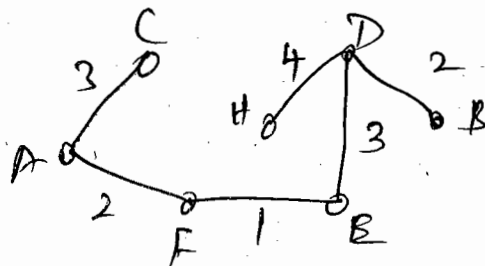
Select the next Shortest edge
 $ED=3$ which does not create a cycle.



Since all the vertices are not connected so Repeat the process until all vertices covered.

Step: 6

Select the next Shortest edge
 $DH=4$ which does not create a circuit.



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Since we covered all the vertices
 $\therefore$ The process will end here. So
we arrived a Shortest Spanning tree.

$$\therefore \text{Shortest path} = 1 + 2 + 2 + 3 + 3 + 4 \\ = 15$$

Note:

① Both algorithms will give
the Shortest Spanning tree with
Same length.

② Prim's algorithm based on
vertex approach

③ Kruskal's Based on edge approach.

④ Running time In Kruskal's
algorithm

$$O(m \log n)$$

where $m \rightarrow$ no. of edges

$n \rightarrow$ no. of vertices

⑤ Running time in Prim's

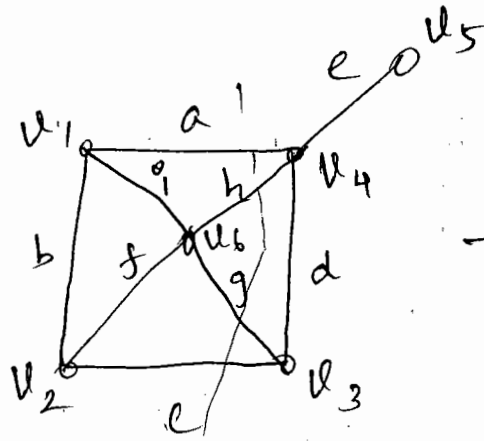
$$O(m + n \log n)$$

Cut sets & cut vertices

Cut set

In a connected graph G , a cutset is a set of edges whose removal from G leaves G disconnected, provided removal of no proper subset of these edges disconnects G .

Eg:



→ Given graph G .

Then the cut sets are

$$S_1 = \{e\}, \quad S_2 = \{a, i, g\},$$

$$S_3 = \{a, h, d\}, \quad S_4 = \{a, h, g, d\}$$

But $S_5 = \{a, h, d, g\}$ is not a cut set

Since $S_3 \subset S_5$

Where removal of S_3 will disconnect the graph. And by defn the removal of no proper subset of these edges

(91)

disconnects ~~it~~.

i.e., In $S_5 = \{a, h, d, g\}$

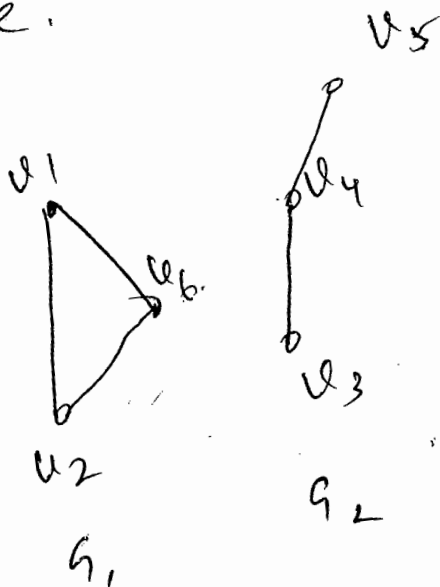
The proper subset $S_3 = \{a, h, d\}$ is a cut set.

$\therefore S_5$ is not a cut set.

Note:

① If we remove S_4 from G , then the graph will be disconnected into two components G_1 & G_2 .

They are.



Note:

② A cut set is a minimal set of edges in a connected graph whose removal disconnects the graph.

③ Removal of any edge from a tree disconnects the graph. Thus every edge of a tree is a cut set.

④ let G be a nontrivial connected graph with vertex set V & let S be a non empty subset of V .

For $\bar{S} = V/S$, let $[S, \bar{S}]$ denote the set of minimal no/- of edges of G that have one end vertex in S & the other in $\bar{S}$.

A set of edges of G of the form $[S, \bar{S}]$ is called as cutset of G .

Theorem: 17

Every cut set in a connected graph G must contain at least one branch of every spanning tree of G .

proof

* let T be the spanning tree of a connected graph G .

* let S be the cutset in G .

* Suppose S not to have any

edge in common with T .

* Then removal of the cutset S from G would not disconnect the graph which is a contradiction to the choice of S (since $T \subseteq G-S$ not disconnect).

Theorem: 18

In a Connected Graph G , any minimal set of edges containing at least one branch of every Spanning tree of G is a cutset.

Proof

- * let G be a Connected Graph.
- * let Q be a minimal set of edges containing at least one branch of every Spanning tree of G .
- * To prove: Q is a Cutset of G .
- * Consider $G-Q$, The Subgraph that remains after removing the edges in Q from G .
- * $G-Q$ contains no Spanning tree of G , $G-Q$ is disconnected.

* Since Q is a minimal set of edges with this property, any edge e from Q & adding e to $G-Q$ will create at least one spanning tree.

* Thus the subgraph $G-Q+e$ will be a connected graph.

* Therefore, Q is a minimal set of edges whose removal from G disconnects G .

* Hence Q is a cutset of G .

Theorem: 18

P.T. Every Circuit has an even no. of edges in common with any cutset.

Proof

* Let S be the cutset in the graph G .

* Let the removal of S partition the vertices of G into two mutually disjoint subsets V_1 & V_2

* Let C be the circuit in G .

* If all the vertices in C are entirely within vertex set V_1 (or) V_2 the no. of edges common to S & C is zero which is an even no.

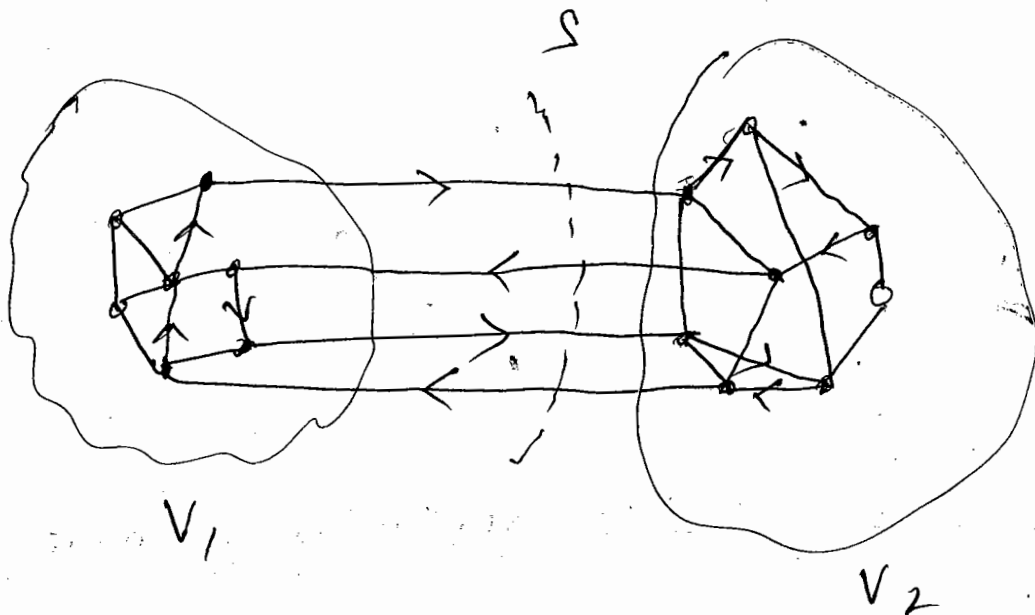
otherwise, Some vertices in C are in V_1 & Some in V_2

The due to the closed nature of the circuit, the no/- of edges of C that traverses between V_1 & V_2 be even.

All the edges must be in S otherwise S cannot destroy paths between V_1 & V_2

which is a Contradiction

Every Circuit has an even no/- of edges in common with any cutset.



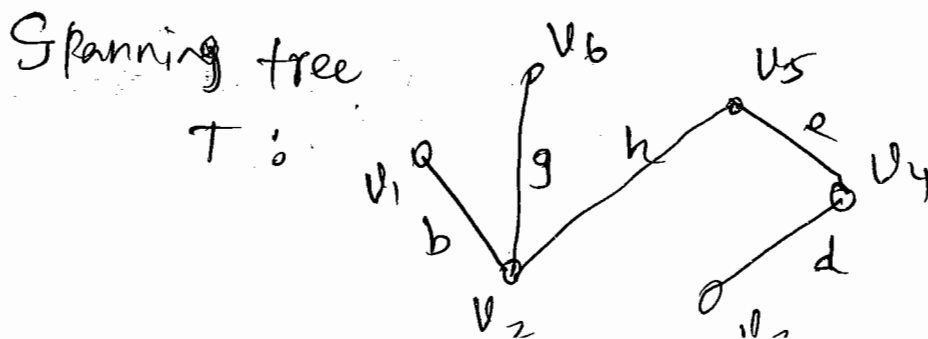
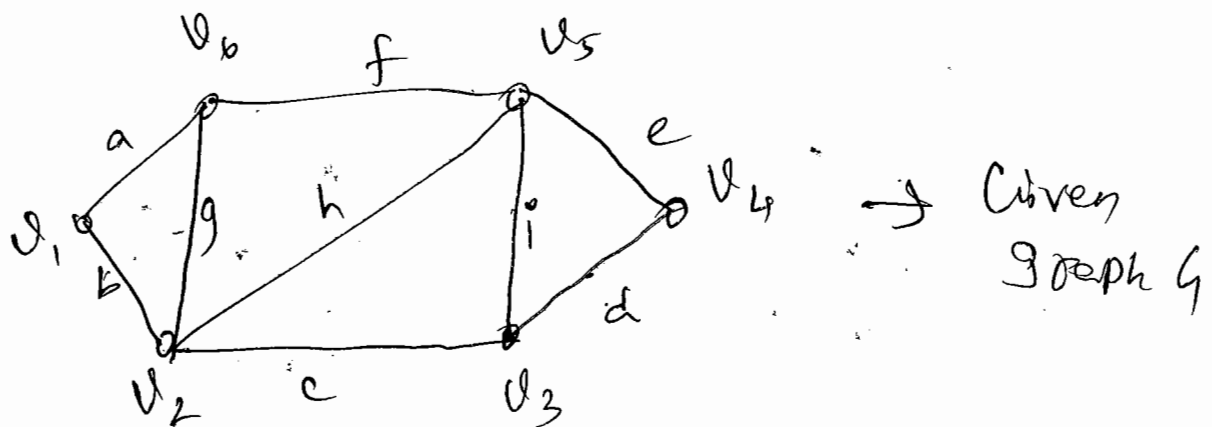
Fundamental cut sets:

A cut-set S will contain only one branch of a spanning tree T & rest of the edges in S are chords w.r. to T . Such a cutset S containing exactly one branch of a tree T is called a fundamental cutset w.r. to T .

Note:

Some times a fundamental cutset is also called a basic cutset.

Example



Branches of T : $\{b, g, h, e, d\}$

Chords of T : $\{a, f, c, i\}$

The possible cut sets are

$$S_1 = \{a, b\}, \quad S_2 = \{a, g, f\}, \quad S_3 = \{f, h, e\}$$

$$S_4 = \{e, d\}, \quad S_5 = \{c, i, d\} \text{ etc.}$$

out of these cutsets,

* S_1, S_2, S_3, S_5 are Fundamental cut sets
(since it includes only one branch of T)

* S_4 is not a fundamental cutset
(since it includes more than one branches of T)

Note:

* For a Spanning tree T in a given connected graph G ,

let c_i be a chord w.r.to T
& let b_k be a branch of T .

* let the fundamental circuit made by c_i be called Γ_i , consisting of k branches $b_1, b_2, \dots, b_k$ in addition to the chord c_i

Let S be the fundamental ^{cutset} associated with b_i (branch) consisting of q chords $c_1, c_2, \dots, c_q$ in addition to the branch b_i

i.e., In General.

Fundamental ^{cutset} is denoted by

$$S = \{ b_i, c_1, c_2, \dots, c_p \}$$

& Fundamental circuit is denoted by

$$\Gamma = \{ c_i, b_1, b_2, \dots, b_k \}$$

Theorem: 19:

with respect to a given spanning tree T , a branch b_i that determines a

(49)

fundamental cutset S is contained in every fundamental circuit associated with the chords in S & in no others.

Proof:

* ~~Let~~ S be the fundamental cutset
i.e. $S = \{b_i, c_1, c_2 \dots c_p\}$

* let Γ_1 be the fundamental circuit determined by the chord c_1 .

i.e. $\Gamma_1 = \{c_1, b_1, b_2 \dots b_q\}$

* Since the no/- of edges common to S & Γ_1 must be even

* i.e., b_i belong to both S & Γ_1 ,
& the chords also satisfies these condition.

* on the other hand,

Suppose that b_i occurs in a fundamental circuit Γ_{p+1} made by a chord other than $c_1, c_2 \dots c_p$.

Since none of the chords $c_1, c_2, \dots, c_p$ is in Π_{p+1} . There is only one edge b_i common to a circuit Π_{p+1} & the cutset S_i which is not possible.

* Hence proved.

Note:

Theorem: 20

with respect to a given spanning tree T , a chord c_i that determines a fundamental circuit Γ occurs in every fundamental cutset associated with the branches in Π & in no other.

Proof:

The similar way what we proved in Theorem-19

By replacing chord c_i as branch b_i vice versa.

Cut vertices & Planar & dual graphs

Connectivity and separability - network flows -
1-isomorphism - 2-isomorphism. Planar and
dual graphs - Combinatorial representation -
Planar graphs - Kuratowski's graph - detection
of planarity - Geometric dual graphs.

Edge connectivity

The edge connectivity of a connected graph can be defined as the minimum number of edges whose removal disconnects the graph.

Vertex Connectivity

The vertex connectivity of a connected graph G is defined as the minimum no. of vertices whose removal from G leaves the remaining graph disconnected.

Separable graph

A connected graph is said to be separable if its vertex connectivity is one.

Note:

① All other Connected graphs are called non separable.

② A Connected Graph G is said to be separable if $\exists$ a subgraph g in G such that $\bar{g}$ (complement of g in G) & g have only one vertex in common.

Cut vertex

In a separable graph a vertex whose removal disconnects the graph is called a cut vertex (or) a cut node (or) an articulation point.

~~Vertex Cut~~

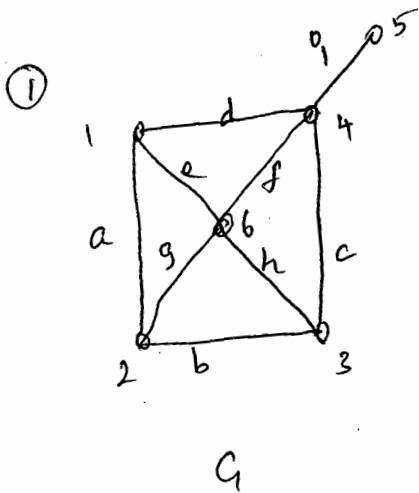
Vertex cut

A subset V' of the vertex set $V(G)$ of a connected graph G is a vertex cut of G if $G - V'$ is disconnected.

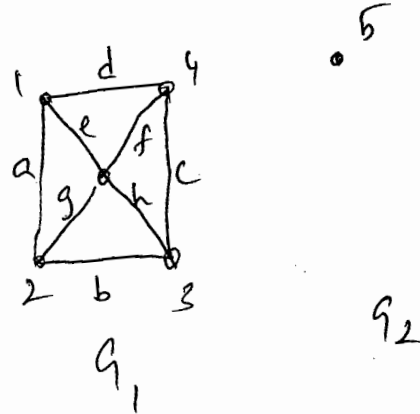
Note:

~~It is~~ It is a k vertex cut if $|V'| = k$ & V' is then called a separating set of

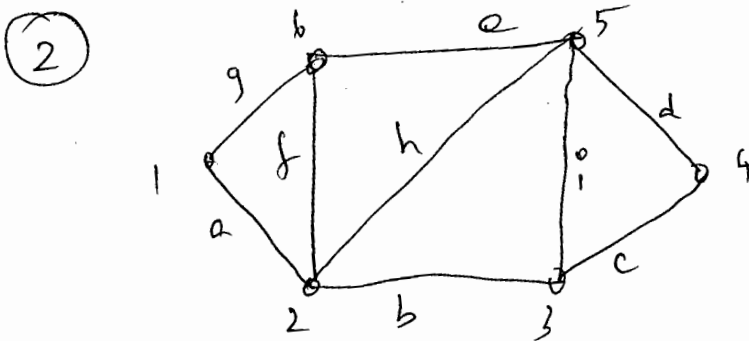
Example:



$\therefore G - \{i\}$



Edge connectivity is one. i.e. $\{i\}$
 Vertex Connectivity is also one i.e., $\{4\}$



Edge Connectivity = {two edges} = $\{d, c\}$ (or) $\{g, a\}$
 Vertex Connectivity = {two vertices} = $\{3, 5\}$ (or) $\{6, 2\}$
 (or) $\{2, 5\}$

Note:

From the above examples,

- ① $\{4\}$ is the cut vertex.
- ② $\{i\}$ is the cut edge.
- ③ $\{i\}$ is the cut edge.
- ④ $\{i\}$ is the cut edge.
- ⑤ $\{i\}$ is the cut edge.
- ⑥ $\{i\}$ is the cut edge.
- ⑦ $\{i\}$ is the cut edge.
- ⑧ $\{i\}$ is the cut edge.
- ⑨ $\{i\}$ is the cut edge.
- ⑩ $\{i\}$ is the cut edge.
- ⑪ $\{i\}$ is the cut edge.
- ⑫ $\{i\}$ is the cut edge.
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- ⑯ $\{i\}$ is the cut edge.
- ⑰ $\{i\}$ is the cut edge.
- ⑱ $\{i\}$ is the cut edge.
- ⑲ $\{i\}$ is the cut edge.
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- ㊼ $\{i\}$ is the cut edge.
- ㊽ $\{i\}$ is the cut edge.
- ㊾ $\{i\}$ is the cut edge.
- ㊿ $\{i\}$ is the cut edge.

Greater than one is a cut vertex.

Theorem: 1

A vertex v in a connected graph G is a cut vertex if and only if there exist two vertices x and y in G such that every path between x and y passes through v .

Proof

* If v is a cut vertex of G
then $G - v$ is disconnected & has at least two components, namely G_1 & G_2 .

* let $x \in V(G_1)$, $y \in V(G_2)$

* Then any path in G must contain v otherwise x & y would belong to the same component of $G - v$.

Conversely,

Suppose that $\exists$ two vertices x & y in G such that every path between x & y passes through v .

To Prove: v is a cut vertex.

* Deletion of v destroys every xy path in G .

* Hence x & y lie in distinct component of $G - v$.

* $G - v$ is disconnected &
 v is a cut vertex of G .

Theorem: 2

P.T. the vertex connectivity of any graph G can never exceed the edge connectivity of G & the edge connectivity of a graph G cannot exceed the degree of the vertex with the smallest degree in G .

i.e., P.T. In any graph G ,

$$\begin{aligned} \text{vertex connectivity} &\leq \text{edge connectivity} \\ &\leq \text{minimum degree.} \end{aligned}$$

i.e., $k \leq k' \leq \delta$

Proof

Case - 1 To prove: $k' \leq \delta$

* let v_0 be the vertex with smallest degree in G .

* let $d(v_i)$ be the degree of v_i & $d(v_i) = \delta$

* Vertex u_i can be separated from G by removing δ edges incident on u_i .

* i.e., δ forms an edge-cut

* Thus $k' \leq \delta$.

Case: 2 $k \leq k'$

* let α denotes the edge connectivity of G .

* Then $\exists$ a cutset S in G with α edges.

* let S partition the vertices of G into subsets V_1 & V_2 -

* By removing at most α vertices from V_1 (or V_2) on which the edges in S are incident, we can effect the removal of S (together with all other edges incident on these vertices) from G .

* Hence proved.

Theorem-3

The maximum vertex Connectivity one can achieve with a graph G of n vertices and e edges ($e \geq n-1$) is the integral part of the number $\frac{2e}{n}$ that is $\lfloor \frac{2e}{n} \rfloor$

Proof

* W.K.T. Every edge in G contributes two degrees & G consists n vertices & e edges.

$$* \sum d(v_i) = 2e \quad \& \quad \delta(G) \leq d(v_i) \leq \Delta(G)$$

where $\delta(G) \rightarrow$ minimum degree of G

$\Delta(G) \rightarrow$ Maximum degree of G

$d(v_i) \rightarrow$ degree of the vertex
for $i=1, 2, \dots, n$

$$n \delta(G) \leq \sum d(v_i) \leq n \Delta(G)$$

$$n \delta(G) \leq 2e \leq n \Delta(G)$$

$$\delta(G) \leq \frac{2e}{n} \leq \Delta(G)$$

* Also W.K.T.

Vertex Connectivity $\leq$ edge Connectivity

$\leq$ minimum degree

$$\text{i.e., } k \leq k' \leq \delta \leq \frac{2e}{n}$$

∴ The maximum Vertex Connectivity

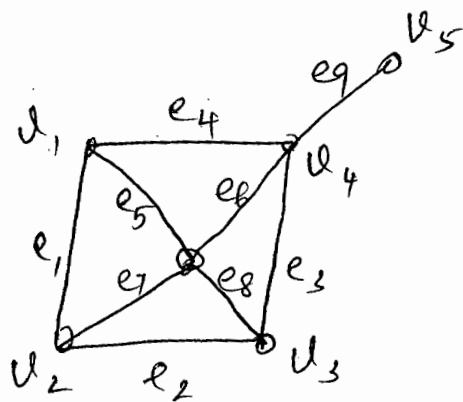
$$\text{Possible} = \left\lfloor \frac{2e}{n} \right\rfloor$$

k Connected Graph

A Graph G is said to be k -connected if the vertex connectivity of G is k , therefore, a 1-Connected Graph is the same as a Separable graph.

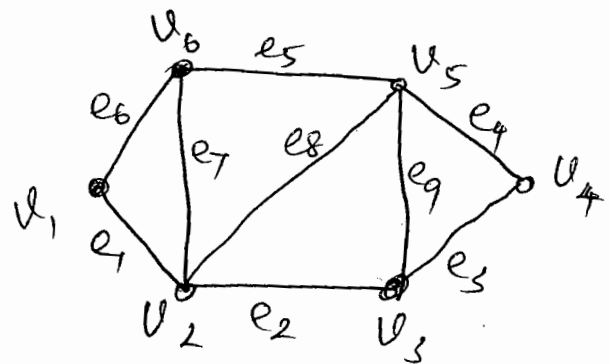
Example

1 - Connected



vertex cut = $\{v4\}$

2 - Connected



Planar graphs

A graph G is said to be planar if there exists some geometric representation of G which can be drawn on a plane such that no two of its edges intersect.

Non Planar graphs

A graph that cannot be drawn on a plane without a cross over between its edges is called nonplanar.

Embedding

A drawing of a geometric representation of a graph on any surface such that no edges intersect is called embedding.

Plane representation:-

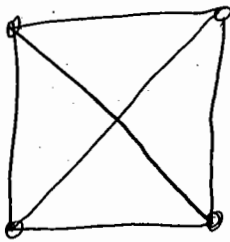
An embedding of a planar graph G on a plane is called plane representation of G .

Plane graph

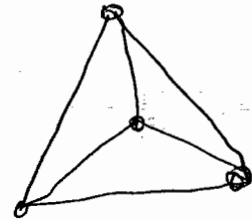
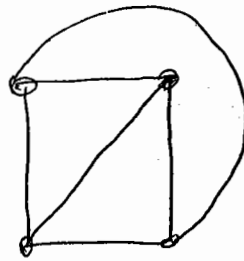
A plane graph is a planar graph that has already been embedded in the plane.

Examples

①



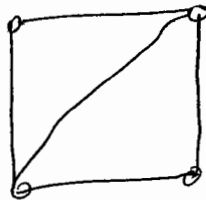
Planar graph K_4



Two plane Embeddings of K_4

②

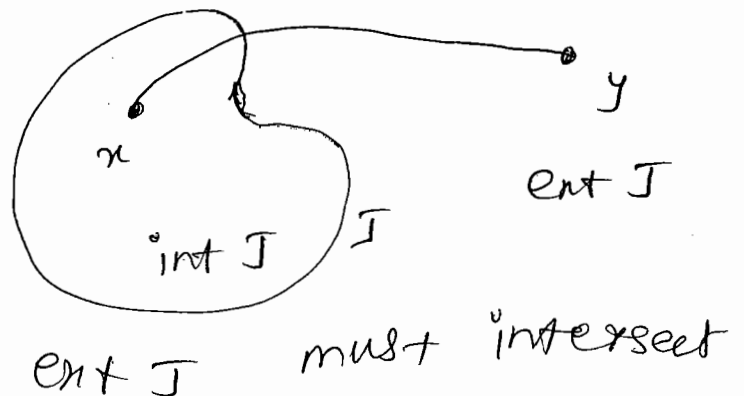
Plane graph



Jordan Curve Theorem

Statement

If J is any closed Jordan curve in the plane any arc joining a point of $\text{int } J$ and a point of $\text{ext } J$ must intersect J at some point.



Theorem:

The complete graph of five vertices is non planar.

Proof

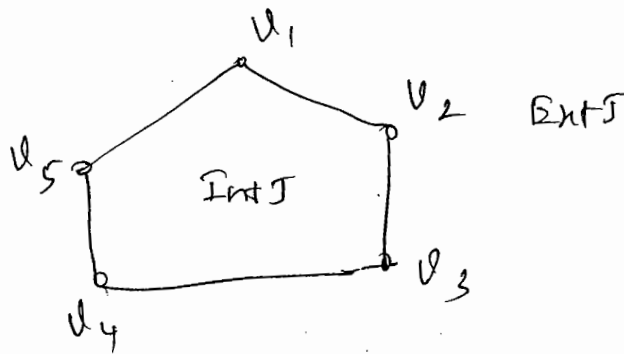
Let K_5 is planar

Let K_5 be the complete graph with 5 vertices $\{v_1, v_2, v_3, v_4, v_5\}$.

Since K_5 is complete also simple,

$\therefore$ every vertex is joined to every other vertex by means of an edge.

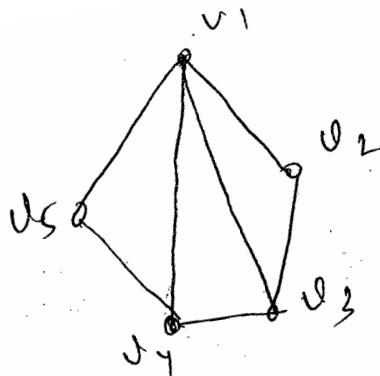
* Consider a circuit v_1 to v_2 to v_3 to v_4 to v_5 to v_1 , i.e., a pentagon.



By Jordan Curve Theorem,

* we have two regions. One is Int J (Interior region of the circuit) & another one is Ext J (Exterior region of the circuit).

* Connect v_1 to v_3 also v_1 to v_4 .



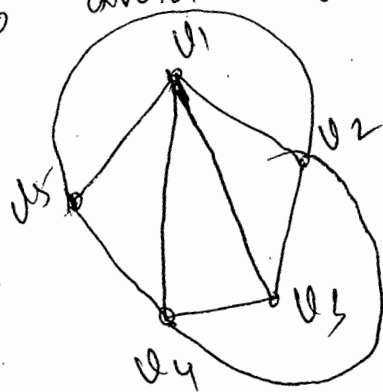
* Since the graph K_5 is complete.

So each vertex is of degree 4.

* Here $\deg(v_1) = 4$.

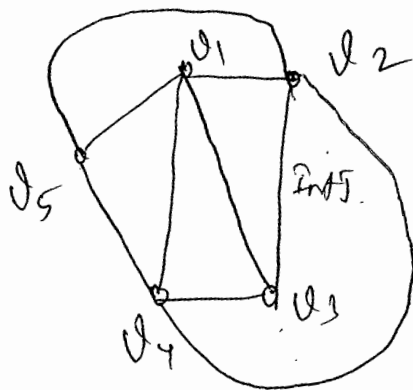
* Connect v_2 to v_4 & v_2 to v_5 in
to avoid edge crossing.

Ext J

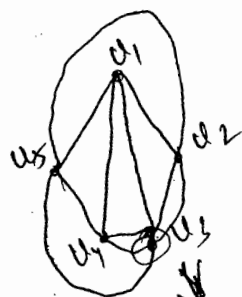


* Now our new Jordan curve

is



Ext J



Intersection point

* now connect v_3 to v_5
then ~~that~~ By Jordan Curve theorem,
 v_3 is an intersection point & v_5 in

* The arc connecting u_2 & u_5 intersect the Jordan curve at some point.

which is a contradiction for K_5 is Planar.

$\therefore K_5$ is non Planar.

Theorem:

P.T. $K_{3,3}$ is non Planar.

Regions (or) faces (or) meshes (or) windows

A plane representation of a graph divides the plane into regions (or) faces.

A region is characterized by the set of edges (or the set of vertices) forming its boundary

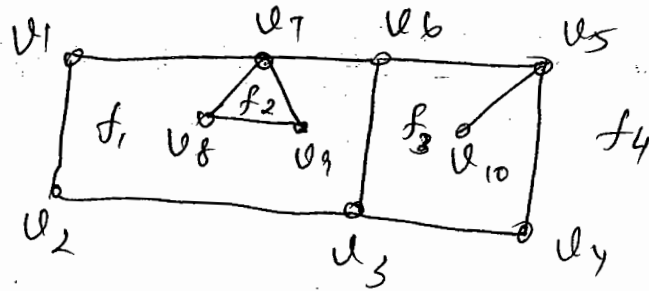
Note:

① Region (or) face is not defined for a non Planar graph.

② Every edge exactly share two faces.

③ ~~if~~ e is a cut edge iff $e \in$ exactly one face.

Example



f_1, f_2, f_3, f_4
→ Regions
(or) faces

Infinite region:

The portion of the plane lying outside a graph embedded in a plane such a region is said to be infinite region.

In the above example,

f_4 → infinite (or) unbounded,
(or) exterior region

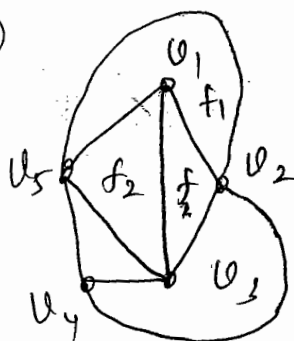
for the particular plane
representation.

Note

① By changing the embedding of a given planar graph, we can change the infinite region into finite region.

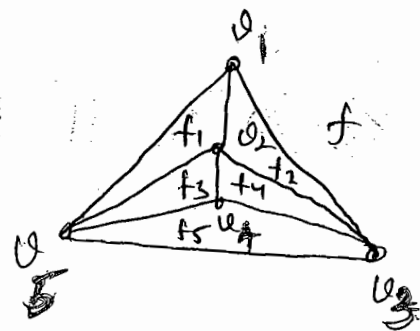
Example

(1)



Here v_1, v_2, v_3 is a finite region - f_1 ,
 $\& v_1, v_3, v_5$ is also a finite region - f_2
 With respect to particular embedding.

(2)



Here v_1, v_2, v_3 is an infinite region - f
 with respect to particular embedding.

Euler's formula:

Theorem: A Connected planar graph with n vertices and e edges has $e - n + 2$ regions (or) faces. i.e., $f = e - n + 2$ i.e., $n - e + f = 2$

Proof

Proof by mathematical induction on f .

* If $f = 1$ then G is a tree &

$e = n - 1$ edges. Vice versa.

$$\begin{aligned} \text{i.e., } f &= e - n + 2 \\ 1 &= e - n + 2 \\ e &= n - 1 \end{aligned}$$

$$\text{Now, } e = n - 1 \text{ \& } f = 1$$

$$\therefore n - (n - 1) + 1 = 2$$

$$n - n + 1 + 1 = 2$$

$$2 = 2$$

* Assume that the result is true for all plane graphs with $f-1$ faces, $f \geq 2$

i.e., $f-1 = e-n+2$

(or) $n-e+(f-1)=2$, $f \geq 2$.

* Suppose that G has f faces. Since $f \geq 2$, G is not a tree.

* It contains a cycle C .

* Let e be an edge of C

* Let $e \in f_1$ & f_2 of G

* $G-e$ is connected, and now we have a single face from f_1 & f_2 .

* Since e is not a cut edge of G .

* No. of faces in $G-e$ is $f-1$.

* By induction to $G-e$,

$$n - (f-1) + (f-1) = 2$$

* By adding an edge e , we get G ,

$$n - e + f = 2$$

* i.e., $f = e - n + 2$

Hence proved.

Note:

① All plane embeddings of a given planar graph have the same no. of faces.

Proof

Since $f = e - n + 2$, the no. of faces depends only on n & e and not on the particular embedding.

Degrees of the face - f

The no. of edges incident with a face f is defined as the degree of f . In counting the degree of a face, a cut edge is counted twice.

Theorem:

P.T. The sum of the degree of the faces is twice the no. of edges.

Proof

* Since the no. of edges incident with the face f is defined as the degree of f with cut edge counted twice.

* Thus, each edge of a plane graph G contributes two to the sum of the degrees of the faces. *

* If F denotes the set of faces of a plane graph G

$$\text{then } \sum_{f \in F} d(f) = 2e$$

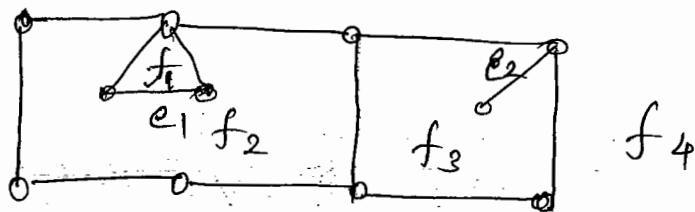
where $d(f)$ denotes the degree of the face f .

Note

① The union of the vertices & edges of G incident with a face f of G is called the boundary of f & it is denoted by $b(f)$

Example

Let us consider the plane graph with 14 vertices, 4 faces, 13 edges



G

Since $G - \{e_2\}$ is a disconnected graph
 $\therefore \{e_2\} \rightarrow$ cut edge.

$$d(f_1) = 3, \quad d(f_2) = 4, \quad d(f_3) = 4 \text{ \& } d(f_4) = 8$$

Note

① A connected graph is a tree if and only if it has only one face.

② Any plane graph has exactly one unbounded face & it is known as the exterior face of the plane graph.

Theorem:

In any simple connected planar graph with f regions, n vertices & e edges

($e \geq 2$), the following inequalities must hold

$$e \leq 3n - 6$$

(*) If G is a simple planar graph with at least 3 vertices, then $e \leq 3n - 6$.

* Since G is simple & $e \geq 2$ & connected.
planar graph with f faces,

$$\text{then } \sum_{f \in F} d(f) \geq 3f$$

where F denotes the set of faces.

* But $\sum_{f \in F} d(f) = 2e$ (x) By Euler's formula
($\because f = e - n + 2$)

$$\therefore 2e \geq 3f$$

By Euler's formula,

$$2e \geq 3(e - n + 2)$$

$$2e \geq 3e - 3n + 6$$

$$3n - 6 \geq 3e - 2e$$

$$e \leq 3n - 6$$

Note:

① The above Result is not valid
for $n = 1$ or 2 (OR) $e = 1$ (or) 2

② The inequality $e \leq 3n - 6$ is
the necessary condition to check whether
the given graph is planar or not.
But not sufficient condition

③ Since the graph K_5 is nonplanar

i.e., In K_5 , $n=5$ & $e=10$

$$e \leq 3n-6$$

$$10 \leq 3(5)-6$$

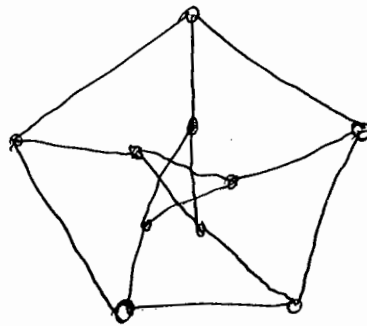
$$10 \leq 15-6$$

$$10 \leq 9$$

which is not possible

$\therefore K_5$ is nonplanar.

④ ~~The~~ Petersen Graph



is not planar
but it satisfies
the inequality

$$e \leq 3n-6$$

Since $e=15$, $n=10$

$$15 \leq 3(10)-6$$

$$15 \leq 30-6$$

$$15 \leq 24$$

but it is nonplanar.

So $e \leq 3n-6$ is not a sufficient

Condition for Planarity.

Girth of a Graph G

The girth of a graph G is the length of the shortest cycle in G .

Theorem: If G be a simple connected planar graph with girth k
Proof: then $e \leq \frac{k(n-2)}{k-2}$

* let F denotes the set of faces f of a planar graph G .

* If $f \in F$ then $d(f) \geq k$

$$\sum_{f \in F} d(f) = 2e$$

~~2~~

* we get, $2e \geq kf$

By Euler's formula, $f = e - n + 2$

$$2e \geq k(e - n + 2)$$

$$2e \geq ke - nk + 2k$$

$$nk - 2k \geq ke - 2e$$

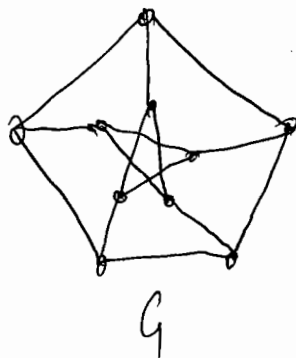
$$k(n-2) \geq e(k-2)$$

$$e \leq \frac{k(n-2)}{k-2}$$

Sum:

P.T. The Peterson graph P is nonplanar.

Proof



* Since the Peterson graph is of girth 5.

* Here $n=10$ & $e=15$.

$$e \leq \frac{k(n-2)}{(k-2)}$$

When $k \rightarrow$ girth, $e \rightarrow$ edges
 $n \rightarrow$ no. of vertices

$$15 \leq \frac{5(10-2)}{5-2}$$

$$15 \leq \frac{40}{3}$$

$15 \leq 13.33$ is not true.

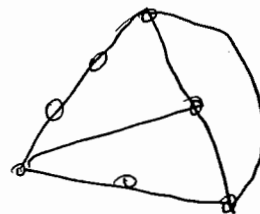
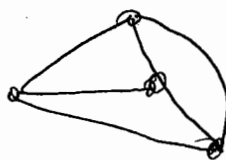
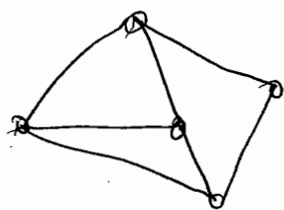
$\therefore$ Peterson graph is not a planar graph.

Homeomorphic Graphs:

Two graphs are said to be homeomorphic if one graph can be obtained from the other by the creation of edges in series (or) by the merger of edges in series.

Example:

These 3 graphs are homeomorphic to each other.



Note:

A graph G is planar if and only if every graph that is homeomorphic to G is planar.

Kuratowski's theorem:

A necessary & sufficient condition for a graph G to be planar is that G does not contain either of Kuratowski's

two graphs or any graph homeomorphic to either of them.

(or) A graph is planar if and only if it has no subgraph homeomorphic to K_5 or $K_{3,3}$.

Note!

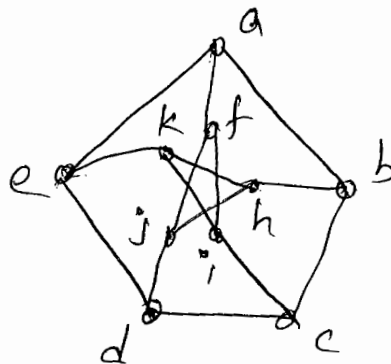
Kuratowski's 2 graphs means that K_5 & $K_{3,3}$.

Sums

P.T. The Petersen graph is non planar.

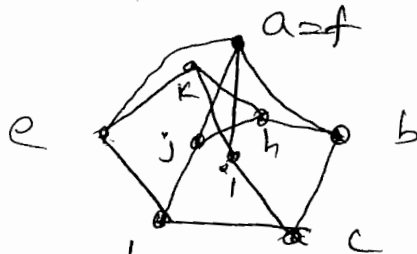
Sol

State the Petersen graph is



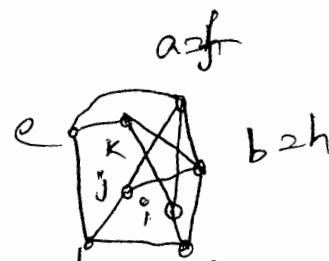
Case -1

Merge a & f

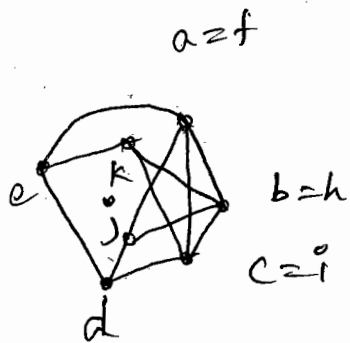


Case -2

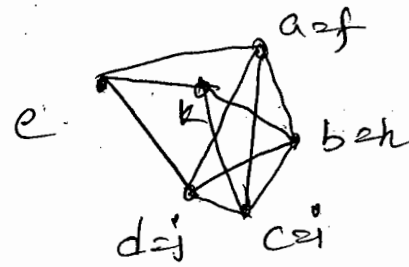
Merge b & h



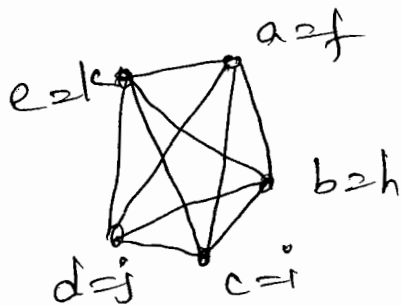
Case-3 Merge $c \& i$



Case-4 Merge $d \& j$



Case : 5 Merge $e \& k$



which is K_5

$\therefore$ Petersen graph is homeomorphic to K_5

$\therefore$ By Kuratowski's theorem
it is non planar.

Dual graphs

IInd part of Unit III - 4

MAT 206 - GT

Geometric Dual

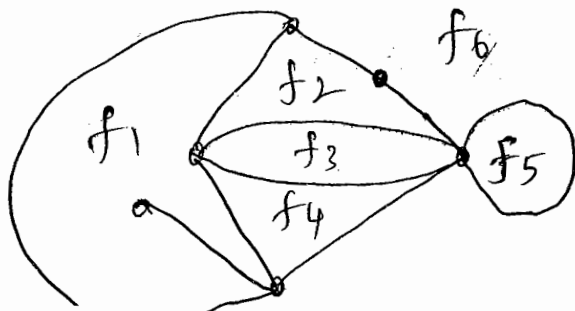
* If two faces f_i & f_j are adjacent (i.e., have a common edge), draw a line joining points p_i & p_j which intersects the common edge between f_i & f_j exactly once.

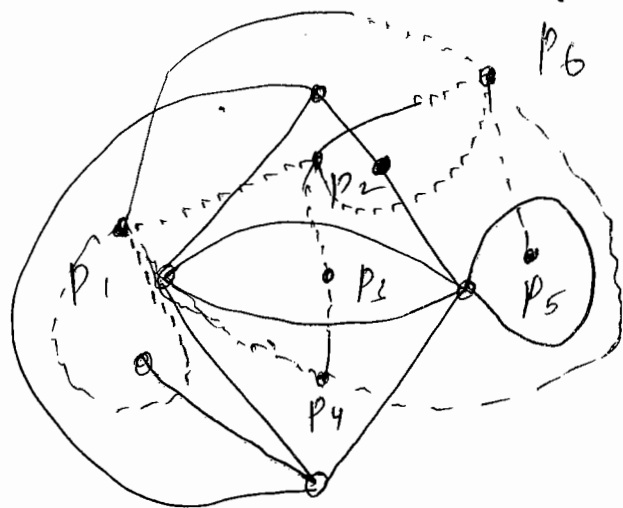
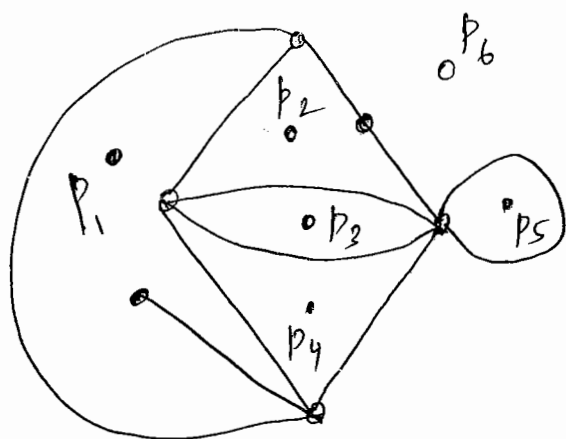
* If there is more than one edge common between f_i & f_j draw one line between points p_i & p_j for each of the common edges.

* For a cut edge, i.e., for an edge e lying entirely in one face f_i , draw a self loop at p_i intersecting that e exactly once.

By using this procedure we get a new graph G^* which is known as Dual of G .

Example





Maximal Planar Graph

A graph G is maximal planar if G is planar but for any pair of non adjacent vertices u & v of G , $G + uv$ is nonplanar

Note:

① Any Planar Graph is a spanning subgraph of a maximal planar graph

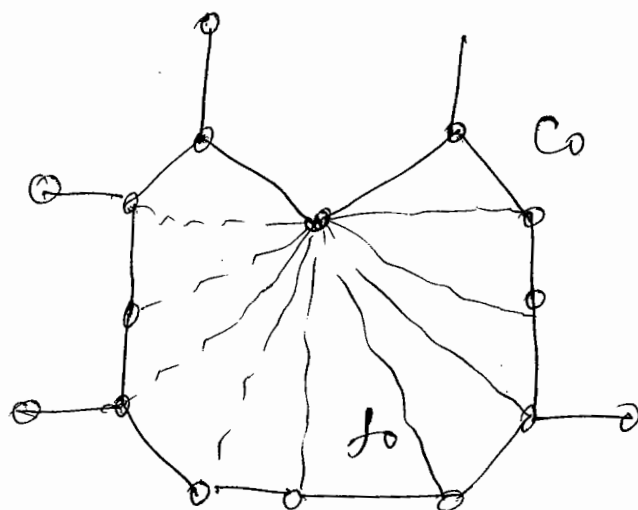
Plane triangulation

A plane triangulation is a plane graph in which each of its faces is bounded by a triangle.

A plane triangulation of a plane graph G is a plane triangulation H s.t G is a spanning subgraph of H .

Example

(29)

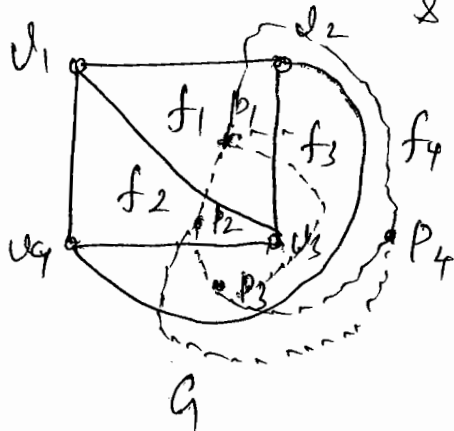


Self Dual Graphs:

If a planar graph G is isomorphic to its own dual, it is called a self dual graph.

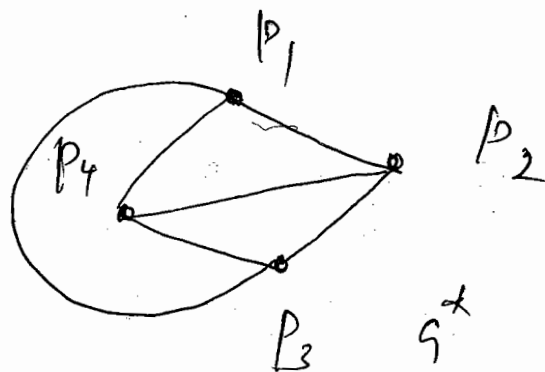
Example

Let K_4 be the complete graph & planar



It has 4 faces f_1, f_2, f_3, f_4

Then its Dual:
(Dual of G)



$$\therefore G^* \cong G$$

$\therefore G$ is a self dual graph

Ques (Exercise)

- ① Show that the complete graph of four vertices is self dual. Give another example of a self dual graph.
- ② Prove that every face in a minimal planar graph is a triangle.
- ③ Find the edge connectivity of the complete graph of n vertices.

Sol

W.K.T,

The vertex Connectivity $\leq$ edge Connectivity
 $\leq \frac{2e}{n}$

For a complete graph,

$$\text{edge Connectivity} = \frac{2e}{n} \quad \text{where } e = \frac{n(n-1)}{2}$$

$$= \frac{2 \left(\frac{n(n-1)}{2} \right)}{n} = n-1$$

-
- ④ Find the ^{rank} nullity of a complete graph with n vertices.

Sol rank = $n-1$ & Nullity = $e - n + 1$
for a connected graph with n vertices & e edges.

(31)
* Since Given graph is a complete graph
with n vertices & $e = \frac{n(n-1)}{2}$ edges

* Rank = $n - 1$

* Nullity = $e - n + 1$

$$= \frac{n(n-1)}{2} - (n-1)$$

$$= (n-1)\left(\frac{n}{2}-1\right) = \frac{(n-1)(n-2)}{2}$$

(5) Prove that every Connected graph
with three (or) more vertices has atleast
two vertices which are not cut vertices.

(6) The edge & vertex connectivity of
a tree are equal to 1.

Proof

* Let T be a tree with n vertices
& $(n-1)$ edges

* Since removal of any one edge
disconnect T .

* Edge connectivity = 1 for a tree

* Also the removal of non pendant vertex
in a tree, ~~the~~ the graph will be disconnected

* vertex connectivity = 1

- ⑦ If a connected graph G is separable, then there exists a subgraph g of G which contains only one vertex in common with its complement $\bar{g}$.

Proof

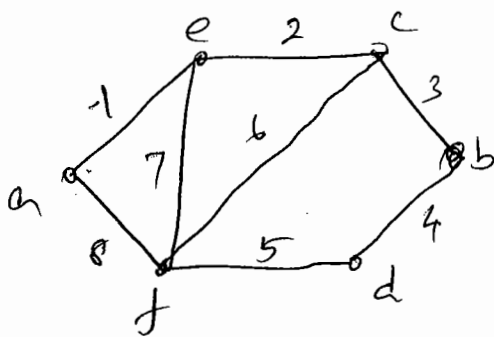
- * Let G be a separable graph
- * Then $K(G) = 1$ & v is a cut vertex in G .
- * G is disconnected into two components G_1 & G_2
- * Since the vertex set V_1 & V_2 of G_1 & G_2 respectively are disjoint.
- * Let $g = G_1 \cup v$ then its edge complement $\bar{g} = G_2 \cup v$
- * g & $\bar{g}$ contains only one vertex v in common.

Network flow:Capacity:

Let G be a Connected weighted graph.
 Each edge $e \in E$ is assigned a positive real no. -
 $c(e)$ called as Capacity of e ,

Cutset:

A Cutset with respect to a pair of vertices a and b in a connected graph G puts a & b into two different components.

Example

Since $a, b, c, d, e, f \rightarrow$ vertices &

$1, 2, 3, 4, 5, 6, 7, 8 \rightarrow$ weight (or) Capacity of the edges.

Note:

* There are two vertices, namely a source S & sink T

* Vertices other than the source & sink are called intermediate vertices.

* The given graph is denoted as a network (N)

Flow function:

A flow function (or) a flow on the network $N = G(V, E)$ is a non negative function $f: E \rightarrow \mathbb{R}$ with the following choice.

- ① For each edge, $e \in E$
- $f(e) \rightarrow$ flow value of e
- $c(e) \rightarrow$ Capacity of e

w.r.to $0 \leq f(e) \leq c(e)$

- ② For every intermediate vertex $v \in V$,
We have,

$$\sum_{u \in V} f(u, v) = \sum_{w \in V} f(v, w), \quad v, w \neq s, t.$$

where $s \rightarrow$ source, $t \rightarrow$ sink

i.e. Inflow = Outflow at each $v \neq s, t$.

(3) The maximum flow through an edge cannot exceed its capacity

Example:

In the networks of telephone-lines, highways, oil (gas (or) water) pipelines;

- * The vertices are stations
- * The edges are lines of commodity flow
- * The maximum amount of flow per unit time along a line of commodity flow is given by the capacity of the edge that represents the flow line.

Theorem (Maximum flow - Minimum cut)
 The maximum flow possible between two vertices a & b in a network is equal to the minimum of the capacities of all cut sets w.r. to a & b .

Proof

- * Let S be any arbitrary cutset with respect to vertices a & b in G

* $G-S$ (The Subgraph left after removing S from G)

is disconnected

* $G-S$ consists of at least two components namely G_1 & G_2

* Since $G-S$ is disconnected

* $a \in G_1$, also $b \in G_2$

therefore there is no path between a & b

* Since Every path in G between a & b contains at least one edge of S .

* Therefore, Total rate of flow between a & b cannot exceed the sum of the edge capacities in S .

* hence cannot exceed the capacity of S .

* This holds for all cutsets w.r. to a & b .

* The flow rate cannot exceed the minimum of the capacities.

1- Isomorphism

Component

A maximal Connected ^{Sub}graph is its Component.

Separable & non separable Graphs

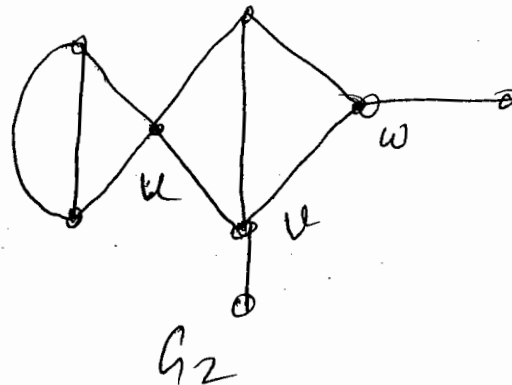
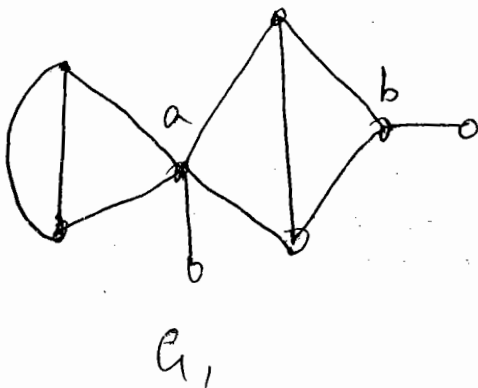
* A graph which admits a cut vertex is called Separable otherwise nonseparable.

* A separable graph consists two or more non separable subgraphs.

Blocks (or) Components

Each non separable subgraphs are said to be Blocks (or) Components

Example



In G_1 , a & b are cut vertices

since $G_1 - \{a\}$ & $G_1 - \{b\}$ gives 4 blocks.

III) $G_2 - \{a\}$, $G_2 - \{b\}$ gives another 4 blocks.

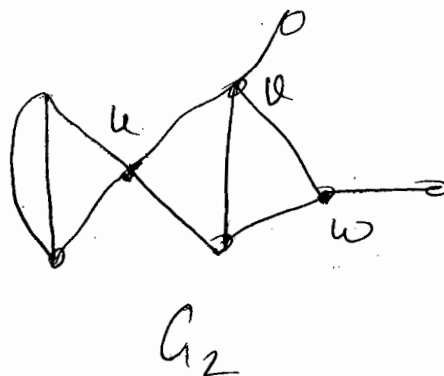
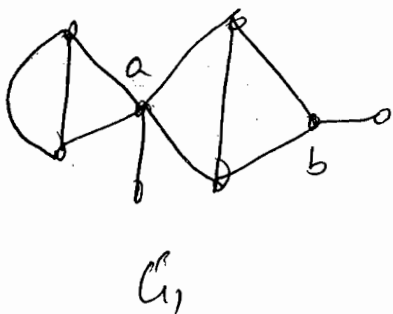
1-isomorphism

Operation: Split a cut vertex into two vertices to produce two disjoint subgraphs which is known as operation-1.

1-isomorphism:

Two graphs G_1 & G_2 ~~be two~~ are said to be 1-isomorphic if each and every block _{in G_1} are isomorphic to all blocks in G_2 by doing repeated application of operation-1.

Example



Theorem: 4.14

If G_1 & G_2 are two 1-isomorphic graphs,
 the rank of G_1 equals the rank of G_2 and
 the nullity of G_1 equals the nullity of G_2 .
 i.e., Rank & Nullity are invariant
 under 1-isomorphism.

Proof:

* By using operation 1,
 A cutvertex of a graph split
 G into connected component.

* Also ~~each~~ cutvertex split into atleast
 two vertices, the no. of components in
 G increase by atleast one.

* By proceeding this way,
 In G_1 with n_1 no. of vertices
 by using operation-1

Case-1

In G_1 the rank

$$= \text{no. of vertices in } G_1 - \text{no. of components in } G_1$$

$$= (n_1 + k) - (w + k)$$

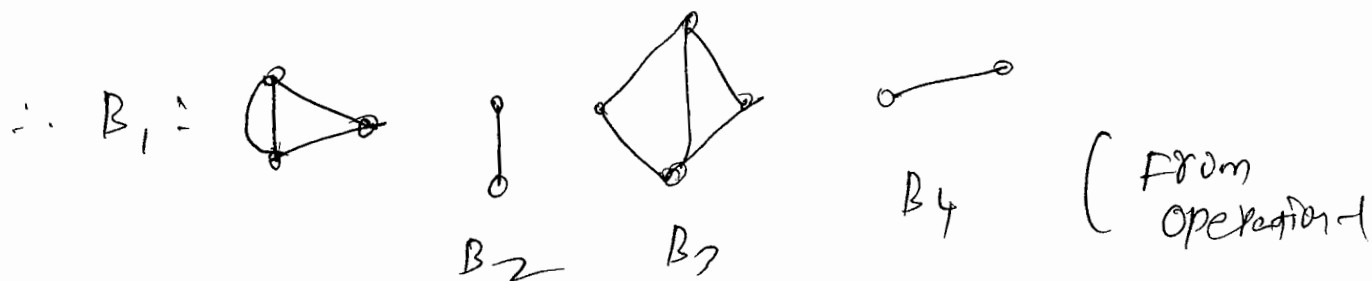
$$= (n_1 - w)$$

where $w \rightarrow$ no. of components
 $n_1 \rightarrow$ no. of vertices

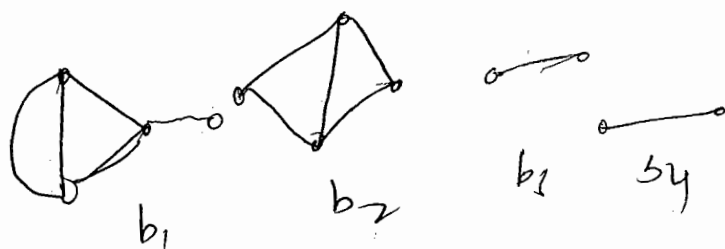
$k \rightarrow$ no. of cutvertices splitted

(39)

In G_1 , The cut vertices are a, b



In G_2 , The cut vertices are u, v, w
& the blocks are (by doing operation-1)



Since $b_1 \cong B_1$, $b_2 \cong B_2$, $b_3 \cong \cancel{B_3}$

& $b_4 \cong B_4$

But $G_1 \not\cong G_2$

& G_1 is 1-isomorphic G_2

Since blocks in G_1 is isomorphic to block in

G_2 .

$\therefore G_1$ & G_2 are 1 isomorphic.

(41)

Case - 2

iii) In G_2 with n_2 no/- of vertices

By using operation 1,

$$\text{Rank of } G_2 = (n_2 + k) - (w + k) \\ = n_2 - w$$

where $w \rightarrow$ no/- of components
 $k \rightarrow$ no/- of ^{times} a ~~split~~ cut vertex splitted into vertices.

$\therefore$ Nullity = no/- of edges - Rank.

$\therefore$ Rank & Nullity is invariant under 1-isomorphism.

Note

① Graphs that are isomorphic are also 1-isomorphic but 1-isomorphic graphs may not be isomorphic.

operation-2 (Twisting)

Let G be a 2 connected graph
(vertex connectivity is two) and (x, y) be
a pair of vertices whose removal from
 G will leave the remaining graph
disconnected.

* Split the vertex x into vertices x_1
& x_2 and the cut-vertex y into
vertices y_1 & y_2 . ~~such that~~

* G is split into two components
 G_1 & G_2 s.t. $x_1, y_1 \in G_1$ & $x_2, y_2 \in G_2$

* Then merge x_1 with y_2 as single
vertex x' & merge x_2 with y_1 as a
single vertex y' .

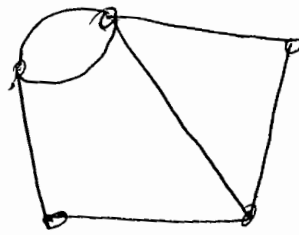
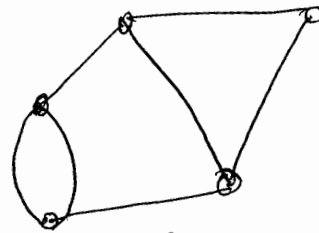
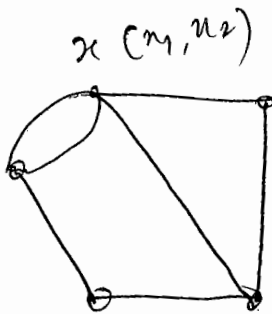
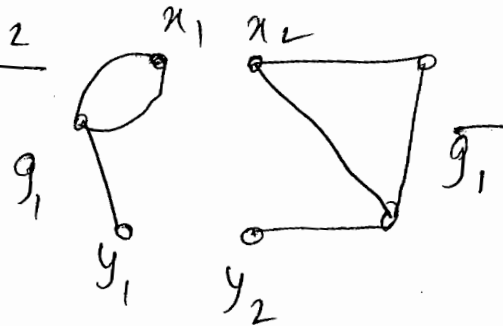
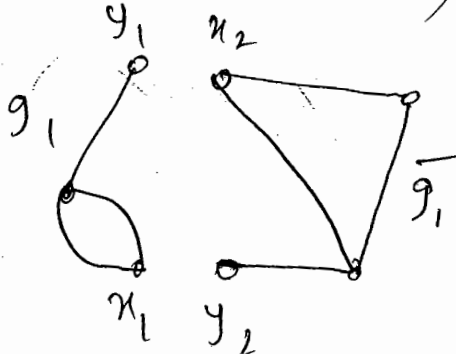
* Then we get a twisted graph G_t .
We get a twisted form of G through
operation 2.

2-Isomorphic graphs

Two graphs G_1 & G_2 are 2-isomorphic if they become isomorphic undergoing the repeated application of operation-1 (or) operation-2 (or) both operations 1 & 2 any no/- of times.

Example

(1)

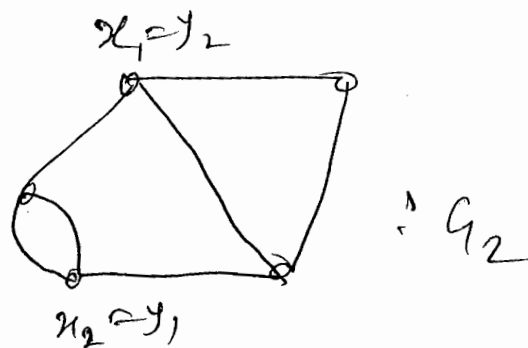
 G_1  G_2 ProofStep-1 G_1 Step-2Step-3

where $g_1 \rightarrow$ Subgraph
 $\overline{g_1} \rightarrow$ Complement of g_1

Step - 4

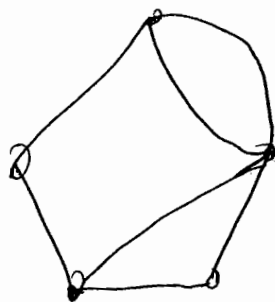
Merge x_1 with y_2

& x_2 with y_1

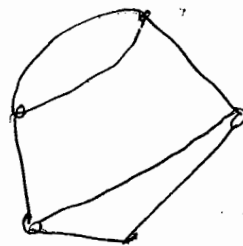


$\therefore G_1$ is 2 isomorphic to G_2

② Check whether the two graphs are 2 isomorphic



G_1



G_2

Note:

① Isomorphism $\Rightarrow$ 1-isomorphism

1 isomorphism $\Rightarrow$ 2-isomorphism converse is not true.

② A Separable graph can be 1-isomorphic to a disconnected graph (but cannot be isomorphic)

- ③ Two non separable graphs are 1-isomorphic if and only if they are isomorphic
- ④ Two nonseparable graphs with vertex connectivity 2 or more are 2-isomorphic if and only if they are 1-isomorphic
- if and only if are 1-isomorphic

Theorem

1-isomorphic graphs have circuit correspondence.

Proof

- * Two blocks of a separable graph G are edge disjoint & contain at most one common vertex
- * Two edges belongs to the same block. if and only if there exist a circuit in B containing both.
- * Since there is no destruction (or) creation of ^{new} edges under operation - 1.
- * Every circuit retains its edges

* Hence 1- isomorphic graphs have circuit correspondence.

Theorem:

2- Isomorphic graphs have Circuit Correspondence.

Proof

* Let C be a circuit made of edges of G_1

* By applying operation 2 to get two components g_1 & $g_2 = \bar{g}_1$.

* If C lies completely in g_1 or g_2 , C will retain same edges.

* If C has edges from both g_1 & g_2 which includes both the cut vertices in x & y then again edges of C are retained but with a twisted path between x & y .

* 2 isomorphic graphs have Circuit Correspondence

Defn of Circuit Correspondence

Two graphs G_1 & G_2 are said to be Circuit Correspondence, If they satisfies the following condition,

i) $\exists$ 1-1 Correspondence between the edges of G_1 & G_2

ii) $\exists$ 1-1 Correspondence between the circuits of G_1 & G_2 such that a circuit in G_1 formed by certain edges of G_1 has a corresponding circuit in G_2 formed by the corresponding edges of G_2 & vice versa.

iii) Isomorphic graphs obviously have circuit correspondence.

Unit -4 - MAT 206 - GT

Matrix Representation of Graphs,
Coloring Covering & Partitioning.

Incidence Matrix - Submatrices - Circuit matrix.

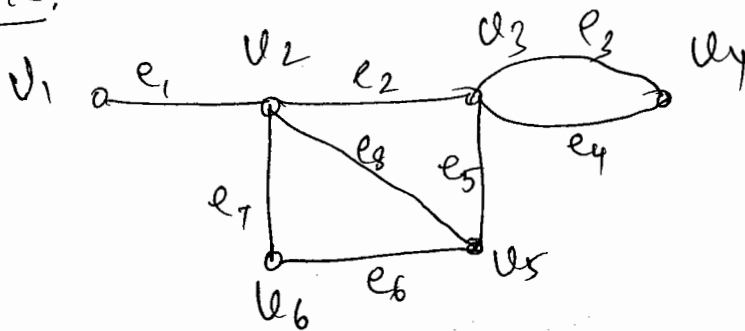
— Path matrix - adjacency matrix - Chromatic
number - Chromatic Partitioning - Chromatic
Polynomial - matching - Covering - four color
Problem.

Incidence Matrix

Let G be a graph with n vertices
& e edges and no self-loops.

Then the incidence matrix is defined
by $A = [a_{ij}]_{n \times e} = \begin{cases} 1, & \text{if } e_j \text{ is incident with } v_i \\ 0, & \text{otherwise.} \end{cases}$

Example:



$$A(G) = \begin{matrix} & e_1 & e_2 & e_3 & e_4 & e_5 & e_6 & e_7 & e_8 \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \\ v_6 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \end{bmatrix} \end{matrix}$$

Note:

- * The incidence matrix contains only two elements 0 & 1.
- * The incidence matrix is also called as a binary matrix.
- * The element of $A(G)$ are only 0 & 1
 $\therefore$ They are the elements of field of integers modulo 2.

Properties of Incidence Matrix:

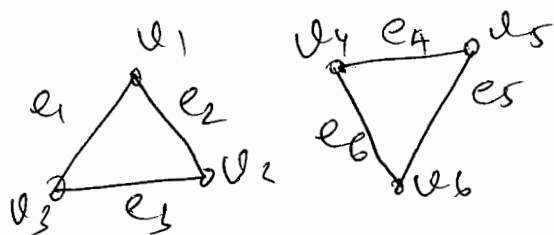
① Since every edge is incident with exactly two vertices, each column of A has exactly two vertices.

② The no. of 1's in each row equals the degree of the corresponding vertex.

③ A row with all 0's represents an isolated vertex.

④ Parallel edges in a graph produce identical columns in its incidence matrix.

⑤ let us consider two components of a disconnected graph G ,



Then $A = [a_{ij}] =$

	e_1	e_2	e_3	e_4	e_5	e_6
v_1	1	1	0	0	0	0
v_2	0	1	1	0	0	0
v_3	1	0	1	0	0	0
v_4	0	0	0	1	0	1
v_5	0	0	0	1	1	0
v_6	0	0	0	0	1	1

* For a disconnected graph which consists two components G_1 & G_2 the incidence matrix

$$A(G) = \begin{bmatrix} A(G_1) & | & 0 \\ \hline 0 & | & A(G_2) \end{bmatrix}$$

where $A(G_1)$ & $A(G_2)$ are the incidence matrices of G_1 & G_2 .

⑥ Permutation of any two rows & two columns in an incidence matrix corresponds to relabelling the vertices & edges of the graph.

Theorem 1
Two graphs G_1 & G_2 are isomorphic if and only if their incidence matrices $A(G_1)$ & $A(G_2)$ differ only by permutation of rows & columns.

Proof:
* Let G_1 & G_2 be isomorphic

then $\exists$ a 1-1 correspondence between the vertices & edges in G_1 & G_2 s.t. incidence relation ship is preserved.

* $A(G_1)$ & $A(G_2)$ are same
(or) differ only by permutation of rows & columns.

* Conversely,

* Let $A(G_1)$ & $A(G_2)$ are same (or) differ only by permutation of rows & columns

* These permutation of rows & columns in an incidence matrix corresponds to relabeling the vertices & edges of the same graph.

* $G_1 \cong G_2$

(5)

Theorem - 2 Rank of the Incidence Matrix.

The rank of the incidence matrix $A(G)$ of a connected graph G with n vertices & e edges is $n-1$.

Proof

* let $A(G) = [a_{ij}]_{n \times e}$ be the incidence matrix of a graph G .

* The no/- of linearly independent row (or column) vectors of $n \times e$ matrix gives its rank.

$$* \quad r(A) \leq \min\{e, n\}$$

* Since G is connected,

$\therefore e \geq n-1$ { for tree (minimally connected) with n vertices we have $(n-1)$ edges }

$$* \quad r(A) = \min\{e, n\} \geq n \text{ (or) } (n-1)$$

* If $\min\{e, n\} = n-1$ then

$$r(A) \leq n-1$$

* Suppose $\min\{e, n\} = n$.

Since each column vector C_i of A has exactly two 1's, each entry of the sum vector $C_1 + C_2 + \dots + C_e$ is an even

$$* \quad c_1 + c_2 + \dots + c_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \pmod{2} \equiv 0 \pmod{2}$$

* This Shows that,

any one of n column vectors is a linear combination of the remaining $(n-1)$ vectors

* The no/- of Linearly independent vectors is less than or equal to $n-1$

$$* \quad r(A(G)) \leq n-1$$

* TO Prove: $r(A(G)) \geq n-1$

* Let $A_1, A_2, \dots, A_n$ be the row vectors of $A(G)$

$$* \quad \text{Then } A(G) = \begin{bmatrix} A_1 \\ A_2 \\ \vdots \\ A_n \end{bmatrix}$$

* Consider any r row vectors $A_1, A_2, \dots, A_r$ where $r < n$

If $A_1 + A_2 + \dots + A_r = 0$ then there is no edge connection any of these r vertices to any of remaining $(n-r)$ vertices.

* which is a contradiction for connectedness.

- * $A_1 + A_2 + \dots + A_r \neq 0 \pmod{2}$
- * Fewer than n row vectors cannot be linearly independent.
- * The sum of n rows is $0 \pmod{2}$
- * The no. of linearly independent row vectors should be at least $(n-1)$
- * $r(A(G)) = n-1$

Reduced Incidence Matrix:

Let $A(G)$ be the incidence matrix $A = [a_{ij}]_{n \times e}$ & $r(A) = n-1$. The removal of any one of n row vectors from $A(G)$ will give an $(n-1) \times e$ submatrix with rank $(n-1)$, called a reduced incidence matrix. A_f {Note: The vertex corresponds to the deleted row A_f is called the reference vertex y .

Theorem:

The reduced incidence matrix of a tree is non-singular.

Proof

* Since every tree is a connected graph with n vertices & $e = n-1$ edges.

* Since The incidence matrix,

$$A(G) = [a_{ij}]_{n \times e} = [a_{ij}]_{n \times n-1} \quad \& \quad \gamma(A) = n-1$$

& The reduced incidence matrix,

$$A_f(G) = [a_{ij}]_{n-1 \times n-1} \quad \text{is a square}$$

matrix & $\gamma(A_f) = n-1$

* $A_f(G)$ is non singular

Theorem: 7.3

Let $A(G)$ be an incidence matrix of a connected graph G with n vertices. An $(n-1) \times (n-1)$ submatrix of $A(G)$ is nonsingular if and only if the $(n-1)$ edges corresponding to $(n-1)$ columns of this matrix constitute a spanning tree in G .

Proof

* Every square submatrix of order $(n-1)$ in $A(G)$ is the reduced incidence matrix of the same subgraph in G with $(n-1)$ edges vice versa.

* Clearly submatrix of $A(G)$ is nonsingular if and only if the corresponding subgraph is a tree.

- * The tree is minimally connected
- & it's a Spanning Subgraph of G
- * Tree is a Spanning tree
- & it contains n vertices with $(n-1)$ edges of G

Definition : Submatrices of $A(G)$

Let g be a Subgraph of G & let $A(g)$ & $A(G)$ be the incidence matrices of g & G respectively.

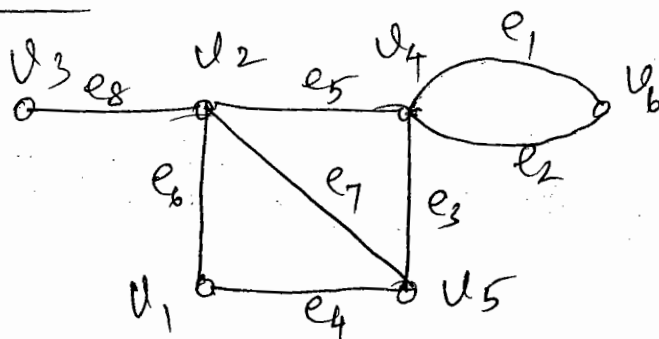
Clearly $A(g)$ is a Submatrix of $A(G)$

Circuit Matrix:

The Circuit matrix of a given graph G is defined as $B = [b_{ij}]$ as $(0,1)$ matrix

$$B = [b_{ij}] = \begin{cases} 1 & \text{if } e_j \in C_i \\ 0 & \text{otherwise} \end{cases}$$

Example :



The graph has 4 different circuits.

$$C_1 = \{e_1, e_2\}, \quad C_2 = \{e_3, e_5, e_7\},$$

$$C_3 = \{e_4, e_6, e_7\} \quad \& \quad C_4 = \{e_3, e_4, e_5, e_6\}$$

Then the circuit matrix is

$$B(G) = \begin{matrix} & e_1 & e_2 & e_3 & e_4 & e_5 & e_6 & e_7 & e_8 \\ \begin{matrix} C_1 \\ C_2 \\ C_3 \\ C_4 \end{matrix} & \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \end{bmatrix} \end{matrix}$$

Properties of circuit matrix:

* A column of all zeros corresponds to a non circuit edge (i.e.) an edge that does not belong to any circuit.

* Each row of $B(G)$ is a circuit vector

* Self loop will be allowed in a circuit matrix & the corresponding row will have a single 1.

* The no. of 1's in a row is equal to the no. of edges in the corresponding circuit.

(11)

* If a graph G is separable (or disconnected) and consists of two blocks (or components) G_1 & G_2 , the circuit matrix $B(G)$ can be written in a block diagonal form as

$$B(G) = \begin{bmatrix} B(G_1) & 0 \\ 0 & B(G_2) \end{bmatrix}$$

where $B(G_1)$ & $B(G_2)$ are the circuit matrices of G_1 & G_2 .

* permutation of any two rows (or) columns in a circuit matrix simply corresponds to relabeling the circuits & edges.

* Two graphs G_1 & G_2 will have same circuit matrix if and only if G_1 & G_2 are isomorphic.

Theorem:

Let B & A be respectively, the circuit matrix & the incidence matrix (of a self loop free graph) whose columns are arranged using the same order of edges.

Then every row of B is orthogonal to every row A . i.e.,

$$A \cdot B^T = B \cdot A^T = 0 \pmod{2}$$

where superscript T denotes the transposed matrix.

Proof

* let v be a vertex & π be a circuit in the graph G .

* Case-1

If $v \in \pi$ then the no. of edges incident with v is exactly two

Case-2

If $v \notin \pi$ then there is no edge incident on v in π .

* let i th row in A & j th row in B can be considered.

* Since the edges are arranged in the same order, the non-zero entries in the corresponding positions occur only if the particular edge is incident on the i th vertex & also in j th circuit

(13)

* If i th vertex is not in j th circuit then there is no such nonzero entry, & the product of the two rows is zero.

* If i th vertex is in j th circuit, there will be exactly two 1's in the sum of the products of individual entries.

* Since $1+1 = 0 \pmod{2}$, the dot product of the two arbitrary rows - one from A & the other from B - is zero.

Hence the theorem.

Theorem:

If B is a circuit matrix of a connected graph G with e edges & n vertices,

$$r(B) = e - n + 1.$$

Proof:

If A is an incidence matrix & B is a circuit matrix then

$$A \cdot B^T = 0 \pmod{2}$$

By Sylvester's theorem,

$$r(A) + r(B) \leq e$$

$$\gamma(B) \leq e - \gamma(A)$$

$$\leq e - (n-1)$$

$$\gamma(B) \leq e - n + 1 \quad \text{--- (1)}$$

$$\text{Also } \gamma(B) \geq e - n + 1 \quad \text{--- (2)}$$

$\therefore$ from (1) & (2)

$$\gamma(B) = e - n + 1$$

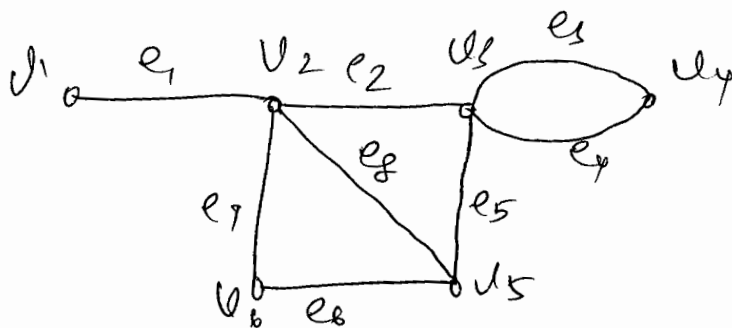
Path Matrix:

The path matrix is defined as

$$P(x,y) = P_{ij} = \begin{cases} 1 & \text{if } j\text{th edge lies in } i\text{th path} \\ 0 & \text{otherwise} \end{cases}$$

where $P(x,y) \rightarrow$ different path between the pair of vertices (x,y)

Example:



$$P(v_1, v_5) = ?$$

Between v_1 to v_5 , the paths are,

$$P_1 = \{e_1, e_2, e_5\}, P_2 = \{e_1, e_7, e_6\}$$

$$P_3 = \{e_1, e_8\}$$

$$\therefore P(v_1, v_5) = P_{ij} = \begin{matrix} & e_1 & e_2 & e_3 & e_4 & e_5 & e_6 & e_7 & e_8 \\ \begin{matrix} P_1 \\ P_2 \\ P_3 \end{matrix} & \begin{bmatrix} 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

Theorem:

If the edges of a connected graph are arranged in the same order for the columns of the incidence matrix A and the path matrix $P(x, y)$ then the product (mod 2)

$$A \cdot P^T(x, y) = M$$

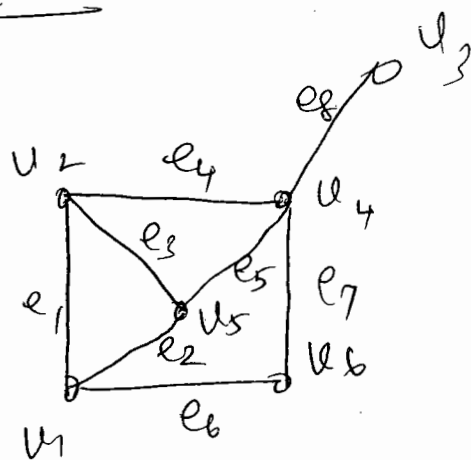
where M is the matrix has 1's in two rows x & y and the rest of the $(n-2)$ rows are all 0's.

Adjacency Matrix:

Let G be a graph with n vertices & no parallel edges then the adjacency matrix is defined as

$$X = [x_{ij}] = \begin{cases} 1 & \text{if } e_i \in v_i \times v_j \\ 0 & \text{otherwise} \end{cases}$$

Example



$$X = \begin{matrix} & \begin{matrix} u_1 & u_2 & u_3 & u_4 & u_5 & u_6 \end{matrix} \\ \begin{matrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \end{bmatrix} \end{matrix}$$

Properties of Adjacency Matrix:

① The entries along the principal diagonal of X are all 0's if and only if the graph has no self loops.

② A self loop at u_i corresponds to $x_{ii} = 1$.

③ The definition of adjacency matrix makes no provision for parallel edges. This is why adjacency matrix X was defined for graphs without parallel edges.

④ If the graph has no self loops (no parallel edges) degree of a vertex equals the no. of 1's in the corresponding row (or) column of X .

- ④ Permutation of rows & the corresponding columns imply re ordering the vertices. Thus, If two rows are interchanged in X , the corresponding columns must also be interchanged.

Hence two graphs G_1 & G_2 with no parallel edges are isomorphic if and only if

$$X(G_2) = P^T X(G_1) \cdot P$$

Where P is a permutation matrix.

- ⑤ A graph G is disconnected & is in two components G_1 & G_2 if and only if its adjacency matrix $X(G)$ can be partitioned as

$$X(G) = \begin{bmatrix} X(G_1) & 0 \\ 0 & X(G_2) \end{bmatrix}$$

$X(G_1) \rightarrow$ Adjacency matrix of the component G_1 ,

$X(G_2) \rightarrow$ " " of the component G_2

which implies that, if $v_i \in g_1$, then $v_i \notin g_2$
 i.e., There exists no edge joining any
 vertex in Subgraph g_1 to any vertex in
 Subgraph g_2

(b) Given any square, symmetric, binary
 matrix A of order n , one can construct
 a graph G of n vertices s.t A is
 the adjacency matrix of G .

Power of X^0

$$X^2 = X \cdot X = \begin{bmatrix} 3 & 1 & 0 & 3 & 1 & 0 \\ 1 & 3 & 1 & 1 & 2 & 2 \\ 0 & 1 & 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 4 & 1 & 0 \\ 1 & 2 & 1 & 1 & 3 & 2 \\ 0 & 2 & 1 & 0 & 2 & 2 \end{bmatrix}$$

The value of an off-diagonal ^{entry} ~~matrix~~
 in X^2 , is i th entry ($i \neq j$) in X^2

= no/- of 1's in the dot product of
 i th row & j th column of X

= no/- of positions in which both i th
 & j th rows of X have 1's.

$\hat{=}$ no/- of vertices in which both i^{th} & j^{th} rows of X have 1's

$\hat{=}$ no/- of vertices that are adjacent to both i^{th} & j^{th} vertices

$\hat{=}$ no/- of different paths of length two between i^{th} & j^{th} vertices

Note:

If the graph has no self loops,
then a matrix commutes with matrices
that are its own power,
 $X \cdot X^2 = X^2 \cdot X = X^3$.

Then

$$X^3 = \begin{bmatrix} 2 & 7 & 3 & 2 & 7 & 6 \\ 7 & 4 & 1 & 8 & 5 & 2 \\ 3 & 1 & 0 & 4 & 1 & 0 \\ 2 & 8 & 4 & 2 & 8 & 7 \\ 7 & 5 & 1 & 8 & 4 & 2 \\ 6 & 2 & 0 & 7 & 2 & 0 \end{bmatrix}$$

let us consider i^{th} entry of X^3

i^{th} entry of $X^3 =$ dot product of i^{th} row of X^2
& j^{th} column of (or row) of
 X

$$= \sum_{k=1}^n i k^{\text{th}} \text{ entry of } X^2 \cdot k_j^{\text{th}} \text{ entry of } X.$$

$$= \sum_{k=1}^n \text{no/- of all different edge sequences of three edges from } i^{\text{th}} \text{ to } j^{\text{th}} \text{ vertex via } k^{\text{th}} \text{ vertex}$$

$$= \text{no/- of different edge sequences of three edges between } i^{\text{th}} \text{ \& } j^{\text{th}} \text{ vertices.}$$

For example.

Row 1 of X^2 . Row 5 of X

$$= (3, 1, 0, 3, 1, 0) \cdot (1, 1, 0, 1, 0, 0)$$

$$= 3 + 1 + 0 + 3 + 0 + 0 = 7$$

$\therefore$ There are seven different edge sequences of three edges between u_1 & u_5 are

$\{e_1, e_1, e_2\}$, $\{e_2, e_2, e_2\}$, $\{e_6, e_6, e_2\}$,
 $\{e_2, e_3, e_3\}$, $\{e_6, e_7, e_5\}$, $\{e_2, e_5, e_5\}$,
 $\{e_1, e_4, e_5\}$.

Theorem:

Let x be the adjacency matrix of a simple graph G . Then the ij th entry in x^r is the no/ of different sequences of r edges between vertices v_i & v_j .

Proof

Proof by mathematical induction on r .
Let us assume that $P(r)$ be the statement.

for $r=1$, $P(1) = x^1$

$\therefore$ The ij th entry in x is the no/ of different paths of length ^{one} between i th & j th vertices.

$\therefore P(1)$ is true

& Each edge sequences will have 1 edge between v_i & v_j

Let us suppose that the result is true for $r = k-1$

$\therefore$ The ij th entry on x^{k-1} is the no/ of different sequences of $k-1$ edges

To Prove: $P(k)$ is true.

For $\forall = k$

$$x^{\forall} = x^k = x^{k-1} \circ x$$

x^{k-1} implies that the i th entry of x^{k-1} is the no/- of different ^(k-1) edge sequences between u_i & u_j

also for x it is true.

$\therefore x^k$ implies that the i th entry of x^k is no/- of different k edge sequence between u_i & u_j .

Note:

Properties of Circuit & Path matrix
[refer the text Book]

Chromatic number - Chromatic partitioning -
 Chromatic polynomial - Matching - Covering -
 four color problem.

Proper Coloring

Painting all the vertices of a graph with colors such that no two adjacent vertices have the same color is called proper coloring.

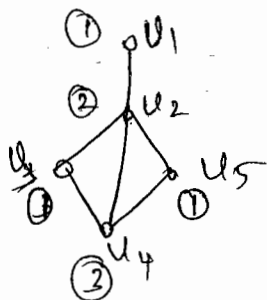
Properly Colored Graph:

A graph in which every vertex has been assigned a color according to a proper coloring is called a properly colored graph.

k-Chromatic Graph & Chromatic number:

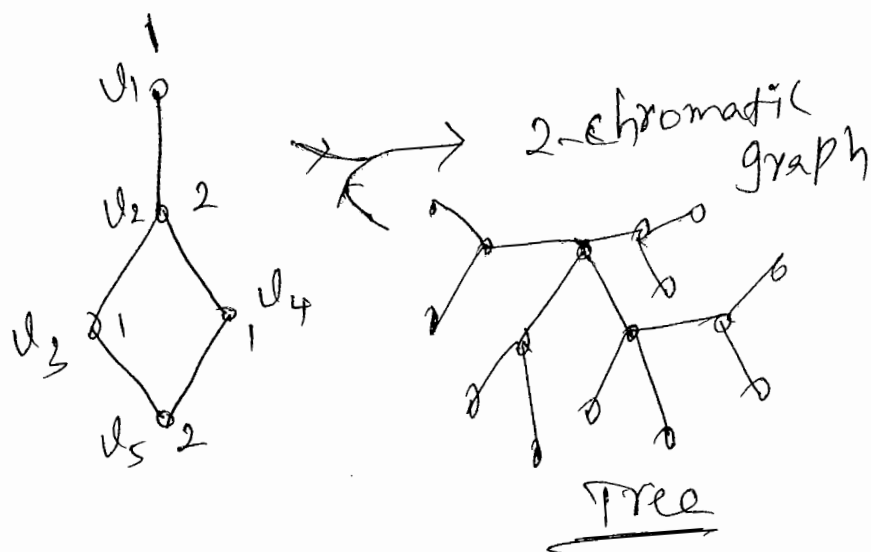
A graph G that requires k different colors for its proper coloring, and no less is called a k -chromatic graph and the number k is called the chromatic number of G .

Example:



— 3 chromatic graph
 (since it is colored
 by color - ①, ② & ③)

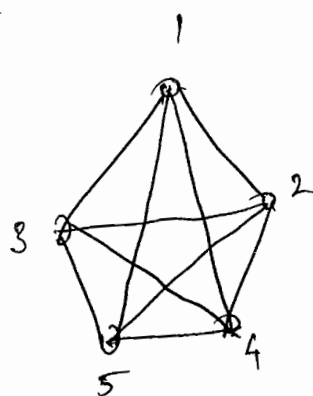
Example - ②



Remarks

- ① A graph consisting of only isolated vertex is 1-chromatic.
- ② A graph with one or more edges is at least 2-chromatic.
- ③ A complete graph with n vertices is n -chromatic.

Example :

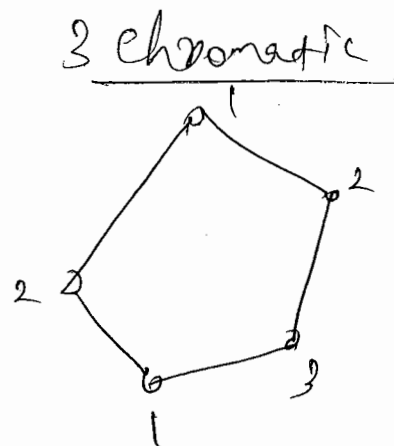
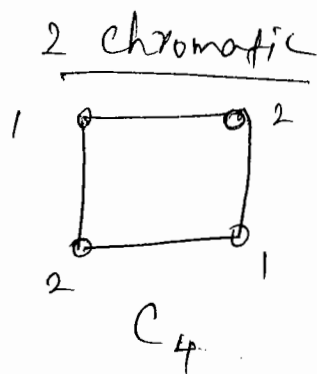


— 5 chromatic graph

Since every pair of vertices are adjacent.

- ④ A graph consisting of simply one circuit with $n \geq 3$ vertices is
 - 2 chromatic if n is even
 - 3 chromatic if n is odd.

Example



Theorem: 8.1

P.T. Every tree with two or more vertices is 2-Chromatic. & check whether the converse is true.

Proof

* let T be a tree with root node as u .

* paint u with color - 1.

* paint all vertices adjacent to u with

color - 2.

* next paint the vertices adjacent to these using color - 1.

* Continuing this process till every vertex in T has been painted.

* Now in T , we find that all vertices at odd distances from u have color 2, while u and vertices at

Even distances from u have color -1 .

* Now Along any path in T the vertices are of alternating colors.

* Since there is one and only ^{one} path between any two vertices in a tree,

* No two adjacent vertices have the same color.

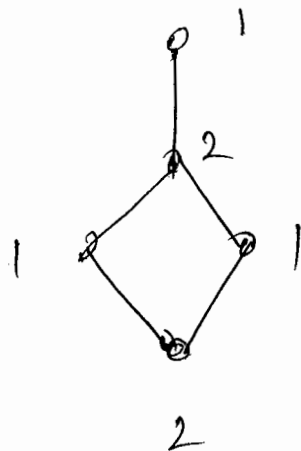
* T has been properly colored with two colors.

* Every tree is 2-chromatic.

Conversely,

The converse is not true.

Since,



This graph is
2-chromatic
but not a tree

∴ Every 2-chromatic graph may not be a tree.

Theorem-2
A graph with at least one edge is 2 chromatic if and only if it has no circuits of odd length.

Proof
* Let G be a connected graph with no circuits of odd length.

to prove: G is 2 chromatic.

* let T be a spanning tree in G
* Then T can be colored properly with

2 colors

* Add the chords to T one by one
* Since G has no circuits of odd length, the end vertices of every chord being replaced are differently colored in T .

* Thus G is colored with 2 colors, with no adjacent vertices having the same color.

* G is 2-Chromatic.
Conversely, G is 2 chromatic,

~~Conversely~~, If G has a Circuit of odd length
then G needs at least 3 Colors for
proper coloring for the circuit.

which is a Contradiction of 2 chromatic.
 $\therefore G$ has no Circuits of odd length.

Chromatic Partitioning

Independent Set:

A Set of vertices in a Graph is said
to be an independent set of vertices (or)
simply an independent if no two vertices
in the set are adjacent.

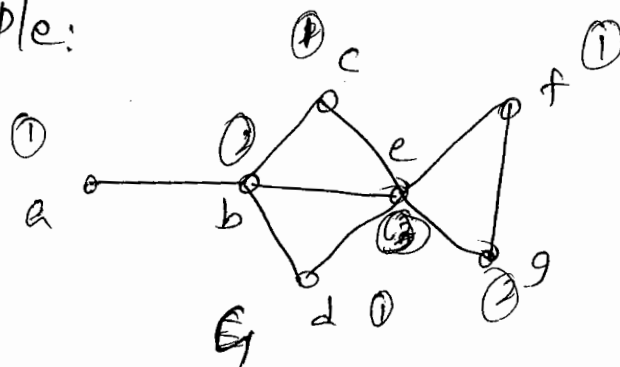
Maximal independent Set:

A maximal independent set is an independent
set to which no other vertex can be added
without destroying its independence property.

Largest independent set:

A Maximal independent set with largest
number of vertices is largest independent
set.

Example:



Independent Set = $\{a, c, d\}$ etc.

Maximal Independent Set = $\{a, c, d, f\}$

Independent no/-

The no/- of vertices in the largest independent set of a graph G is called as the independence number $\beta(G)$

Note:

* While using proper color on a graph, the vertices with same color vertices is also an independent set

Example: From the above graph we can identify

$$\beta(G) = 4.$$

Note:

① Always $\beta(G) \geq \frac{n}{k}$

where $n \rightarrow$ no/- of vertices
" " " " " " " " " " " "

② Minimum no/- of these (maximal independent) Set which includes all the vertices of G Gives the Chromatic no/- of G .

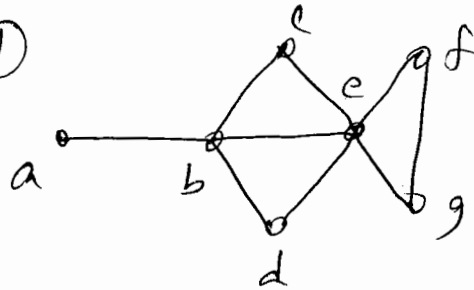
②.

Chromatic partitioning:

For a given simple, Connected graph, G partition all vertices of G into the smallest possible no/- of disjoint, independent sets.

Which is known as Chromatic partitioning of graphs.

Example: ①



The partitions are,

$$\{ (a, c, d, f), (b, g), (e) \},$$

$$\{ (a, c, d, g), (b, f), (e) \},$$

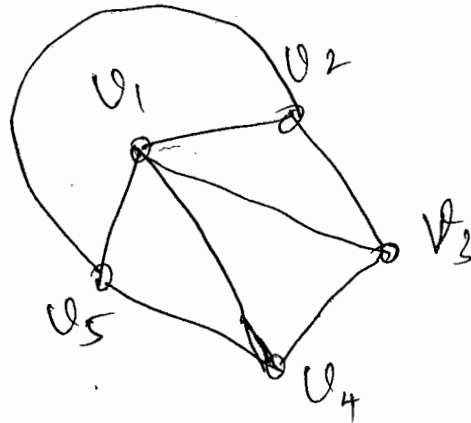
$$\{ (c, d, f), (b, g), (a, e) \},$$

$$\{ (c, d, g), (b, f), (a, e) \}.$$

So the above graph doesn't have a chromatic partitioning

⑨

Example : ②



3 Chromatic Graph.

Which has only one chromatic partition.

Uniquely Colorable Graph

A Graph that has only one chromatic partition is called uniquely colorable graph.

From the examples ① & ②
Example ② is uniquely colorable graph.

Note:

A concept related to that of the independent set and chromatic partitioning is the dominating set.

Dominating set:

A dominating set in a graph G is a set of vertices that dominates

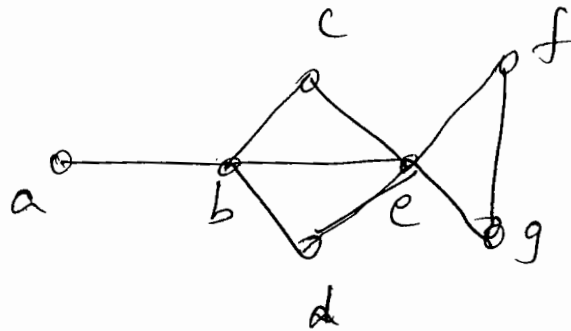
every vertex v in G in the following sense:

- ① Either v is included in the dominating set (or) is adjacent to one or more vertices included in the dominating set.

Minimal dominating set:

A minimal dominating set is a dominating set from which no vertex can be removed without destroying its dominance property.

Example



- ① Dominating sets are:

$\{b, g\}, \{a, b, c, d, f\}$

- ② Minimal dominating set:

$\{b, g\}$

Note:

- ① Any one vertex in a complete graph constitutes a minimal dominating set.
- ② Every dominating set contains at least one minimal dominating set.
- ③ A graph may have many minimal dominating sets & with different sizes.
- ④ The no/- of vertices in the smallest minimal dominating set of a graph G is called the domination number $\alpha(G)$.
- ⑤ A minimal dominating set may or may not be independent.
- ⑥ Every maximal independent set is a dominating set.
- ⑦ An independent set has the dominance property only if it is a maximal independent set.
i.e., An independent dominating set is the same as a maximal independent set.
- ⑧ For any graph G ,
$$\alpha(G) \leq \beta(G)$$

Chromatic polynomial

No. of ways of coloring properly of a complete graph with n vertices & λ colors is called a Chromatic Polynomial.

& it is denoted by,

$$P_n(\lambda) = \lambda(\lambda-1)(\lambda-2) \dots (\lambda-n+1)$$

Theorem:

A graph of n vertices is a complete graph if and only if its chromatic Polynomial is $P_n(\lambda) = \lambda(\lambda-1)(\lambda-2) \dots (\lambda-n+1)$.

Proof:

- * Choose any vertex & paint with colors.
- * The no. of ways of different coloring for the selected vertex is λ .
- * Then choose a second vertex which can be colored properly in exactly $(\lambda-1)$ ways.
- * Similarly 3rd vertex will have $(\lambda-2)$ ways
- & 4th vertex will have $\lambda-3$ ways
- and so on.

* n^{th} vertex will be colored in
 $\lambda - n + 1$ ways if and only if every
 vertex is adjacent to each other.
 * i.e., if and only if the graph is
 complete.

Theorem:

P.T An n -vertex graph is a tree if and
 only if its chromatic polynomial is

$$P_n(\lambda) = \lambda(\lambda-1)^{n-1} = f(G, \lambda)$$

Proof

* Let G be a tree.

* To prove: $f(G, \lambda) = P_n(\lambda) = \lambda(\lambda-1)^{n-1}$
 by induction on ' n '.

* for $n=1$, The result is trivial.

* Assume the result is true for a tree
 with at most $(n-1)$ vertices, $n \geq 2$.

* Let G be a tree with n vertices,
 and e be a pendent edge of G .

$$\begin{aligned} * \quad P_n(\lambda) &= \lambda(\lambda-1)^{n-2} \cdot \lambda-1 \\ &= \lambda(\lambda-1)^{n-1} \end{aligned}$$

Conversely,

let G be a simple graph with

$$\begin{aligned}P_n(\lambda) &= f(G, \lambda) = \lambda(\lambda-1)^{n-1} \\&= \lambda \left(\lambda^{n-1} - (n-1)\lambda^{n-2} + \dots + (-1)^{n-1} \right) \\&= \lambda^n - (n-1)\lambda^{n-1} + \dots + (-1)^{n-1}\lambda\end{aligned}$$

which is a monic polynomial with n vertices (highest degree of the polynomial)

* Also the last term $(-1)^{n-1}\lambda$ ensures that

G is connected

* The next highest degree is $n-1$ & the coefficient of λ^{n-1} is $n-1$ which gives the no. of edges of the graph.

$\therefore G$ is a tree

Theorem:

Prove that the chromatic polynomial of a ~~complete~~ graph G

is $P_n(\lambda)$ of $G = P_n(\lambda)$ of $G' + P_{n-1}(\lambda)$ of G'

Where let a & b be two non adjacent vertices in a graph G & let G' be a graph obtained by adding an edge between a & b .

let G'' be a simple graph obtained from G by fusing the vertices a & b together & replacing sets of parallel edges with single edges, ~~then~~

Proof

* The no/- of ways of properly coloring G can be grouped into two cases, one such that a & b are of the same color & other such that a & b are of different colors.

* Since the no/- of ways of properly coloring G such that a & b have different colors = no/- of ways of properly coloring of G' & no/- of ways of properly coloring G s.t. a & b have the same color.

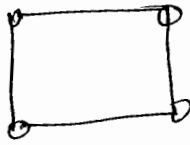
= no/- of ways of proper coloring G''

$$\therefore P_n(\lambda) \text{ of } G = P_n(\lambda) \text{ of } G' + P_{n-1}(\lambda) \text{ of } G''$$

Ques

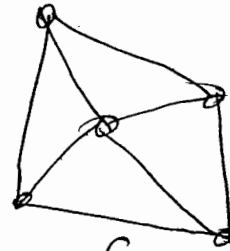
① Find the chromatic polynomial of the given graphs

i)



G_1

ii)



G_2

Solution

i)

$$f(G_1, \lambda) = \left\{ \begin{array}{c} d \quad c \\ \diagup \quad \diagdown \\ a \quad b \end{array} \right\}$$

$$= \left\{ \begin{array}{c} d \quad c \\ \diagup \quad \diagdown \\ a \quad b \end{array} \right\} + \left\{ \begin{array}{c} d \\ | \\ a=c \\ | \\ b \end{array} \right\} \rightarrow \text{merging } a \& c$$

$$= \left\{ \begin{array}{c} d \quad c \\ \diagup \quad \diagdown \\ a \quad b \end{array} \right\} + \left\{ \begin{array}{c} d \\ | \\ a \\ | \\ b \end{array} \right\} \rightarrow \text{deleting multiple edges \& converting as a single edge}$$

$$= \left\{ \begin{array}{c} d \quad c \\ \diagup \quad \diagdown \\ a \quad b \end{array} \right\} + \left\{ \begin{array}{c} b=d \\ \diagup \quad \diagdown \\ a \quad d \end{array} \right\} + \left\{ \begin{array}{c} d \\ | \\ a \\ | \\ b \end{array} \right\} + \left\{ \begin{array}{c} d \\ | \\ a \\ | \\ b \end{array} \right\}$$

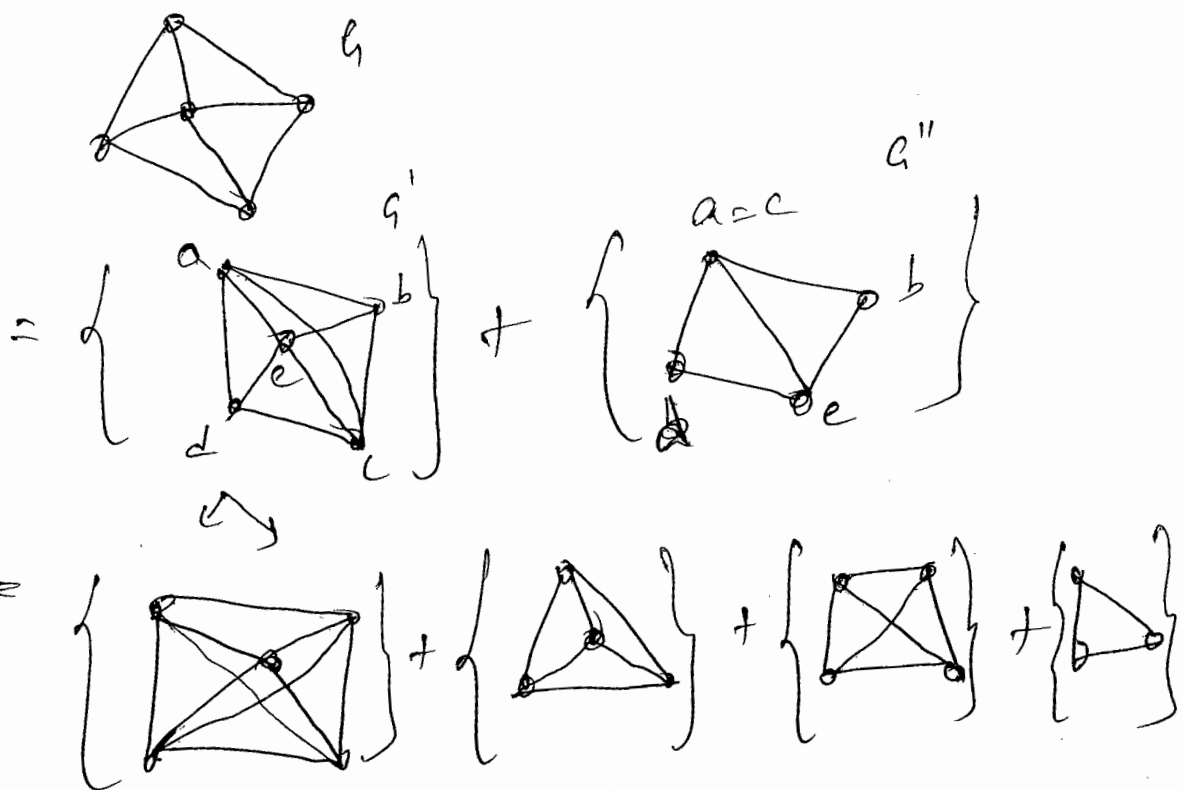
$$= \left\{ \begin{array}{c} d \quad c \\ \diagup \quad \diagdown \\ a \quad b \end{array} \right\} + \left\{ \begin{array}{c} d \\ \diagup \quad \diagdown \\ a \quad b \end{array} \right\} + \left\{ \begin{array}{c} d \\ | \\ a \\ | \\ b \end{array} \right\} + \left\{ \begin{array}{c} d \\ | \\ a \\ | \\ b \end{array} \right\}$$

$$= f(K_4, \lambda) + f(K_3, \lambda) + f(K_2, \lambda) + f(K_2, \lambda)$$

$$\begin{aligned}
 &= f(K_4, \lambda) + 2f(K_3, \lambda) + f(K_2, \lambda) \\
 &= \lambda(\lambda-1)(\lambda-2)(\lambda-3) + 2\lambda(\lambda-1)(\lambda-2) + \lambda(\lambda-1) \\
 &= \lambda^4 - 4\lambda^3 + 6\lambda^2 - \lambda
 \end{aligned}$$

which is a monic polynomial of degree 4

$\therefore G$ will have 4 vertices & the coefficient of λ^2 is 6 which gives the ^{no. of} edges in the given graph.



$$\begin{aligned}
 &= P_5(G) + 2P_4(G) + P_3(G) \\
 &= \lambda(\lambda-1)(\lambda-2)(\lambda-3)(\lambda-4) + 2\lambda(\lambda-1)(\lambda-2)(\lambda-3) \\
 &\quad + \lambda(\lambda-1)(\lambda-2)
 \end{aligned}$$

Matchings

A matching in a graph is a subset of edges in which no two edges are adjacent.

Note

A single edge in a graph is obviously a matching.

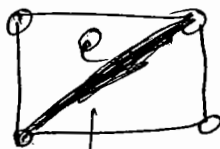
Maximal Matching

A maximal matching is a matching to which no edge in the graph can be added.

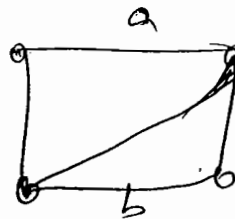
Example: ①

In a complete graph of 3 vertices any single edge is a maximal matching

Example - 2



$\{e\}$ - maximal matching



$\{a, b\}$ → maximal matching

Largest Maximal Matching:

The maximal matchings with the largest number of edges are called
Maximum matchings

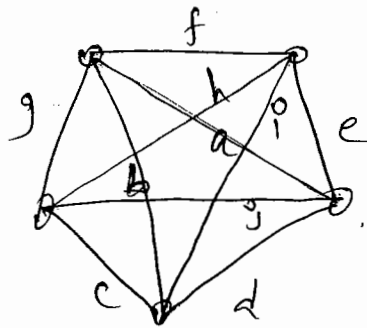
Matching Number:

The no/- of edges in a largest maximal matching is called the matching number of the graph.

Complete matching:

A matching which covers all the vertices of the graph & it's a largest maximal matching

Example



Find the

Complete Matching?



Complete matching in a Bipartite graph

In A Bipartite graphs having a vertex partition V_1 & V_2 , a Complete matching of vertices in set V_1 into those in V_2 is a matching in which there is one edge incident with every vertex in V_1 .

i.e., If every subset of r vertices in V_1 is collectively adjacent to r or more vertices in V_2 for all values of r .

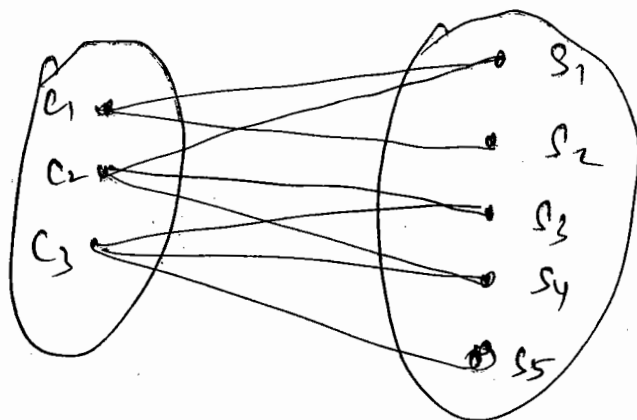
Theorem:

A complete matching of V_1 into V_2 in a bipartite graph exist if and only if every subset of r vertices in V_1 is collectively adjacent to r (or) more vertices in V_2 for all values of r .

Proof

Refer the text book.

Example for a complete matching in a Bipartite Graph



Theorem

In a Bipartite graph a complete matching of V_1 into V_2 exists if there is a +ve integer m for which the following condition is true.

deg of every vertex in $V_1 \geq m \geq$
deg of every vertex in V_2 .

Proof:

* Consider a subset of r vertices in V_1 ,
* These r vertices have at least $m \cdot r$ edges incident on them.

* Each $m \cdot r$ edges is incident to some vertex in V_2 .

* Since the deg of every vertex in set V_2 is no greater than m , these $m \cdot r$ edges are incident on at least $(m \cdot r) / m = r$ vertices in V_2 .

* Thus any subset of r vertices in V_1 is collectively adjacent to r (or) more vertices in V_2 .

* $\exists$ a complete matching of V_1 into V_2 .

Coverings

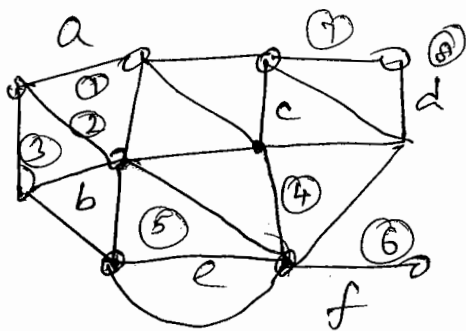
Cover:

In a graph G , a set g of edges is said to cover G if every vertex in G is incident on at least one edge in g .

edge covering:

A set of edges that covers a graph G is said to be an edge covering.

~~Def~~ Examples



Two minimal coverings are

$\{a, b, c, d, e, f\}$

$\Delta \{1, 2, 3, 4, 5, 6, 7, 8\}$

Minimal covering:

A covering in which no edge can be removed without destroying its ability to cover the graph.

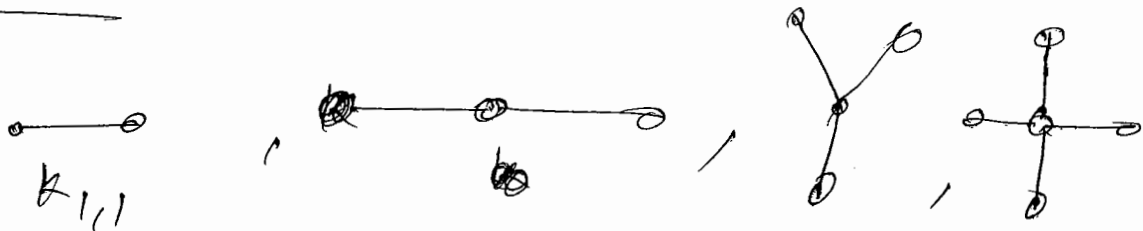
Remarks:

- ① A covering exists for a graph if & only if the graph has no isolated vertex.
- ② A covering of an n vertex graph will have at least $\lceil n/2 \rceil$ edges.
- ③ Every pendant edge in a graph is included in every covering of the graph.
- ④ Every covering contains a minimal covering.
- ⑤ No. of edges in a minimal covering of the smallest size is called covering number of the graph.

⑥ Star graph

$K_{1,n}$ is said to be a star graph and it will have n edges.

Examples:



Star graphs of one, two, three & four edges,

Theorem:

A covering g of a graph is minimal if and only if g contains no paths of length three or more.

Proof

* let g be a ^{minimal} covering of a graph G & it contains a path of length 3.

To prove: g is minimal covering.

* let the path be,

$v_1, e_1, v_2, e_2, v_3, e_3, v_4$

* Removal of e_2 from g

Still g is a covering

* which is a contradiction for minimal covering

Theorem

A Covering $\mathcal{G}$ of a Graph is minimal if and only if $\mathcal{G}$ contains no paths of length three or more.

Proof

* Let $\mathcal{G}$ be a minimal covering of a graph G .

To prove: $\mathcal{G}$ contains no paths of length three or more.

* Let $\mathcal{G}$ contains a path of length three & it is

$$v_1 e_1 v_2 e_2 v_3 e_3 v_4.$$

* Removal of e_2 , still $\mathcal{G}$ is covering

* Which is a contradiction of minimal covering.

* $\mathcal{G}$ contains no paths of length three or more.

Conversely,

* If a Covering $\mathcal{G}$ contains no path of length three or more then all its components must be Star Graph.

* Removal of single edge from Star graph won't be a Covering

+ $\mathcal{G}$ is minimal covering.

Four color problem

Theorem: (i)

P.T. The vertices of every planar graph can be properly colored with five colors.

Proof

Refer the text book.

Unit 5 - Digraphs

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1 Elementary Concepts

Digraph or Oriented Graph: A graph G in which the edges are directed line segments or arcs.

If G is a digraph and $e = (u, v)$ is an arc from u to v , we say that the arc e is incident out of the vertex u and incident into the vertex v . Further, u is called the initial vertex and v the terminal vertex of the arc e . An arc whose terminal vertex coincides with the initial vertex is called a self-loop. Two arcs are called parallel or multiple arcs if they are mapped onto the same pair of vertices, which agree in their direction as well.

Out-degree and In-degree of a vertex: Let v be a vertex in a digraph G . The number of arcs going out of v is called the out-degree or out-valence $d^+(v)$ and the number of arcs incident into v is called the in-degree or in-valence $d^-(v)$ of v .

- Isolated vertex: A vertex v with $d^+(v) = d^-(v) = 0$
- Transmitter: A vertex v with $d^+(v) > 0, d^-(v) = 0$
- Receiver: A vertex v with $d^-(v) > 0, d^+(v) = 0$
- Carrier: A vertex: A vertex v with $d^+(v) = d^-(v) = 1$
- Pendent vertex: A vertex v with $d^+(v) + d^-(v) = 1$

A nonincreasing sequence of in-degrees/out-degrees of a digraph G is called an in-degree sequence/out-degree sequence of G .

Lemma 1 (Euler's Hand-shaking lemma). *The sum of in-degrees of the vertices equals the sum of out-degrees of the vertices in a digraph G and both the sums equal the number of arcs in it, that is*

$$\sum_i d^+(v_i) = \sum_i d^-(v_i) = \text{the number of arcs in } G \quad (1)$$

Proof. Let q be the the number of arcs in a digraph $G = (V, A)$, where V is the vertex set and A is the arc-set. Suppose that $b = \sum_i d^+(v_i)$ and $c = \sum_i d^-(v_i)$. An arc $e = (u, v)$ contributes 1 to the out-degree of u and one to in-degree v of v . Hence each arc contributes 1 to the sum b and 1 to the sum c . Therefore, $\sum_i d^+(v_i) = \sum_i d^-(v_i) = \text{the number of arcs in } G$. $\square$

Orientation: Given a digraph G , if we disregard the orientations of the arcs, we get a corresponding unique undirected graph G' . Conversely, given an undirected graph G' , a digraph G obtained by assigning each edge an arbitrary direction, is called an orientation of G' . An undirected graph may have different orientations.

Isomorphism: Two digraphs G_1 and G_2 are said to be isomorphic if

- the undirected graphs corresponding to G_1 and G_2 are isomorphic, and
- the directions of the corresponding arcs agree.

Isomorphic digraphs are indistinguishable if their labels are removed. While two digraphs with different in-degree sequence (or out-degree sequence) are not isomorphic.

2 Types of digraphs

Simple Digraph: A digraph without parallel arcs and self-loops

Balanced and Regular Digraphs: A digraph in which for each vertex v , the in-degree is equal to the out-degree is called a balanced digraph or isograph. A balanced digraph in which each vertex v has the in-degree and the out-degree as every other vertex is called a regular digraph.

Asymmetric Digraph: A digraph that has at most one arc exists between every pair of vertices, with self-loops allowed

Symmetric Digraph: A digraph in which every pair of vertices corresponds to a pair of oppositely directed arcs

Complete Digraphs: A simple digraph in which there is exactly one arc from every vertex to every other vertex is called a complete symmetric digraph. An asymmetric digraph in which there is exactly one arc between from every pair of vertices is called a complete asymmetric digraph, also called a complete tournament or simply tournament if there is no confusion.

Theorem 1. A complete symmetric graph with n vertices will have $\frac{n(n-1)}{2}$ arcs, while a tournament with n vertices will have $n(n-1)$ arcs, for $n \geq 2$.

3 Connectedness in Digraphs

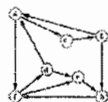
An alternating sequence of vertices and arcs in a digraph of the form:

$$v_0 e_1 v_1 e_2 v_2 \cdots v_{k-1} e_k v_k$$

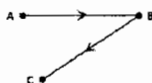
is called a directed path with initial vertex v_0 and terminus v_k . Each arc is oriented from the vertex preceding it to the vertex that follows it. The number of arcs in a directed path is its length. Similarly a directed circuit and its length can be defined.

Connected Digraphs

- **Strongly connected:** There exists at least one directed path from every vertex to any other vertex, that is every vertex is reachable from any other vertex.



- **Weakly connected/connected:** A digraph which is not strongly connected but its underlying undirected graph is connected.

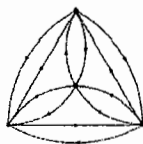


There is no directed path from C to A

Ex. 1 Draw a strongly connected digraph with three vertices.

Ex. 2 Draw a strongly connected complete symmetric digraph on four vertices.

Ans. In the following digraph G , there is exactly one edge directed from every vertex to every other vertex. So it is complete symmetric.



4 Tournaments and Strong connectedness:

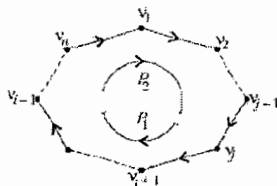
We recall that a digraph with no self-loops in which any two distinct vertices are joined by exactly one arc is a complete asymmetric digraph, also called a *tournament*. Thus a tournament is an orientation of a complete graph. The score of a vertex v in a tournament is the out-degree $d^+(v)$ of v . The out-degree sequence in a tournament is its score sequence.

Hamiltonian Graph: A directed circuit H in a digraph G is a circuit which visits each vertex exactly once is called a *Hamiltonian circuit*. A digraph G which has a Hamiltonian circuit is a *Hamiltonian digraph*.

Theorem 2. A tournament T is Hamiltonian if and only if it is strongly connected.

Proof. Let T have n vertices. If T is strongly connected, then it has a cycle of length n . Such a cycle is a Hamiltonian cycle, since it includes every vertex of T . Hence T is Hamiltonian.

Conversely, let T be Hamiltonian with Hamiltonian cycle $C = v_1v_2\dots v_nv_1$ as shown below:



Then given any v_i, v_j with $i \geq j$, in the vertex set of T , $v_jv_{j+1}\dots v_i$ is a path P_1 from v_j to v_i while $v_iv_{i+1}\dots v_{n-1}v_nv_1\dots v_{j-1}v_j$ is a path P_2 from v_i to v_j . Thus each vertex is reachable from any other vertex and so T is strongly connected. $\square$

Hamiltonian Path: A directed path in a digraph G which visits every vertex of G exactly once (A spanning path).

Theorem 3. Every complete tournament D has a directed Hamiltonian path.

The proof is by induction on the number n of vertices. If $n = 1$ the statement is vacuously satisfied. Let D consist of two $n = 2$ vertices. since D is complete, there exists exactly one arc between them, which spans D , and the statement holds good.

Assume that a tournament with $n-1$ vertices contains a directed Hamilton path, say $(v_1, v_2, \dots, v_{n-1})$ as shown below:

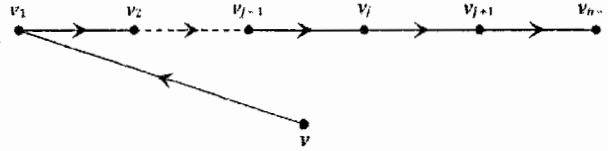


Consider a complete tournament D with n vertices and v be a vertex in D . Then the vertex-removal $D_1 = D - v$ of D is a subtournament of D with $n-1$ vertices. By induction hypothesis, D_1 contains a directed Hamilton path say but not belonging to D_1 .

We should prove that D contains a spanning path. Since D is a complete tournament with n vertices, the vertex v has a directed edge either to or from each of the other vertices $v_1, v_2, \dots, v_{n-1}$.

Note that $v \notin D_1$. We have three cases:

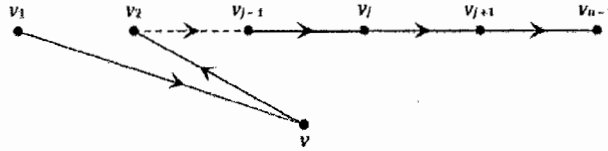
Case (a). (v, v_1) is an arc in D . That is the edge between v and v_1 is directed towards v_1 . Then the path $(v, v_1, v_2, \dots, v_{n-1})$ in D is a directed Hamilton path in D .



Case (b). Suppose the edge between v and v_1 is directed from v_1 towards v . Then we choose the arc (v, v_2) and

$$(v_1, v, v_2, \dots, v_{j-1}, v, v_j, v_{j+1}, \dots, v_{n-1})$$

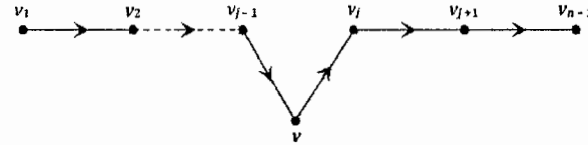
is a directed Hamilton path in D .



In general, choose the smallest j ($2 \leq j \leq n-1$) such that (v, v_j) is an arc in D . Since D is complete,

$$(v_1, v_2, \dots, v_{j-1}, v, v_j, v_{j+1}, \dots, v_{n-1})$$

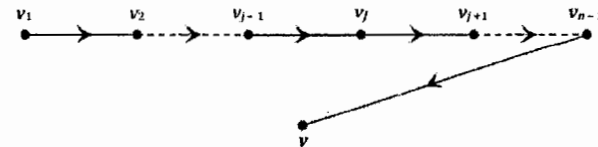
is a directed Hamilton path in D .



Case (c). There is no j ($1 \leq i \leq n-1$) such that the edge is not directed from v to v_j . In particular, the edge between v and v_{n-1} is directed from v_{n-1} to v . But then

$$(v_1, v_2, \dots, v_{n-1}, v)$$

is a directed Hamilton path in D .



5 Eulerian Digraphs

A closed directed walk which traverses every arc of a digraph G exactly once is called a directed **Eulerian circuit**. A digraph G is an Euler digraph if it contains an Eulerian circuit.

- Every Euler digraph is strongly connected (There exists at least one directed path from every vertex to any other vertex, that is every vertex is reachable from any other vertex in a strongly connected digraph)
- A strongly connected graph need not be Eulerian.

6 Necessary and sufficient condition to be Eulerian

Theorem 4. A digraph is Eulerian if and only if it is strongly connected and balanced.

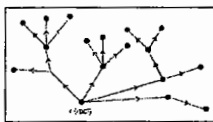
7 Arborescence

A digraph G is said to be an **arborescence** if

- G contains neither directed circuit nor semicircuit
- there exists exactly one vertex of in-degree zero (usually called the **root**)

Thus

- an arborescence is a rooted tree in which all the arcs are pointed away from the root r , where $d^-(r) = 0$
- In an arborescence, every vertex other than the root has the in-degree exactly one.



Theorem 5. A digraph is an arborescence if and only if there is a directed path from the root r to every other vertex.

Proof. Since the sum of the outdegrees of the vertices is equal to the number of arcs in a digraph G , an arborescence with n vertices can have at most $n - 1$ arcs. Therefore, the sum of in-degrees of all vertices in G is

$$d^-(v_1) + d^-(v_2) + \cdots + d^-(v_n) \leq n - 1.$$

Of the n terms on the left-hand side of this equation, only one is zero, others must all be positive integers. Therefore, they must all be 1's. Now since there are

exactly $n - 1$ vertices of in-degree one and one vertex of in-degree zero, digraph G has exactly $n - 1$ arcs. Since G is also circuitless, it must be connected, and hence a tree. $\square$

7 Spanning Arborescence

A spanning tree in an n -vertex connected digraph, analogous to a spanning tree in an undirected graph, consists of $n - 1$ directed edges.

A spanning arborescence in a connected digraph is a spanning tree that is an arborescence.

Theorem 6. *Every Eulerian digraph contains a spanning arborescence.*

Proof. Let G be an Eulerian digraph with n vertices and m arc. Then G contains a directed Euler circuit, which traces out each arc exactly once, say $W = (e_1, e_2, \dots, e_m)$ which starts and ends at a vertex v (that is, v is the initial vertex of e_1 and the terminal vertex of e_m). Among the m edges in W there are $n - 1$ arcs that enter each of $n - 1$ vertices, other than v , for the first time. The subdigraph g of these $n - 1$ directed edges together with the n vertices is a spanning arborescence of G , rooted at vertex v . In fact, in the subgraph g , vertex v is of in-degree zero, and every other vertex is of in-degree one; for g includes exactly one arc going to each of the $n - 1$ vertices, and no edge going to v . Moreover, the way g is defined in W , g is connected and contains $n - 1$ directed edges. Therefore, g is a spanning arborescence in G and is rooted at v . $\square$

8 Shortest Path Problem-Dijkstra's Algorithm

Consider a weighted network (V, E, W) with

- vertex set V , directed-edge set E and
- the weight set W specifying weights $c_{ij} \geq 0$, also called *costs* for the arcs $(i, j) \in E$.

Given a starting node $a \in V$, the problem is to determine the shortest path from a to all the other nodes in the network. The weight of a path is the sum of the weights of the path traversed.

In a networking or telecommunication applications, Dijkstra's algorithm has been used for solving the min-delay path problem (which is the shortest path problem). For example,

- In data network routing, the goal is to find the path for data packets to go through a switching network with minimal delay.
- In plant and facility layout, robotics, transportation, and VLSI design

9 General Description of Dijkstra's Algorithm

Suppose we wish to find the length $l(a, z)$ of the shortest path from a starting node a to another vertex z in a connected weighted graph with arc-weights $c_{ij} > 0$.

Denote by a the starting node. Dijkstra's algorithm starts by assigning some initial values for the distances from node a to every other node in the network. It operates in steps, where at each step the algorithm improves the distance values. At each step, the shortest distance from node a to another node is determined. Each node x is characterized by its status label (Permanent or Temporary) and the distance value $l(a, x)$ from a .

- If $l(a, x)$ is equal to the shortest distance to node a , x is labeled *permanent*, otherwise *temporary*.
- Encircle the Permanent vertices.

10 Step-wise Procedure

Step 1: Assign $l(a, a) = 0$, labeled as permanent (p) and $l(x) = \infty$ for all other nodes x , labeled as temporary (t).

Step 2: Choose the nodes (temporary) adjacent to a , say $a_1, a_2, \dots$ and find their distances through in terms of their weights from a . Then choose the node a_1 which corresponds to the least distance from a and change its status as permanent (p). This is now the *current* node. In case of a tie, select any one of the nodes for permanent labeling.

Step 3: If all nodes that can be reached from node a have been permanently labeled, then stop and we are done. If we cannot reach any temporary labeled node from the current node, then all the temporary labels become permanent and then we are done. Otherwise, go to Step 2.

If x_i is a current node and the distance value $l(a, x_j)$ of node x_j is updated as follows

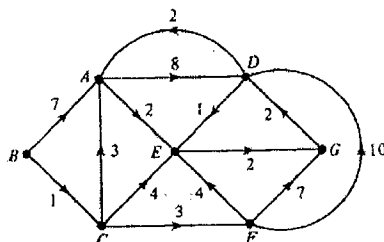
$$\text{new } l(a, x_j) = \min \{ \text{existing } l(a, x_j), l(a, x_i) + c_{ij} \}.$$

The node once becomes permanent, cannot be used again.

Step 4: Repeat Step 2 and Step 3 until the destination vertex z gets permanent label.

The union of the shortest paths from a to all other nodes which are accessible from it is the *shortest-distance arborescence* rooted at a .

Find the shortest path from the vertex B to the vertex G in the following weighted graph:



Solution.

Step 1: Assign $l(B,B) = 0$, labeled as permanent. For the remaining (temporary) vertices assign

$$l(B,A) = l(B,C) = l(B,D) = l(B,E) = l(B,F) = l(B,G) = \infty.$$

Step 2: The (temporary) nodes adjacent to B are C and A . Now

$$\text{new } l(B,C) = \min\{\infty, 0 + 1\} = 1,$$

$$\text{new } l(B,A) = \min\{\infty, 0 + 7\} = 7.$$

Of these two, Node C has the smallest distance value. Therefore node C becomes permanent (fixed), while A remains temporary.

Step 3: The (temporary) nodes adjacent to C are A , E and F . Now

$$\text{new } l(B,A) = \min\{\infty, 0 + 1 + 3, 0 + 7\} = 4,$$

$$\text{new } l(B,E) = \min\{\infty, 0 + 1 + 4\} = 5,$$

$$\text{new } l(B,F) = \min\{\infty, 0 + 1 + 3\} = 4.$$

Of these, Nodes both A and F have the smallest distance values. We can choose either of them. Now we choose F as the current permanent node, while A remains temporary.

Step 4: The (temporary) nodes adjacent to F are E , G and D . Now we can reach these through F with

$$\text{new } l(B,E) = \min\{\infty, 0 + 1 + 3 + 4\} = 8,$$

$$\text{new } l(B,G) = \min\{\infty, 0 + 1 + 3 + 7\} = 11,$$

$$\text{new } l(B,D) = \min\{\infty, 0 + 1 + 3 + 10\} = 14.$$

Of these, Node E has the smallest distance value. But A is not labeled permanent in Step 2 with $l(B,A) = 4$, which is smaller than $l(B,E) = 8$. Hence we choose A as the current permanent node, while G , E and D remain temporary.

Step 5: The (temporary) nodes adjacent to A are D and E . Now we can reach each of these from B through A with the following distances

$$\text{new } l(B, D) = \min\{\infty, 0 + 1 + 3 + 8, 0 + 7 + 8\} = 12,$$

$$\text{new } l(B, E) = \min\{0 + 1 + 3 + 2, 7 + 2\} = 6.$$

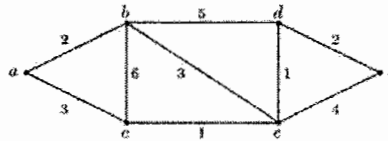
Of these, Node E has the smallest distance value from B . But already $l(B, E) = 5$ from Step 3. Thus we choose E as the current permanent node with $l(B, E) = 5$, while G and D remain temporary.

Step 6: The (temporary) node adjacent to E is G , which is the destination vertex and

$$\text{new } l(B, G) = \min\{\infty, 0 + 1 + 3 + 2 + 2, 0 + 1 + 4 + 2\} = 7.$$

This is the final shortest path from B to G with the weight 7.

Use Dijkstras algorithm to find the cost of the cheapest path between a and z in the following weighted graph.:



Solution. The iterations of Dijkstra's algorithm are described in the following table:

S	$L(a)$	$L(b)$	$L(c)$	$L(d)$	$L(e)$	$L(z)$
$\emptyset$	0	∞	∞	∞	∞	∞
$\{a\}$		2	3	∞	∞	∞
$\{a, b\}$			3	7	5	∞
$\{a, b, c\}$				7	4	∞
$\{a, b, c, e\}$				5		8
$\{a, b, c, e, d\}$						7
$\{a, b, c, e, d, z\}$						

At the last iteration, $z \in S$ and $L(z) = 7$. We conclude that the cheapest path from a to z has a cost of 7.

Find the shortest path from the vertex a to the vertex z in the following weighted graph:

